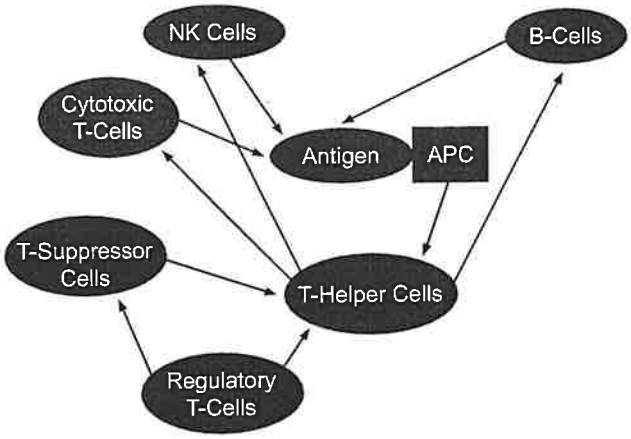
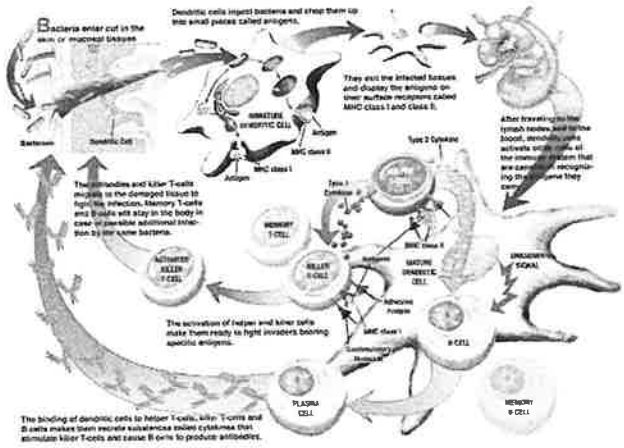
used to stimulate the innate immunity. These cytokines increase the production and activity of natural killer cells and cytotoxic T-cells to destroy the antigen. T-helper cells also produce interleukin 4 and 10 to increase B-cell activity and production of antibodies against the antigen. As soon as the antigen is recognized by the macrophage, it is attacked.

Shortly after attack by the APC, the APC sends out signals (IL-1) to T-helper cells, which then send out signals (IL-2 & IFN) to bring in natural killer cells and cytotoxic T-cells to also attack the antigen. As soon as messages are sent to T-helper cells from the APC, signals are also sent (IL-4 & IL-10) to B-cells to produce antibodies. After a few days, if the antigen has not been destroyed by the TH-1 system, antibodies are produced by B-cells and start attaching to the antigen. Once the antigen is tagged with an antibody protein, it is easily recognized and destroyed by natural killer cells and cytotoxic T-cells.



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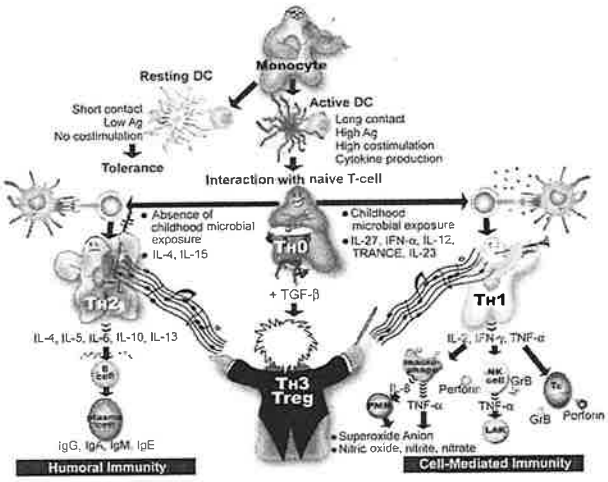


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**Regulation of the Immune System by T-Reg Cells**

## Infectious Agents Associated with Thyroid Disease

"Classical autoimmune thyroid disease (AITD) has been shown to be associated with a variety of infectious agents, while infections of the thyroid gland have been shown to be associated with thyroid autoimmune phenomena. However, the causative role of infectious agents in AITD has not been definitively demonstrated in humans, although AITD can be induced in experimental animals by certain viral infections. Infectious agents may induce thyroid autoimmunity by a variety of diverse mechanisms, such as inducing modifications of self-antigens, mimicking self-molecules, inducing polyclonal T-cell activation (for example by superantigens), altering the idiotypic network, forming immune complexes, and inducing expression of MHC molecules on thyroid epithelial cells. While indirect data suggesting involvement of the infecting organisms in the pathogenesis of human AITD is abundant, only a limited number of studies have employed direct approaches."[246]

These infections are not a conclusive list and in many cases a specific infectious agent cannot be attributed as the sole cause of an autoimmune response.247 Autoimmune thyroid disorders are triggered by a complex set of phenotypic influences with susceptible genotypes. However, the field of immunology has certainly identified specific antigens that promote thyroid dysfunction.248 Additionally, management of the infectious agent does not typically turn off the autoimmune response after it has been triggered, but in certain cases, it may dampen the intensity of the reaction.

# Infectious Agents Associated with Thyroid Disease Mechanism:

## Cross-reaction between thyroid-stimulating hormone receptors and infectious agents

- Rubella
- Rubeola
- EBV
- Retrovirus
- Influenza B virus
- Coxsackievirus
- *Yersinia*

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

### Rubella and Rubeola

Rubella is known as "German measles" and rubeola is known just as "measles." They are viruses. "The name 'rubella' is derived from the Latin, meaning *little red*. Rubella is also known as German measles because German physicians first described the disease in the mid-eighteenth century. This disease is often mild and attacks often pass unnoticed. The disease can last one to three days. Children recover more quickly than adults. Infection of the mother by the rubella virus during pregnancy can be serious; if the mother is infected within the first 20 weeks of pregnancy, the child may be born with congenital rubella syndrome (CRS), which entails a range of serious incurable illnesses."

"Rubella is a common childhood infection usually with minimal systemic upset, although transient arthropathy may occur in adults. Serious complications are very rare. Apart from the effects of transplacental infection on the developing fetus, rubella is a relatively trivial infection."

"Acquired (ie, not congenital) rubella is transmitted via airborne droplet emission from the upper respiratory tract of active cases. The virus may also be present in the urine, feces, and on the skin. There is no carrier state; the reservoir exists entirely in active human cases. The disease has an incubation period of 2 to 3 weeks."

"In most people, the virus is rapidly eliminated. However, it may persist for some months postpartum in infants surviving the CRS. These children are a significant source of infection to other infants and, more importantly, to pregnant female contacts."

"After an incubation period of 14-21 days, the primary symptom of rubella virus infection is the appearance of a rash (exanthema) on the face that spreads to the trunk and limbs and usually fades after three days. Other symptoms include low-grade fever, swollen glands (postcervical lymphadenopathy), joint pains, headache, and conjunctivitis.[249] The swollen glands or lymph nodes can persist for up to a week, and the fever rarely rises above 38°-C (100.4°-F). The rash disappears after a few days with no staining or peeling of the skin. Forchheimer's sign occurs in 20% of cases, and is characterized by small, red papules on the area of the soft palate. Rubella can affect anyone of any age and is generally a mild disease, rare in infants or those over the age of 40. The older the person is, the more severe the symptoms are likely to be. Up to two-thirds of older girls or women experience joint pain or arthritic type symptoms with rubella. The virus is contracted through the respiratory tract and has an incubation period of 2 to 3 weeks. During this incubation period, the carrier is contagious but may show no symptoms." (Please refer to News-Medical.net [www.news-medical.net/health/what-is-Rubella.aspx] for more information.)

"Rubella virus–specific IgM antibodies are present in people recently infected, but these antibodies can persist for over a year, and a positive test result needs to be interpreted with caution.[250] The presence of these antibodies along with, or a short time after, the characteristic rash confirms the diagnosis.[251] There is no specific treatment for rubella; however, management is a matter of responding to symptoms to diminish discomfort."

Several studies have demonstrated associations with rubella and thyroid dysfunction.[249,252-253] Listed in the following two paragraphs are some additional studies.

In one study, "201 deaf adolescents with congenital rubella syndrome and 83 age-matched deaf control subjects were evaluated for the presence of organ-specific antibodies directed against thyroid microsomes, thyroglobulin, pancreatic islets, adrenal cortex, and gastric parietal cells. Positive thyroid microsomal or thyroglobulin antibodies were found in 23.3% (47/201) of the rubella group and in 12.0% (10/83) of control subjects. Nine of 46 (19.6%) in the rubella group and two of nine (22.2%) control subjects with thyroid autoimmunity had thyroid gland dysfunction as indicated by elevated serum TSH concentrations. Neither islet cell nor adrenal cortical antibodies were detected in any subject tested; parietal cell antibodies were detected in 5.5% (8/146) of those in the rubella group and 8.8%

(6/68) of control subjects tested, but occurred most frequently in subjects with thyroid autoimmunity (6/36, 16.7% vs 8/178, 4.5%; P less than 0.05)." The study recommended "that all patients with congenital rubella syndrome be screened for thyroid autoimmunity and that those with positive antibody titers be evaluated for the presence of thyroid dysfunction."[254]

In a case study, "a 40-year-old housewife was referred to [the] hospital for evaluation of a thyrotoxic state. A month after rubella infection, she developed heat intolerance. Physical examination revealed a hard and non-tender goiter with no signs or symptoms of inflammation. Free thyroid hormones were high and TSH was undetectable. 123I-thyroidal uptake was suppressed. Anti-thyroglobulin and anti-microsomal antibodies were negative throughout the course. A serologic study revealed high levels of anti-rubella antibodies. After a month without any treatment, she became euthyroid. Free thyroid hormones and TSH gradually became normal and the antibodies to rubella decreased. 123I-thyroidal uptake increased. From the clinical course, the patient was diagnosed as having silent thyroiditis." The study suggested "that viral infection such as rubella could cause the development of silent thyroiditis."[255]

### Epstein-Barr Virus

The Epstein-Barr virus (EBV) is a virus of the herpes family. This is also called human herpesvirus 4 (HHV-4) and is one of the most common viruses in humans. Most people become infected with EBV, which is often asymptomatic, but infection commonly causes infectious mononucleosis (also known as glandular fever).

Epstein-Barr virus occurs worldwide. Most people become infected with EBV sometime during their lives and therefore gain adaptive immunity, preventing repeated sickness from reinfection through EBV antibodies. In the United States, as many as 95% of adults between 35 and 40 years of age have been infected. "Infants become susceptible to EBV as soon as maternal antibody protection (present at birth) disappears. Many children become infected with EBV, and these infections usually cause no symptoms or are indistinguishable from the other mild, brief illnesses of childhood. In the United States and in other developed countries, many persons are not infected with EBV in their childhood years. When infection with EBV occurs during adolescence or young adulthood, it causes infectious mononucleosis 35% to 50% of the time." (Please refer to the OC Integrative Medical Center [www.ocimc.com/Lyme-Disease-EBV.html] for more information.)

"It is postulated that following natural infection with EBV, the virus executes some or all of its repertoire of gene expression programs to establish a persistent infection. Given the initial absence of host immunity, the lytic cycle produces large amounts of viruses to infect other (presumably) B-lymphocytes within the host. The latent programs reprogram and subvert infected B-lymphocytes to proliferate and bring infected cells to the sites at which the virus presumably persists. Eventually, when host immunity develops, the virus persists by turning off most (or possibly all) of its genes, only occasionally reactivating to produce fresh virions. A balance is eventually struck between occasional viral reactivation and host immune surveillance removing cells that activate viral gene expression. The site of persistence of EBV may be bone marrow."[256]

Several studies have linked Epstein-Barr virus to the activation of autoimmune thyroid dysfunction and thyroiditis.[257-259] "Elevated titers of antibodies against different herpes virus antigens have been reported in some immunodeficient and systemic autoimmune disorders." A study was conducted "to examine if herpes simplex virus (HSV), Epstein-Barr virus (EBV), and cytomegalovirus (CMV) IgG and IgM antibodies are detected more frequently in children with autoimmune thyroid disease (AITD) compared to controls. Thirty-four children with AITD, aged 9.62 +/- 2.35 years, and 31 matched controls, aged 9.24 +/- 2.98 years, were studied." The results found that the percentage of EBV IgG+ children with AITD was statistically higher than "the percentage of EBV IgG+ controls (82.35% versus 51.61%, P = 0.008). The percentage of EBV IgG+ children with AITD and hypothyroidism was statistically higher than the percentage of EBV IgG+ children with AITD, without hypothyroidism (100% versus 70%, P = 0.024). No other statistically significant differences were observed in HSV-1+2, and CMV IgG or IgM antibodies between the subgroups of children studied." The conclusion of the study was that "EBV seroprevalence is higher in children with AITD compared to controls and the underlying pathology remains to be elucidated."[260]

"The frequency of cell precursors producing Ig of different classes and Ag-binding activities were determined, using EBV-infection and limiting dilution assays, in healthy subjects and patients with autoimmune disease. A large proportion of circulating B cells from healthy subjects were committed to the production of IgM antibodies that were polyreactive and bound a variety of self- and exogenous Ag, i.e., IgG Fc fragment, ssDNA, thyroglobulin, thyroid microsomal Ag, insulin, and tetanus toxoid. Similar frequencies of these polyreactive antibody-producing cells were found in patients with Hashimoto's disease and SLE. In contrast, significantly higher frequencies of cell precursors producing monoreactive IgG autoantibodies to thyroid Ag (thyroglobulin and thyroid microsomal Ag) and ssDNA were found in Hashimoto's disease and SLE patients, respectively. Calculation of the Kd revealed that monoclonal polyreactive antibodies were in general low affinity (Kd, 10(-3) to 10(-7) mol/liter), whereas monoclonal monoreactive autoantibodies were high affinity (Kd, 10(-9) to 10(-11) mol/liter). The detected frequency and high affinity of the monoreactive autoantibodies in Hashimoto's disease and SLE patients were comparable to those of anti-tetanus toxoid and anti-insulin IgG mAb produced by B-cell clones from vaccinated healthy subjects and insulin-treated patients with IDDM, respectively. These findings support the hypothesis that the autoimmune B-cell repertoire in patients with organ-specific and systemic autoimmunity is shaped by Ag-driven responses rather than merely reflecting a polyclonal B cell activation."[261]

A case study of "a unique 16-year old female patient presented after acute Epstein-Barr virus (EBV) infection with severe primary hypothyroidism. Her thyroid test results were thyrotropin level of 198 mU/L (normal, 0.4-4 mU/L), free thyroxine [FT(4)], 2.5 pmol/L (normal, 10-25 pmol/L), total triiodothyronine (TT(3)) > 19.5 nmol/L (normal, 1.3-2.7 nmol/L), and free triiodothyronine (FT(3)), 0.77 pmol/L (normal, 3.3-6.3 pmol/L). She had high titers of thyroglobulin and thyroid peroxidase autoantibodies. In vitro triiodothyronine (T(3))-binding measured by radioimmunoprecipitation was 86% (normal, up to 8.5%) and thyroxine (T(4))-binding 8.2% (normal, 6.4%). Serum immunoglobulin G (IgG) absorption, achieved by protein-G Sepharose beads, decreased TT(3) toward normal. Levothyroxine treatment normalized the low baseline FT(4) and FT(3) values, and suppressed TSH to normal. However, TT(3) remained highly elevated and returned to normal after 20 months, while T(3)binding gradually decreased. Thus, her severe hypothyroidism was masked by this unusual phenomenon. Thirty-four patients with EBV infection (15 with acute disease and 19 with previous infection) were tested for thyroid hormone levels. EBV antibodies (early antigen immunoglobulin M [IgM] and IgG and anti-Epstein-Barr virus nuclear antigen [EBNA] IgG) were measured by enzyme-linked immunosorbent assay (ELISA). In 15 patients with acute EBV, the mean TT(3) level was 2.47 +/- 0.39 nmol/L (5 had TT(3) values above normal) compared to a mean TT(3) of 1.70 +/- 0.53 nmol/L in 19 subjects with previous infection ($p < 0.0005$; only 1 had a TT(3) result above normal), with no differences in FT(4) and TSH concentrations between the two groups." It was concluded that acute EBV infection may be associated with transient mild to severe TT(3) elevation as a result of assay interference by anti-T(3) autoantibodies."[262]

### Retrovirus

Retrovirus "is an RNA virus (human T-lymphotropic virus) that is replicated in a host cell via the enzyme reverse transcriptase to produce DNA from its RNA genome. The DNA is then incorporated into the host's genome by an integrase enzyme. The virus thereafter replicates as part of the host cell's DNA. Retroviruses are enveloped viruses that belong to the viral family *Retroviridae*. The virus itself stores its nucleic acid in the form of a mRNA (including the 5'cap and 3'PolyA inside the virion) genome and serves as a means of delivery of that genome into cells it targets as an obligate parasite and constitutes the infection. Once in the host's cell, the RNA strands undergo reverse transcription in the cytosol and are integrated into the host's genome, at which point the retroviral DNA is referred to as a provirus. It is difficult to detect the virus until it has infected the host."

"When retroviruses have integrated their own genome into the germ line, their genome is passed on to a following generation. These endogenous retroviruses (ERVs), contrasted with exogenous ones, now make up 5-8% of the human genome. Most insertions have no known function and are often referred to as 'Junk DNA.' However, many endogenous retroviruses play important roles in host biology, such as control of gene transcription, cell fusion during placental development in the course of the germination of an embryo, and resistance to exogenous retroviral infection. Endogenous retroviruses have also received special attention in the research of related immunology-related pathologies,

such as autoimmune diseases like autoimmune thyroid, although endogenous retroviruses have not yet been proven to play any causal role in this class of disease."

"While transcription was classically thought to only occur from DNA to RNA, reverse transcriptase transcribes RNA into DNA. The term 'retro' in retrovirus refers to this reversal (making DNA from RNA) of the central dogma of molecular biology. Reverse transcriptase activity outside of retroviruses has been found in almost all eukaryotes enabling the generation and insertion of new copies of retrotransposons into the host genome. These inserts are transcribed by enzymes of the host into new RNA molecules that enter the cytosol. Next, some of these RNA molecules are translated into viral proteins. For example, the *gag* gene is translated into molecules of the capsid protein, the *pol* gene is transcribed into molecules of reverse transcriptase, and the *env* gene is translated into molecules of the envelope protein. It is important to note that a retrovirus must "bring" its own reverse transcriptase in its capsid otherwise it is unable to utilize the enzymes of the infected cell to carry out the task, due to the unusual nature of producing DNA from RNA."

Industrial drugs that are designed as protease and reverse transcriptase inhibitors can quickly be proved ineffective because the gene sequences that code for the protease and the reverse transcriptase can undergo many substitutions. These substitutions of nitrogenous bases, which make up the DNA strand, can make either the protease or the reverse transcriptase difficult to attack. The amino acid substitution enables the enzymes to evade the drug regimens because mutations in the gene sequences can cause physical or chemical change that makes them harder to detect by the drug. When the drugs that are supposed to attack enzymes—such as protease—are designed, the manufacturers target specific sites on the enzyme. One way to attack these targets can be through hydrolysis of molecular bonds, which means that the drug will add molecules of $H_2O$ (water) to specific bonds. By adding molecules of water at a site on the virus, the drug breaks the previous bonds that were linked to each other. If several of these breaks occur, the result can lead to lysis, the death of the virus. "Because reverse transcription lacks the usual proofreading of DNA replication, a retrovirus mutates very often. This enables the virus to grow resistant to antiviral pharmaceuticals quickly, and impedes the development of effective vaccines and inhibitors for the retrovirus."

Several studies have linked the associations with retrovirus with autoimmune thyroid disorders.[263-264] "Human T-lymphotropic virus type I (HTLV-I) protein and messenger RNA (mRNA) for HTLV-I were examined in thyroid tissues from two patients with Hashimoto's thyroiditis and serum anti-thyroid antibody. The virus envelope protein and signals for the mRNA were detected in many of the follicular epithelial cells of the thyroid tissue from one of the patients, respectively, by immunohistochemistry and in situ hybridization. PCR-Southern blotting revealed the presence of HTLV-I DNA in the thyroid tissue, in which the viral protein and mRNA were detected, although no virus particles were found in the epithelial cells by electron microscopy. HTLV-I virus was not present in the thyroid tissue from the second patient. The present findings suggest that infection of thyroid tissue with HTLV-I is associated with the pathogenesis of Hashimoto's thyroiditis in some patients."[265]

A study was conducted to "describe two HTLV-I virus carriers who have biopsy-proven Hashimoto's thyroiditis. The first patient, a 64-year-old female, has had goiter and hypothyroidism since the age of 56. The second patient, a 66-year-old male, developed hyperthyroidism and goiter at the age of 44, but at present he is hypothyroid. Both patients are positive for anti-thyroid antibodies and anti-HTLV-I virus antibody. Findings of the thyroid biopsy specimens were consistent with Hashimoto's thyroiditis. These data suggested that Hashimoto's thyroiditis develops in HTLV-I carriers who have no clinical evidence of HAM/TSP."[266]

### Influenza B Virus

"Influenzavirus B is a genus in the virus family Orthomyxoviridae. The only species in this genus is called 'Influenza B virus.' Influenza B viruses are only known to infect humans and seals, giving them influenza. This limited host range is apparently responsible for the lack of Influenzavirus B–caused influenza pandemics in contrast with those caused by the morphologically similar influenzavirus A, as both mutate by both antigenic drift and reassortment."

Influenza, commonly referred to as the flu, is an infectious disease caused by RNA viruses of the family of Orthomyxoviridae (the influenza viruses), which affects birds and mammals. The name influenza comes from the Italian influenza, meaning "influence" (Latin: influentia). "The most common symptoms of the disease are chills, fever, sore throat, muscle pains,

severe headache, coughing, weakness, and general discomfort. Fever and coughs are the most frequent symptoms. In more serious cases, influenza causes pneumonia, which can be fatal, particularly for the young and the elderly. Although it is often confused with other influenza-like illnesses, especially the common cold, influenza is a much more severe disease than the common cold and is caused by a different type of virus. Influenza may produce nausea and vomiting, particularly in children, but these symptoms are more common in the unrelated, gastroenteritis, which is sometimes called 'stomach flu' or '24-hour flu.'"

"Typically, influenza is transmitted through the air by coughs or sneezes, creating aerosols containing the virus." Influenza can also be transmitted by bird droppings, saliva, nasal secretions, feces, and blood. Infection can also occur through contact with these body fluids or through contact with contaminated surfaces. "Airborne aerosols have been thought to cause most infections, although which means of transmission is most important is not absolutely clear. Influenza viruses can be inactivated by sunlight, disinfectants, and detergents. As the virus can be inactivated by soap, frequent hand washing reduces the risk of infection." (Please refer to OmniCanna Health Solutions [www.omnicanna.com/products/critical-ailments/influenza] for more information.)

It appears the leading theories associated with influenza and autoimmune thyroid suggests that the elevation of TH-1 cytokines can contribute to the phenotypic expression of autoimmune thyroid.[267-270]

### Coxsackie Virus

"Coxsackie A virus is a cytolytic coxsackie virus of the *Picornaviridae* family, an enterovirus (a group containing the polioviruses, coxsackieviruses, and echoviruses). The most well known Coxsackie A disease is hand, foot, and mouth disease (unrelated to hoof-and-mouth disease) a common childhood illness which affects mostly children aged 5 or under, often produced by Coxsackie A16. In most cases infection is asympotomatic or causes only mild symptoms. In others, infection produces short-lived (7-10 days) fever and painful blisters in the mouth (a condition known as herpangina), on the palms and fingers of the hand, or on the soles of the feet. There can also be blisters in the throat, or on or above the tonsils. Adults can also be affected. The rash, which can appear several days after high temperature and painful sore throat, can be itchy and painful, especially on the hands/fingers and bottom of feet."

Coxsackie B viruses also cause infectious myocarditis, infectious pericarditis, pleurodynia, and immune complex-mediated glomerulonephritis. "Treatment is dependent on the disease process initiated by the virus. There is no known cure or vaccine against the coxsackie."

Several studies have linked possible infectious mechanisms of coxsackie virus and autoimmune thyroid disorder such as Hashimoto's and Graves' disease. "Graves' disease (GD) is an autoimmune thyroid disorder that is associated with the human leucocyte antigens HLA-DR3 and DQA1* O501 in Caucasians." A research study "explored the possibility that some patients with certain HLA specificities develop anti-HLA antibodies, which are correlated with environmental factors that may contribute to the development of GD. We studied 40 GD patients and 157 healthy individuals (controls). Serology was used to type HLA-A, -B, -Cw, and -DR antigens. The frequencies of these antigens in relation to lymphocytotoxic anti-HLA-A-B-Cw-DR antibodies and two environmental factors (*Yersinia enterocolitica* and coxsackie B virus) were determined. The frequencies of HLA-B15, -B21 and DR3 antigens were increased, whereas HLA-DR5 antigen was decreased in GD patients. A significant association between HLA-DR3 antigen and lymphocytotoxic antibodies was observed, i.e., IgGs from GD patients were cytotoxic to HLA-DR3+ normal B cells. Following absorption with *Yersinia enterocolitica* or Coxsackie-B-virus, only Coxsackie-B virus completely inhibited the lymphocytotoxic reactions against HLA-DR3+ B cells. Besides confirming the association of HLA-DR3 with GD, the study also suggested the role of Coxsackie-reactive HLA-DR3 antibodies as contributing factors to the pathogenesis of the disease."[271]

A study of "15 untreated patients aged 16 to 55 with diffuse toxic goiter had morphologically confirmed diagnosis of the presence of antibodies to Coxsackie virus (group B); titers of group A (1:128 and more) were revealed in 3 of them. It indicated recent (not more than 2 yrs.) Coxsackie virus infection. Such titers were absent in the control group (18 healthy persons). Difference in the study and control groups was statistically significant (p less than 0.05). The role of Coxsackie virus in the breakdown of natural tolerance to autologous thyroid tissue was discussed" as a possibility.[272]

### Yersinia

"*Yersinia* is a genus of bacteria in the family Enterobacteriaceae. *Yersinia* are gram-negative rod shaped bacteria, a few micrometers long and fractions of a micrometer in diameter, and are facultative anaerobes. Some members of *Yersinia* are pathogenic in humans; in particular, *Y. pestis* is the causative agent of the bubonic plague. Rodents are the natural reservoirs of *Yersinia*; less frequently other mammals serve as the host. Infection may occur either through blood (in the case of *Y. pestis*) or in an alimentary fashion, occasionally via consumption of food products (especially vegetables, milk-derived products and meat) contaminated with infected urine or feces. *Y. pestis* is the causative agent of plague. The disease caused by *Y. enterocolitica* is called Yersiniosis. *Yersinia* may be associated with Crohn's disease, an inflammatory autoimmune condition of the gut." Consistent with Yesinia's unusual ability to thrive at low temperatures, it becomes a threat with unrefrigerated foods. "*Yersinia* is implicated as one of the causes of reactive arthritis worldwide."

"Hashimoto's thyroiditis is a common autoimmune disorder of the thyroid gland. It has been linked to infections with hepatitis C, EBV, HTLV-1, and *Yersinia enterocolitica*. As parvovirus B19 has been associated with a wide spectrum of autoimmune diseases", a study "investigated the potential role of B19 infection in inducing Hashimoto's thyroiditis. Serum samples derived from 73 children and adolescents with Hashimoto's thyroiditis and from 73 age-matched controls were included in the study. The mean age of disease manifestation was 10 y 7 mo. All samples were analyzed for the presence of viral DNA and for antibodies against VP1, VP2, and NS1 proteins. VP1- and VP2-specific antibodies were present in 38 patients (52%) and 43 controls (59%; N.S.). NS1-specific antibodies were detectable in 23 patients (32%) and 19 controls (26%; N.S.). Parvovirus B19 DNA was detectable in 9 patients (12%) and 2 controls (3%; $p < 0.03$), indicating recent B19-infection. A negative correlation between disease duration and the detection of viral DNA was seen. The mean disease duration in B19-DNA-positive patients was 6 mo, compared to 29 mo in the remainder ($p < 0.01$)." The study concluded that "there is strong evidence that acute parvovirus B19 infections are involved in the pathogenesis of some cases of Hashimoto's thyroiditis."[273]

"Recent reports have suggested that *Yersinia enterocolitica* proteins encoded by a 72-kilobase virulence plasmid (known as release proteins and now identified as YOP2-5) are antigens recognized specifically by patients with Graves' disease and of potential etiological importance in this disorder. To examine this hypothesis [researches] evaluated immune responses to YOP in patients with autoimmune thyroid disease and in normal controls. Humoral responses to *Yersinia* were assessed using Western blots of crude *Y. enterocolitica* membrane proteins, *Yersinia* release proteins (YOP2-5), and human thyrocyte membranes. Twenty-four of 25 Graves' and 10 of 18 Hashimoto's patients showed reactivity with the release proteins, primarily the 67-, 46-, 36-, and 25-kilodalton bands. However, 17 of 24 normal subjects also demonstrated serological reactivity to the release proteins, and the pattern of reactivity of these sera was similar to that in the thyroid patients. No correlation was noted between serological reactivity to the release proteins and thyroid hormone levels. Patients and controls with serological reactivity to YOP also showed reactivity with *Yersinia* membranes. In addition to the serological studies, cellular immune responses were determined by peripheral blood mononuclear cell proliferation assays. Cellular reactivity to the release proteins was present in four of five Graves' and both Hashimoto's patients tested, but also in two of six nonthyroid illness patients with serological immunity to the release proteins. Intrathyroidal lymphocytes obtained from two Graves' patients demonstrated marked proliferation in response to the release proteins. The results indicated that there was no unique pattern of serological reactivity against *Yersinia* membranes or the release proteins in patients with autoimmune thyroid diseases and suggest that any causal relationship between *Yersinia* infection and Graves' disease may be related to T-cell immunity." [274]

"Infections have been implicated in the pathogenesis of a number of autoimmune diseases, and *Yersinia enterocolitica* (YE) might play a role in the development of autoimmune thyroid disease (AITD). Clinical evidence in support of this hypothesis has been inconclusive." A study was conducted that "reasoned that looking earlier in the natural course of AITD might enhance chances of finding evidence for YE infection. Consequently, we determined seroreactivity against YE in subjects at risk of developing AITD, i.e. in 803 female relatives of AITD patients in self-proclaimed good health. As a comparison group we used 100 healthy women who

participated in a program for reference values. IgG and IgA antibodies to virulence-associated outer membrane proteins (YOPs) of YE were measured by a specific assay. Serum thyroid peroxidase antibodies (TPO-Ab) as indicators of AITD were considered to be positive at levels of> 100 kU/l. The prevalence of YOP IgG-Ab was higher in AITD relatives than in controls (40.1% vs. 24%, P = 0.002), and the same was true for YOP IgA-Ab (22% vs. 13%, P < 0.05). Of the 803 AITD relatives, 44 had an increased or decreased plasma TSH, and 759 were euthyroid as evident from a normal TSH; the prevalence of YOP-Ab did not differ between these three subgroups. TPO-Ab were present in 10% of controls and in 27% of the AITD relatives (P < 0.001). The prevalence of TPO-Ab in the euthyroid AITD relatives was not different between YOP IgG-Ab positive and negative subjects (23.3% vs. 24.7%, NS), nor between YOP IgA-Ab positive and negative subjects (21.2% vs. 24.9%, NS)." The study concluded that "healthy female relatives of AITD patients have an increased prevalence of YOP antibodies, which, however, is not related to the higher prevalence of TPO antibodies in these subjects. The findings suggest a higher rate of persistent YE infection in AITD relatives. Susceptibility genes for AITD may also confer a risk for YE infection."[275]

## Cytokines – Immune Cell Messengers

Cytokines are cellular protein messengers that are released by cells and allow communication between physiological systems. Cytokines have been classified into four major groups that include the colony stimulating factors (CSF), interferons (IFN), tumor necrosis factors (TFN), and interleukins (IL).

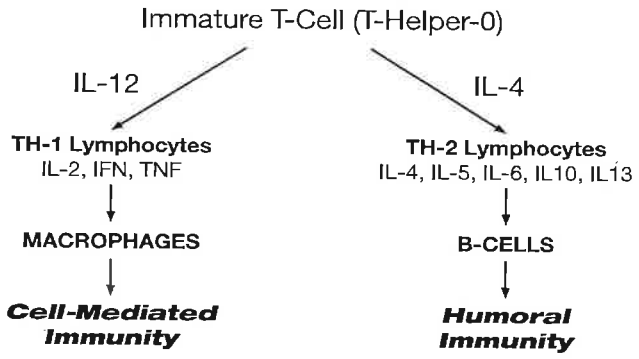
Colony stimulating factors (CSF) cause the proliferation and differentiation of immature bone marrow cells. The three main types of CSFs are granulocyte CSF (G-CSF), which stimulates the growth of mature granulocytes; macrophage CSF (M-CSF), which stimulates the growth of monocytes and macrophage colonies; and finally granulocyte-macrophages (GM-CSF), which stimulate the formation of both mature granulocytes and monocyte/macrophage colonies.

Interferons are glycoproteins produced in response to viral infections, immune stimulation, and/or chemical sensitivities. There are three main types of interferons, which include the alpha, beta, and gamma subsets. Interferon alpha is produced by leukocytes, interferon beta is produced by fibroblasts, and interferon gamma is produced by T-cells. Interferon alpha and beta play important roles in the immune response against viral infections. The alpha and beta interferons inhibit viral replication, increase the expression of MHC Class I molecules on virally infected cells, and activate NK cells. The impact of these influences allows the immune system to become more effective in recognizing viral infected cells and also by enhancing the production of cells such as cytotoxic T-cell and natural killer cells that can destroy viral infected cells.

Tumor necrosis factors can be divided into the alpha and beta subsets. TNF-alpha is produced by macrophages/monocytes, lymphocytes, and natural killer cells in response to antigen stimulation (viral, bacterial, parasitic, fungal) or by complement 5a stimulation. TNF-alpha stimulates the production of T- and B-cells and directly kills some human tumor cells. Clinically, chronic activation of TNF-alpha has been shown to promote anemia (normocytic normochromic anemia) because it suppresses hematologic progenitors of myeloid and erythrocyte cell lines and increases the rate of differentiation of the existing progenitors present. TNF-beta was formerly known as lymphotoxin because it exhibits direct cytotoxic effects for selected cells. TNF-beta is synthesized by helper and cytotoxic T-cells after exposure to an antigen.

The fourth group of cytokines are interleukins (IL), which are primarily produced by leukocytes and act on cells of the neuroendocrine-immune system. There are many different interleukins that are classified by number.

### Development of T-Helper Cells



### Interleukin-1

Interleukin-1 (IL-1) has many roles in immunity and inflammation and exists in two forms, which include the alpha and beta forms. It is produced and released with antigenic stimulation from immune cells (macrophages, T-lymphocytes, fibroblasts, monocytes). IL-performs many functions to modulate the immune response, which include promotion of T- and B-cell differentiation, the release of other cytokines (IL-2, IL-3, IL-4, interferon-gamma), the promotion of IL-2 receptors on helper T-cells, and the release of arachidonic acid and prostaglandins.

IL-1 acts on the central neuraxis to induce fever, sleep, and suppression of appetite. IL-1 stimulates the release of norepinephrine and serotonin in the hypothalamus and also modulates the release of adrenaline from the adrenal medulla. Interleukin-1 elevates the release of ACTH, beta endorphin, and the glucocorticoids, and has also been shown to exhibit stimulatory and inhibitory influences on the central release of TSH and LH, but has no influence on FSH, prolactin, or growth hormone. IL-1 beta has been shown to activate the HPA axis due to its excitatory nature on the paraventricular cells of the hypothalamus, from its influence on prostaglandin and serotonin production. IL-1 is also released from the median eminence of the hypophyseal portal vessels to regulate the release of ACTH from the pituitary directly. IL-1 not only exhibits central influences, but it peripherally influences the release of hormones by acting directly on the adrenal, gonadal, and thyroid gland.

It appears that IL-1 plays an important role in the activation and suppression of the immune system from its influences on central and peripheral glucocorticoid production. In the initial acute responses to antigens, the immune system activates the inflammatory stress response, which is later attenuated by the release of glucocorticoids. IL-1 may therefore play an important role via a negative feedback loop with the central attenuation of the immune response via glucocorticoid release and the initial activation of the HPA axis with acute antigenic stimulation.

### Interferon Gamma

Interferon gamma (IFN) is produced and released by T-lymphocytes after antigen exposure. It stimulates the differentiation and proliferation of B-cells and also increases the production of Interleukin-2 and its receptor sites on helper T-cells. The alpha and beta forms of interferon are produced in other immune cells to inhibit the activity of viral, bacterial, and protozoan infections.

### Interleukin-2

Interleukin-2 (IL-2) was originally named "T-cell growth factor" because of the role it plays in promoting the generation of T-cells and the differentiation of B-cells. It is produced by helper T-cells after the antigen presenting cells have encountered a foreign invader and have released IL-1. IL-2 also increases the cytotoxicity of macrophages.

IL-2 modulates the neuroendocrine system by modulating the release of acetylcholine from the magnocellular cells of the hypothalamus. IL-2 promotes the elevation of ACTH and beta-endorphin release from the anterior pituitary and stimulates the release of vasopressin from the PVN and thus increases water retention.

### Interleukin-4

Interleukin-4 (IL-4) is produced by activated helper T-cells and macrophages and stimulates the proliferation and differentiation of B-cells, mast cells, and bone marrow hematopoietic cells.

### Interleukin-6

Interleukin-6 (IL-6) is produced by macrophages, helper T-cells, B-cells, and fibroblasts. IL-6 upregulates the productions of IL-2 receptors on T-cells, promotes B-cell differentiation, and stimulates the parvocellular cells of the paraventricular nucleus of the hypothalamus to activate the HPA axis. IL-6 often times acts synergistically with IL-1 in its role in modulating the neuroendocrine-immune axis. It promotes the release of ACTH, GH, LH, and TRH release, but its most expressive influence on central structures appears to be the stimulation of glucocorticoids from its response on ACTH activation. The half-life of interleukin-6 is one hour.

### Interleukin-10

Interleukin-10 inhibits the production of interferon gamma, antigen presentation, and macrophage production of IL-1. IL-12 is considered to be the direct antagonist of IL-10 because it performs the opposite actions. IL-12 promotes macrophages and natural killer cells and increases the production of IFN gamma.

TH-1 cells release the pro-inflammatory cytokines such as IL-2, IFN-gamma and TNF-alpha, which are used

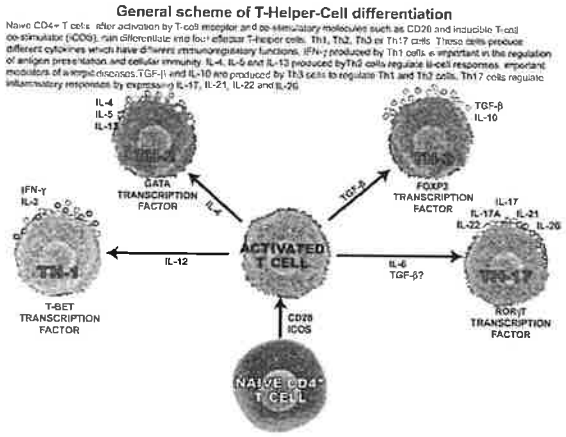
in cell-mediated immunity. TH-2 cells release anti-inflammatory cytokines such as IL-4, IL-5, IL-10 and IL-13, which are used in humoral immunity. In a normal immune response, there is balance between TH-1 and TH-2 systems, and these systems work in collaboration together to destroy antigens, cancer cells, and senescent cells. In autoimmune states, either the TH-1 or TH-2 system can become dominant and promote dysregulated attacks against self-tissue.

There are different variables that can promote dominance. It is suggested that the influence of certain cytokine production can be the causative factor. For example, physically overtraining increases IL-6, parasitic infections increase IL-4, allergies increase IL-4, the third trimester of pregnancy increases IL-10, -4, and -6. So you may have an individual who is overtraining during pregnancy with non-symptomatic parasitic infections and active allergies become TH-2 dominant. These cytokine shifts in conjunction with a certain genotype may trigger a TH-2–based autoimmune disease, such as multiple sclerosis.

On the other hand, you may have an individual who develops chronic hepatitis C infection, gets exposed to benzene compounds with an immune response against the chemical, has a susceptible TH-1 genotype, and will develop a TH-1–dominant autoimmune disease such as Graves' disease. Typically, once the genes turn on to express self-tissue attack, they rarely turn off. The goal of management is to decrease their cytokine dominance, improve immune coordination, and improve quality of life and physiology. At times, the modulation of various factors may be curative, but improvement of quality of life and diminished autoimmune disease expression are clinically realistic. It should also be noted at times, despite all clinical attempts made, that the autoimmune response remains unchanged. The response to any form of treatment is unpredictable.

*Evid Based Complement Alternat Med*, 2011;2011:927294.

The role of Th17 in neuroimmune disorders: Target for CAM therapy.

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## Immunology Laboratory Interpretation

In this section of the manual, a review of laboratory testing and analysis will be discussed. Although it is common to test markers such as ANA and rheumatoid factor, these markers provide little information regarding the specifics of the immune response. This section of the manual will focus on identifying the immune response that is present, despite the tissue or tissues involved. Or to say it differently, an understanding of how and why the body is attacking self-tissue will be explored. Typically, with an autoimmune expression, it is important to evaluate all the specific immune cells, natural killer cell activity, and cytokines as a baseline minimum.

### TH-1/TH-2 Production Panel

This panel includes interleukin-2, interferon, interleukin-4, and interleukin-10. These cytokines are used to stimulate the TH-1 and TH-2 systems. When evaluating an autoimmune response, it is first important to identify if the patient is TH-1 or TH-2 dominant. TH-1 is the initial cell-mediated response and TH-2 is the B-cell–mediated antibody response. Autoimmune diseases today are classified in such a manner. Many compounds may be stimulatory to one system versus another. Remember, the TH-1 response is the initial response to an antigen. In the TH-1 response, T-helper cells are up-regulated as they release cytokines such as interferon and interleukin-2 to activate natural killer and cytotoxic

T-cells. Therefore, if the TH-1 system is dominant, these cytokines (IFN and IL-2) will be elevated. In the TH-2 response, the immune antigen presenting cells are sending messages by the production of IL-4 and IL-10 to the B-cells to activate antibody production against the antigen. If the individual is TH-2 dominant, these cytokines (IL-4 and IL-10) may be elevated.

It should be noted that not all individuals with active autoimmune disease will demonstrate a TH-1 or TH-2 cytokine dominance when only limited to testing IFN, IL-2, IL-10 and IL-4. As a matter of fact, many individuals will have a totally normal TH-1/ TH-2 cytokine panel, despite the presentation of a very active autoimmune response as represented by elevated tissue antibodies and clinical presentation. In these individuals, the TH-1/ TH-2 production panel is still important to conduct in the future, because certain treatments (botanicals) may increase the expression of their self-immune attack and demonstrate a T-helper dominance.

At times, the patient's immune system may be so suppressed from a chronic overzealous response that the cytokines may appear depressed or within normal limits. Nutritional and botanical treatments may increase their response. This may be interpreted as a bad response, since the individual's immune system is attacking self-tissue, but the elevation of the cytokine response may be important in re-establishing healthy immune modulation. For example, if a patient has a normal or depressed TH-1/TH-2 cytokine panel, you initiate some type of therapy; after four weeks, a repeat test demonstrates TH-1 dominance, the patient's T-helper/T-suppressor ratio gets worse (increases in number) and clinical symptoms increase, the treatment may be adverse. On the other hand, if repeat testing demonstrates a TH-1 dominance, a depressed T-helper/ T-suppressor ratio and clinical symptoms improve, the treatment may be effective in supporting healthy immune regulation. To summarize, the TH-1/TH-2 cytokine production levels are not substantial enough to monitor an immune response. They must be compared with the rest of the immune system, especially with T-cells, B-cells, natural killer cells, and T-helper/T-suppressor ratios.

### TH-1 Dominance

**TH-1 Dominance Due to Active Infection:** TH-1 dominance is identified if IL-2 and/or TNF-alpha is elevated. It is important to determine whether TH-1 dominance is due to immune dysregulation or active infection. If the patient has an active infection that is resulting in TH-1 dominance, the following markers may be demonstrated: low T-suppressor, elevated T-helper, elevated T-helper/suppressor, and elevated B-cells. The natural killer cell activity needs to be closely examined. With active infections, the natural killer cell activity will be elevated. With chronic infections, the total lymphocytes and the natural killer cell activity may be depressed. It is not possible to identify the antigen driving TH-1 dominance by looking at cytokines and immune cells. Further testing may be required. The management of these patterns involves identifying and eliminating the antigen.

**TH-1 Dominance Due to Immune Dysregulation:** At other times, with a TH-1 dominance, the immune system does not appear to have a pattern of fighting an antigen. For example, although the IL-2 and/or TNF-alpha may be elevated, the pattern of elevated T-helper cells and T-suppressor cells is not present. A pattern that does not indicate an active infection denotes dysregulation of the immune system. Environmental toxicity should be strongly investigated.

*TH-1 Dominance Due to Immune Dysregulation*
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- The immune system

### TH-2 Dominance

**TH-2 Dominance Due to Active Infection:** TH-2 dominance is identified if IL-4 and/or IL-10 are elevated. It is important to determine whether TH-2 dominance is due to immune dysregulation or active infections. If the patient has an active infection that is resulting in TH-2 dominance, look for demonstrated low T-suppressor, elevated T-helper, elevated T-helper/suppressor and elevated B-cells. The natural killer cell activity needs to be closely examined. With active infections, the natural killer cell activity will be elevated. With chronic infections, the total lymphocytes and the natural killer cell activity may be depressed. It is not possible to identify the antigen driving TH-2 dominance by looking at cytokines and immune cells. Further testing may be required. The management of these patterns involves identifying and eliminating the antigen. If a TH-2 dominance pattern is found related to active infection, antigens such as a virus, environmental hapten, and cancer must be ruled

out since they are strongly associated with elevation of the TH-2 system. An elevated TH-2 system with increased natural killer cells is a pattern that strongly suggests the presence of cancer cells.

**TH-2 Dominance Due to Immune Dysregulation:** At other times, with a TH-2 dominance, the immune system does not appear to have a pattern of fighting an antigen. For example, although the IL-4 and/or IL-10 may be elevated, the pattern of elevated T-helper cells and T-suppressor cells is not present. A pattern that does not indicate the pattern seen with an active infection indicates dysregulation of the immune system. Environmental toxicity should be strongly investigated.

*TH-2 Dominance Due to Immune Dysregulation*
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- The immune system

### Suppressed TH-1 and TH-2 Cytokines

Suppressed cytokines may be present if the patient is exposed to immunosuppressive medications such as cortisone, or is in a "severe stress" response. At times, chronic infections that wipe out the immune system may be causative. It is important in these cases to carefully examine the total and percent lymphocytes and WBCs. Although it may seem paradoxical, it is not uncommon to see individuals with both immune suppression and active autoimmune disease. These patients are usually in very poor health. These patterns are typically seen with chronic viral infections and infections with ticks (Lyme disease) and immune-specific (not systemic) autoimmune responses in clinical practice. If this pattern is identified, a viral antibody panel should be conducted. If antibodies are found to be elevated for viral infections (especially IgM), a viral PCR will be useful in measuring the active viral load. The viral load can be used as an effective marker to monitor effective viral suppression with management. If Lyme Disease is suspected, a Lyme Panel by Immunosciences Labs is strongly recommended. Their unique panel has many markers that may help identify the condition that other types of testing for Lyme disease may miss.

### Normal Cytokine Response

If the cytokines are normal, several possibilities exist. The patient may have a combination of both active and suppressed TH-1 and TH-2 systems balancing out the influence upon each other. If this scenario is present, you may see something that presents as elevated T- and B-cells. These are not uncommon in active infections. It should also be noted that the TH-1/TH-2 cytokine panel is in no way exhaustive when looking at the cytokines involved with immune regulation. Therefore, other cytokines related to these systems need to be investigated. Despite a normal TH-1/TH-2 cytokine panel, the panel is worth repeating anytime treatment is provided in order to identify if the patient's immune response is increasing. If a patient has an autoimmune disease and the TH-1/TH-2 cytokine panel is normal before some type of treatment and then becomes dominant after some type of therapy, the therapy may be promoting overzealousness of the immune response and be promoting autoimmune expression.

### Comprehensive Immune Panel

Please note that Immunoscience's comprehensive immune panel that is used to evaluate the immune response in autoimmune dysfunctions is no longer available. This test will be available from Cyrex Laboratories in the future. This panel will include: total WBC, %WBC, total T-cell, % T-cell, total T-helper cell, %T-helper cell, total suppressor cell, % suppressor cell, T-helper/T-suppressor ratio, total B-cell, % B-cell, total natural killer cell, % natural killer cell, total immunocompetent, % immunocompetent natural killer HT3+, total natural killer HT3-, % natural killer HT3-, total CD3+ CD26+, %CD3 CD26+, total T3 positive, natural killer cell activity, natural killer cell activity/cell, % natural killer cells, % immunocompetent natural killer HT3+, T3 positive cells, phytohemagglutinin, concanavalin A, pokeweed mitogen, lipopolysaccharide.

A brief description and clinical significance of the markers will be discussed. It should be noted that the panel first lists the total and percentage for all the WBCs and their breakdown into specific immune cells. The total number of the immune cells (example total T-cells) signifies the immune system's ability to produce cells. Increased production of immune cells generally indicates that the immune system is working harder to respond to something. Decreased total numbers of immune cells are generally seen with immune system compromise. Unlike the total number, the percentages (example %

T-cell) indicate the quality of the immune response. To put it simply, the total amounts of immune cells indicates quantity and the percentages indicate quality. With most autoimmune dysfunctions that are chronic, the total cells are usually depressed and therefore the immune cell percentages are a better marker to assess immune function.

### Lipopolysaccharides (Adjuvants)

Lipopolysaccharides released from gram-negative bacteria are considered endotoxins that may be used as adjuvants in the immune response. Adjuvants are agents that cannot potentiate an immune response alone but, with the orchestration of other cellular agents, can influence physiological responses. Lipopolysaccharides have been shown to enhance the IgM response.

### Lymphocyte Transformation Mitogens (phytohemagglutinin, concanavalin A, and pokeweed mitogen)

In this test, cells are challenged with plant and bacterial mitogens, such as phytohemagglutinin (PHA), concanavalin A (Con A), and pokeweed mitogen (PWM), to stimulate mitosis. These mitogens will stimulate both T- and B-lymphocytes. When normal lymphocytes are exposed to any of these substances, the cell will begin to divide. Lower counts indicate decreased lymphocyte responses.

### T-Helper/T-Suppressor Ratio

The T-helper/T-suppressor ratio is an evaluation of the intricate balance between the activation and suppression of immune signals. T-helper cells release cytokines that activate T-cells and B-cells. T-suppressor cells decrease the activity of cellular messenger activation. In order to have a healthy immune response, a coordinated response between T-helper and T-suppressor cells must take place. The coordination of these activities is evaluated clinically by this ratio. Functionally, the ratio is considered to be 1.2-2.0 although most labs will list it as 1.0-2.5. When individuals have levels above 2.5, it establishes the physiological shift into autoimmune states. In these cases, the body is not demonstrating a functional ability to suppress immune activation. As individuals increase their ratios to about 2.0, it strongly suggests that they are susceptible to autoimmune disease.

Depressed T-helper/T-suppressor ratios indicate an immune compromise and the inability of the immune system to regulate and respond to antigen activation. Many times depressed levels are a consequence of chronic antigenic loads with antigens, chemical exposure, or immunodeficiency deficiency diseases.

### T-Cell and B-Cell Lymphocyte Panel:

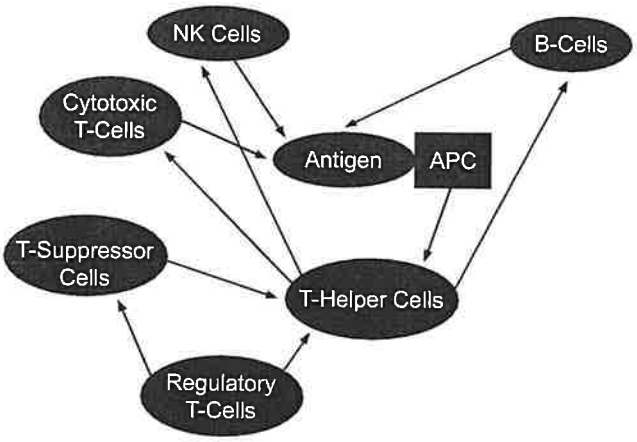
- Total WBC
- % WBC
- Total T-cell
- % T-cell
- Total T-helper cell
- % T-helper cell
- Total suppressor cell
- % Suppressor cell
- T- helper/T-suppressor ratio
- Total B-cell
- % B-cell
- Total natural killer cell
- % Natural killer cell

## Immune Response to an Active Antigen

### Pattern Demonstrated with Active Antigen Stimulation (Infection)

When the immune system triggers a response against an antigen, it must synchronize an attack by all immune cells. With acute infections, the total and percent of WBCs and lymphocytes will be elevated. With chronic infection, the total and percent of white blood cells and lymphocytes are usually depressed. At times, a patient is transitioning from an acute infection to a chronic infection, and the total and percent of white blood cells may be normal. *Natural killer cells and cytotoxic T-cells are the immune system's soldiers. If they are elevated, this strongly suggests the presence of an active antigen.* For the immune system to trigger an attack, the T-helper cells must be elevated in order to call for help, and the T-suppressor cells must be depressed. This may present as an elevated T-helper/T-suppressor ratio. The B-cells may also be elevated in an active infection response, and if they show the possibility of viral infection, cancer and chemical hapten should be investigated. Listed below are a summary of the markers that may be found.

Immune Response Against An Antigen.




### TH-1 and TH-2 Pathway Challenge

| TH-1 Pathway Challenge | TH-2 Pathway Challenge |
|---|---|
| Astragalus Extract | Pine Bark Extract |
| *Echinacea purpurea* Extract | Green Tea Extract |
| Licorice Extract | Grape Seed Extract |
| Maitake Mushroom | Resveratrol |
| Lemon Balm | Pycnogenol® |
| Pomegranate | |

Please see registered trademark ownership in Appendix 1.

The TH-1/TH-2 pathway challenge should only be considered if the individual has some degree of health, is not attacking self-tissue, had unclear laboratory results, and his/her history could not establish a specific issue. Each product is given at a different time to see if either one creates an exaggeration of the symptomatology. The challenge should never be performed with someone who is hyperfunctioning. The healthcare professional must have the skills in examination, laboratory analysis, and history taking to know when it is clinically appropriate to challenge.

Individuals should keep a journal of their symptoms when the challenge is performed. They should supplement exclusively with nutrients from the TH-1 category for three days and then nutrients from the TH-2 category for three days. If a reaction is noted, the individual is instructed to stop and not continue with any further challenge of that side of the T-helper system. The other side is then challenged. There are four possible responses for the challenge and related clinical strategies listed below.

Four Possible Responses with the TH-1/TH-2 Challenge

1. TH-1 challenge reactions
2. TH-2 challenge reactions
3. Both TH-1 and TH-2 challenge reactions
4. Neither TH-1 nor TH-2 challenge reactions

### 1. TH-1 Challenge Reactions

In these cases, the immune system is having an immune response, potentially against self-tissue. This does not necessarily mean that the individual is TH-1 dominant. He/she may be TH-1 dominant with immune dysregulation, or he/she may be TH-2 dominant with an active antigen response. In any case, the use of TH-1 challenge supplements should be avoided for life. The potential for TH-2 support to dampen the immune response can be monitored by evaluating cytokines, T- and B-lymphocytes, CD4 and CD8 ratios, and total white blood cells. If the immune response is being supported effectively with TH-2 support the immune cells should be normal, the cytokines should be normal.

Consider diet, nutrition, and lifestyle support for:

- TH-2 system
- Immune system T-cells (TH-3)

### 2. TH-2 Challenge Reactions

In these cases, the immune system is having an immune response, potentially against self-tissue. This does not necessarily mean that the individual is TH-2 dominant. He/she may be TH-2 dominant with immune dysregulation, or he/she may be TH-1 dominant with an active antigen response. In any case, the use of TH-2 challenge supplements should be avoided, possibly for life. The potential for TH-1 support to minimize the immune response can be monitored by evaluating cytokines, T- and B-lymphocytes, CD4 and CD8 ratios, and total white blood cells. If the immune response is being supported effectively with TH-1 support the immune cells and cytokines should be normal.*

Consider diet, nutrition, and lifestyle support for:

- TH-1 system
- Immune system T-cells (TH-3)

### 3. Both TH-1 and TH-2 Challenge Reactions

In these cases, the potential immune trigger is typically a hapten, and active IgM antibodies against environmental

compounds are usually observed with laboratory testing. The clinical battle in these cases is to evaluate and support the barrier systems of the body. Additionally, the individual should totally avoid compounds that support the TH-1 or TH-2 pathways. Interestingly, a polarity to TH-1 or TH-2 may become apparent after the barrier system is improved.

Consider diet, nutrition, and lifestyle support for:

- Gastrointestinal system
- Blood-brain barrier
- Regulatory T-cells
- Body under temporary stress

### 4. Neither TH-1 nor TH-2 Challenge Reactions

In these cases, the individual is typically responding to a chronic antigen, such as a virus that is not causing immune dysregulation, and the immune cells are depressed from chronic activity. In these instances, supporting the TH-1, TH-2, and TH-3 pathways seems to be most effective. The immune response can be monitored by evaluating cytokines, T- and B-lymphocytes, CD4 and CD8 ratios, and total white blood cells. These markers should be normal if the person is responding effectively to care.

Consider diet, nutrition, and lifestyle support for:

- TH-1 and TH-2 systems

### TH-3 Cells, or Regulatory T-Cell, Activation

TH-3 cells, or regulatory T-cells, are a subpopulation of cells that down-regulate the immune system, maintain tolerance to self-antigens, and down-regulate autoimmune disease. Regulatory cells are a component of the immune system that suppresses the immune responses of other cells. This is an important "self-check" built into the immune system to prevent excessive reactions. Regulatory T-cells come in many forms, with the best understood being those that express CD4, CD25, and Foxp3 (CD4+CD25+ regulatory T cells). These cells are involved in shutting down immune responses after they have successfully eliminated invading organisms and also in preventing autoimmune TH-1 and TH-2 polar shifts. Several nutrients have been shown to support the regulatory T-cells, such as vitamin D and glutathione support.

*TH-3 Irregularity*
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Immune system

### TH-17 Activation and Dampening

T-helper-17 (TH-17) is a subset of T-helper cells producing IL-17. These cells are considered developmentally distinct from TH-1 and TH-2 cells, and healthy levels of TH-17 may support the immune system.

There is evidence that curcumin minimizes the activity of TH-17. A recent paper demonstrated positive support with curcumin. The researchers concluded that curcumin, a dietary spice from turmeric, supports the immune system and neurosupportive effects.

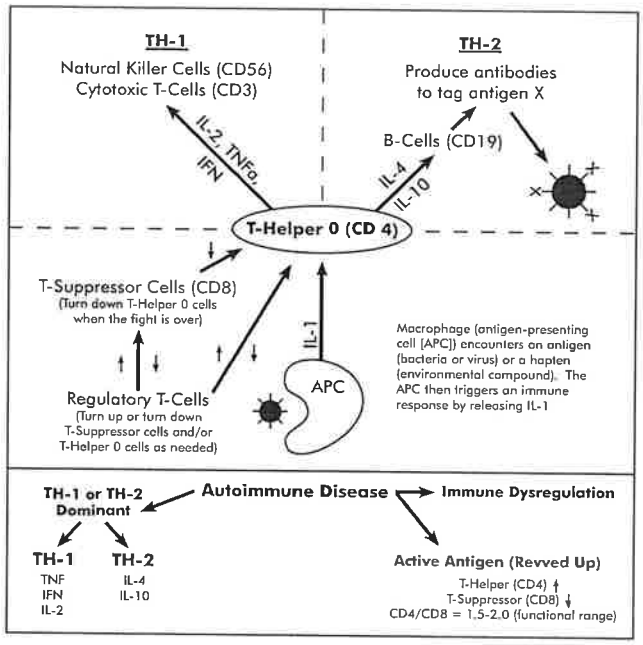
Another study found that providing curcumin supplementation to Lewis rats provided immune system support. Furthermore, the mRNA expression of the cytokine profiles was assessed by quantitative reverse-transcription polymerase chain reactions (qRT-PCR), revealing support of healthy IL-17, TGF-beta, IL-6, IL-21, STAT3, and RORgammat levels in curcumin-treated groups.

Several papers have also been published on the role of resveratrol and its ability to support a healthy balance of TH-17 activity. It appears that the use of both resveratrol and curcumin may provide IL-17 minimizing activity and provide the nutritional support necessary to support immune responses. It should also be noted that both curcumin and resveratrol are potent natural NF-kB minimizers.

*TH-17 Irregularity*
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Immune system




*Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.*

## Case Examples

Case 36

***TPO Ab: Positive***

***IL-2: Elevated***

***IFN-Gamma: Elevated***

IL-10: Normal Range

IL-13: Normal Range

IL-6: Normal Range

IL-17: Normal Range

***Total Lymphocytes: Depressed***

% Lymphocytes: Normal Range

***Total CD3 (T-Cells): Depressed***

% CD3 (T-Cells): Normal Range

***Total CD4 (Helper): Elevated***

***% CD4 (Helper): Elevated***

***Total CD8 (Suppressor): Depressed***

***% CD8 (Suppressor): Low Normal***

***CD4:CD8 ratio: >2.0***

Total CD19 (B-Cells): Normal Range

% CD19 (B-Cells): Normal Range

Total CD56 (Natural Killer Cells): Normal Range

% CD56 (Natural Killer Cells): Normal Range

Case 37

***TPO Ab: Positive***

Antithyroglobulin Ab: Negative

IL-2: Normal Range

***IFN-Gamma: Elevated***

***IL-10: Depressed***

IL-13: Normal Range

***IL-6: Elevated***

***IL-17: Elevated***

Total Lymphocytes: Normal Range

% Lymphocytes: Normal Range

Total CD3 (T-Cells): Normal Range

% CD3 (T-Cells): Normal Range

Total CD4 (Helper): Normal Range

***% CD4 (Helper): Elevated***

Total CD8 (Suppressor): Normal Range

***% CD8 (Suppressor): Low Normal***

***CD4:CD8 ratio: >2.0***

Total CD19 (B-Cells): Normal Range

% CD19 (B-Cells): Normal Range

***Total CD56 (Natural Killer Cells): Elevated***

***% CD56 (Natural Killer Cells): Elevated***

Case 38

TPO Ab: Negative

Antithyroglobulin Ab: Negative

IL-2: Normal Range

IFN-Gamma: Normal Range

***IL-10: Depressed***

IL-13: Normal Range

IL-6: Normal Range

IL-17: Normal Range

***Total Lymphocytes: Depressed***

***% Lymphocytes: Depressed***

Total CD3 (T-Cells): Normal Range

***% CD3 (T-Cells): Depressed***

***Total CD4 (Helper): Depressed***

***% CD4 (Helper): Depressed***

Total CD8 (Suppressor): Normal Range

% CD8 (Suppressor): Normal Range

CD4:CD8 Ratio: <1.5

***Total CD19 (B-Cells): Depressed***

***% CD19 (B-Cells): Depressed***

Total CD56 (Natural Killer Cells): Normal Range

% CD56 (Natural Killer Cells): Normal Range

Case 39

***TPO Ab: Positive***

Antithyroglobulin Ab: Negative

IL-2: Normal Range

IFN-Gamma: Normal Range

***IL-10: Elevated***

IL-13: Normal Range

***IL-6: Elevated***

IL-17: Normal Range

Total Lymphocytes: Normal Range

% Lymphocytes: Normal Range

Total CD3 (T-Cells): Normal Range

% CD3 (T-Cells): Normal Range

Total CD4 (Helper): Normal Range

% CD4 (Helper): Normal Range

***Total CD8 (Suppressor): Elevated***

***% CD8 (Suppressor): Elevated***

CD4:CD8 Ratio: Normal Range

Total CD19 (B-Cells): Normal Range

% CD19 (B-Cells): Normal Range

Total CD56 (Natural Killer Cells): Normal Range

% CD56 (Natural Killer Cells): Normal Range

Case 40

***TPO Ab: Positive***

Antithyroglobulin Ab: Negative

***IL-2: Depressed***

IFN-Gamma: Normal Range

***IL-10: Elevated***

IL-13: Normal Range

***IL-6: Elevated***

***IL-17: Elevated***

Total Lymphocytes: Normal Range

% Lymphocytes: Normal Range

Total CD3 (T-Cells): Normal Range

***% CD3 (T-Cells): Depressed***

Total CD4 (Helper): Normal Range

***% CD4 (Helper): Elevated***

***Total CD8 (Suppressor): Elevated***

***% CD8 (Suppressor): Elevated***

CD4:CD8 ratio: Normal Range

Total CD19 (B-Cells): Normal Range

% CD19 (B-Cells): Normal Range

Total CD56 (Natural Killer Cells): Normal Range

% CD56 (Natural Killer Cells): Normal Range

# 24 Thyroid Patterns

Both thyroid and NTI patterns can promote defects in thyroid physiology. Thyroid illness includes hypothyroidism, hyperthyroidism, Graves', and Hashimoto's. In these conditions, there is actual illness and pathology in thyroid tissue. Non-thyroid illness or euthyroid, consists of defects in thyroid metabolism that are not related to actual pathology in the thyroid tissue, but rather a defect in thyroid physiology. The NTI thyroid disorders promote the symptoms of hypothyroidism clinically. It is important to note that the specific pattern of NTI may be identified and monitored with specific laboratory tests in a clinical setting. Non-thyroid illness patterns are typically overlooked or dismissed in the traditional healthcare system. However, a detailed history and specific lab testing can be of a great value in treating the chronic patients with unrelenting thyroid symptoms who do not have thyroid illness, but rather NTI. Listed below are the sites of potential thyroid physiological defect of both thyroid illness and non-thyroid illness.

1. *Hypothalamic paraventricular defect related to central nervous system deficiency of serotonin leading to low TSH* [67-69,118-119]

These patients suffer from classical thyroid symptoms and serotonin deficiency symptoms. Serotonin deficiency symptoms include loss of pleasure in hobbies and interests, feel overwhelmed with ideas to manage, feelings of inner rage, feelings of paranoia, feelings of depression, feeling like not enjoying life, lack of artistic appreciation, depression with lack of sunlight, loss of enthusiasm for favorite activities, not enjoying favorite foods, not enjoying relationships, unable to fall into deep restful sleep, feelings of dependency on others, feeling more susceptible to pain, and feelings of unprovoked anger.

In this pattern, it is theorized that low serotonin activity at the level of the paraventricular cells of the hypothalamus leads to a less-than-ideal output of TSH by the pituitary. These patients will have a TSH below the functional range of 1.8, but typically not below the traditional laboratory range of TSH, and their T4 will be below the functional range of 6. They will usually be told their thyroid is normal since they do not have primary hypothyroidism; however, they will fit our category of hypothyroidism secondary to functional pituitary hypofunction.

Since testing serotonin in the brain (not urine) is not established, the diagnosis of this pattern is always a working diagnosis that is suggested with a low TSH (<1.8), a low serum T4 (<6), hypothyroid symptoms, and serotonin deficiency symptoms. The diagnosis of this pattern cannot be confirmed until clinical strategies to improve serotonin activity improve clinical outcomes of symptoms, TSH, and T4. It may take several months to see changes in TSH, but thyroid and serotonin subjective indications should respond rather quickly.

Since this pattern is potentially due to serotonin deficiency, a critical look at dysglycemia (insulin resistance, diabetes, and hypoglycemia) should be considered. Dysglycemia is the most common clinical cause for serotonin defects. Nutritional and dietary management to address dysglycemia should be conducted, if necessary. The typical symptoms of a hypoglycemic are fatigue, sugar-cravings, and irritability between meals that are normalized after a meal. The typical symptoms of insulin resistance include fatigue and sugar cravings after meals.

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Serotonin levels
- Sugar metabolism
- Neurotransmitters
- The pituitary

2. *Hypothalamic paraventricular defect related to central nervous system deficiency of dopamine leading to low TSH*[70-73]

These patients suffer from the classical thyroid symptoms and dopamine deficiency symptoms. Dopamine deficiency symptoms include feelings of worthlessness, feelings of hopelessness, self-destructive thoughts, inability to handle stress, anger and aggression under stress, desire to isolate from others, unexplained lack of concern for family and friends, distracted easily, inability to finish tasks, need for caffeine to feel mentally alert, low libido, and loss of temper for minor reasons.

In this pattern, it is theorized that low dopamine activity at the level of paraventricular cells of the hypothalamus leads to a less-than-ideal output of TSH by the pituitary. These patients will have a TSH below the functional range of 1.8, but typically not below the traditional laboratory range of TSH, and their T4 will be below the functional range of 6. They will usually be told their thyroid is normal since they do not have primary hypothyroidism; however, they will fit our category of hypothyroidism secondary to functional pituitary hypofunction.

Since testing dopamine in the brain (not urine) is not established, the diagnosis of this pattern is always a working diagnosis that is suggested with a low TSH (<1.8), a low serum T4 (<6), hypothyroid symptoms, and dopamine deficiency symptoms. The diagnosis of this pattern cannot be confirmed until clinical strategies to improve dopamine activity improve clinical outcomes of symptoms, TSH, and T4. It may take several months to see changes in TSH, but thyroid and dopamine subjective indications should respond rather quickly.

Since this pattern is potentially due to dopamine deficiency, a critical look at dysglycemia (insulin resistance, diabetes, and hypoglycemia) should be considered. Dysglycemia is the most common clinical cause of dopamine defects. Nutritional and dietary management to address dysglycemia should be conducted, if necessary. The typical symptoms of a hypoglycemic are fatigue, sugar cravings, and irritability between meals that are normalized after a meal. The typical symptoms of insulin resistance include fatigue and sugar cravings after meals.

If the patient has major stress in his/her life, he/she should support healthy blood sugar balance with herbal adaptogens, which have shown the ability to support the activity of neurotransmitters in the central nervous system. Additionally, phosphatidylserine has demonstrated the ability to support neurotransmitter activity in the central nervous system. Phosphatidylserine supports neurotransmitter activity in the hypothalamus, which is associated with minimizing cortisol, but it is also very effective for these patterns, despite the patient's adrenal status.

Chronic use of caffeine, nicotine, marijuana, and other recreational drugs may have powerful depletive influences on dopamine. Although these compounds initially elevate dopamine, long-term use may promote the loss of dopamine regulation of the paraventricular cells of the hypothalamus, which are used to coordinate endocrine function.

The importance of methylation cannot be overlooked in these patterns. Methyl donors are crucial for dopamine synthesis. Supplementing with liquid vitamin B12, (methylcobalamin) would be suggested for these patterns. Methyl donor deficiency should especially be considered if the patient is hypochlorhydric, is taking antacid medication, or is taking estrogen replacement therapy. These scenarios promote depletion of methyl donors.

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Dopamine levels
- Sugar metabolism
- Neurotransmitters
- The pituitary
- Methyl donors

3. *Hypothalamic paraventricular defect promoted by cytokines leading to low TSH*[7-12]

These patients suffer from thyroid symptoms and have an associated pattern of an immune challenge. The immune patterns that elevate cytokines to the degree that TSH becomes suppressed include autoimmune disorders or a chronic antigen pattern. Chronic antigen patterns typically involve a chronic virus (hepatitis C, cytomegalovirus, Epstein-Barr, etc), Lyme disease, Chlamydia, pneumonia, chronic pulmonary mold infection, etc. The history will typically clarify these associated patterns, but the connection to TSH suppression has been suggested to these patients. Usually, depressed serum total white blood cells will be discovered in these patients.

In these patterns, cytokines such as IL-1, IL-2, TNF-alpha, IL-6 suppress the hypothalamus-pituitary axis. These cytokines are elevated as a consequence of immune activity, and modulation of immune activity is critical in optimizing the hypothalamus-pituitary axis. These patients will have a TSH below the functional range of 1.8, but typically not below the traditional TSH laboratory range, and their T4 will be below the functional range of 6. They will usually be told that their thyroid is normal since they do not have primary

hypothyroidism; however, they will fit our category of hypothyroidism secondary to functional pituitary hypofunction.

Testing cytokines, such as IL-2, IL-1, TNF-alpha, IL-6 is easily conducted with serum testing at any major laboratory. It is best to screen with the series of cytokines to identify the elevated cytokines. These cytokines can then be used as a source to evaluate the patient's progress. Appropriate management of the patient should normalize the cytokine elevations and improve the percent and total of white blood cells and lymphocyte differentials.

Modulating cytokine dominances that are secondary to infections requires effective eradication of the source. Modulating cytokines secondary to autoimmune patterns should start with regulatory T-cell support. Regulatory T-cells help balance TH-1/TH-2 shifts. Natural compounds that support healthy regulator T-cell function include high-dose vitamin D, EPA, DHA, glutathione, and super oxide dismutase (SOD).

Some patients with this pattern will only tolerate support for either TH-1 or TH-2. Sometimes, the patient's history will reveal which T-helper pathway support they can tolerate. Botanicals, such as echinacea, astragalus, reishi, and beta-glucans, support the TH-1 pathway. Other botanicals, such as grape seed extract, pine bark extract, caffeine, and green tea extract, support the TH-2 pathway.

The management of elevated cytokines is complex and is beyond scope of this manual. However, the point is, immune-related disorders that chronically elevate cytokines can dysregulate the hypothalamic-pituitary-thyroid axis. Cytokine and immune cell measurements can be used effectively to monitor these patterns.

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Regulatory T-cell mechanisms
- The TH-1 system
- The TH-2 system
- The gastrointestinal system and intestinal barrier

4. *Hypothalamic paraventricular imbalance promoted by elevated prolactin*[91-94]

Prolactin is very sensitive to shifts in dopamine, thyroid hormones, progesterone, and serotonin. There is a very tight physiological loop between these hormones and neurotransmitters. A shift in one of these neurotransmitters and hormones can cause shifts in the others.

Depressed dopamine and/or progesterone can promote elevations of prolactin that may functionally suppress TSH. On the other hand, a prolactinoma (benign pituitary tumor) may produce high amounts of prolactin in hypothyroid patterns.

In either case, patients who have elevated prolactin may have a history of discharge from their breasts (both male and female). In females, elevated prolactin promotes suppression of luteinizing hormone (LH) and progesterone, leading to infertility, dysmenorrhea, and amenorrhea. In men, elevated prolactin suppresses LH, leading to depressed testosterone and symptoms of impotence and andropause.

If the elevated prolactin is caused by a tumor, it may promote severe headaches and visual disturbances (scotomas). Anytime serum prolactin is elevated, a prolactinoma must be ruled out. Imaging may be helpful, but many tumors cannot be visualized, and not all prolactinomas are enlarged. Nutritional support should be focused on supporting LH and dopamine, but it is not a substitute for medical attention.

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Dopamine levels
- Female reproductive hormones
- Male reproductive hormones

5. Pituitary suppression from cortisol leading to low TSH[7-12]

Suppression of TSH may be caused by activation of the hypothalamic-pituitary axis. The central integrative state of paraventricular (PVN) cells of the hypothalamus dictate the cortisol response. The cytokine messengers stimulate the PVN cells for cortisol responses, but they also exhibit suppressive influences on TSH. Therefore, it is not clinically uncommon to see patients with

altered cortisol (hyperadrenal, hypoadrenal, and adrenal maladaptation) present with suppressed TSH. The functional range of TSH is 1.8. Patients in this pattern will present with cortisol-related symptoms, a TSH below 1.8, a depressed functional T4 (<6), and thyroid symptoms. An adrenal salivary index demonstrates adrenal dysfunction. The main causes for this pattern are active infection, dysglycemia, or unrelenting stress. The most common cause is an uncontrolled blood sugar imbalance, such as insulin resistance or hypoglycemia. If the patient feels shaky, irritable, and has sugar cravings between meals, suspect reactive hypoglycemia. If the patient has fatigue and sugar cravings after meals, suspect insulin resistance.



For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Cortisol/PVN levels
- The pituitary
- Sugar metabolism

6. *Thyroid tissue imbalance related to thyroid peroxidase autoimmune response*[276]

Thyroid peroxidase (TPO) is the enzyme in the thyroid that is responsible for the production of thyroid hormones. It liberates iodine for addition onto tyrosine for the production of T4 and T3. Thyroid peroxidase is a common marker for autoimmune thyroid disease and can be measured as TPO autoantibody. Positive TPO autoantibodies suggest the presence of an autoimmune response against the thyroid. When the immune response against TPO is very aggressive, it may promote an active thyroiditis that presents as hyperthyroidism. In many cases, the immune response against TPO is gradual and slow and leads to hypothyroidism. In early stages of the disorder, the patient may have normal thyroid markers (TSH, T3, T4); however, a positive TPO autoantibodies test is a predictor of future thyroid disease. Positive TPO autoantibodies suggest the presence of Hashimoto's disease. In aggressive hyperthyroid responses, it is typically classified as Hashimoto's thyroiditis. In less aggressive hypothyroid responses, it is classified as Hashimoto's hypothyroidism. Typically, TPO autoantibodies is found positive along with Tg Ab.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician. (If hormone levels are low, treatment may include: thyroid hormone replacement.)

Consider diet, nutrition, and lifestyle support for:

- Regulatory T-cells
- The TH-1 system
- The TH-2 system

7. *Thyroid tissue imbalance related to thyroglobulin autoimmune response* [238 277]

Thyroglobulin is produced by the thyroid epithelial cells called thyrocytes and is used by the thyroid gland to produce thyroid hormones. The enzyme thyroperoxidase covalently binds iodine and tyrosine molecules into thyroglobulin, forming MIT and DIT. Thyroxine (T4) is produced by combining two moieties of DIT, and triiodothyronine (T3) is produced by combining one molecule of MIT and one molecule of DIT. Thyroglobulin is a common site for autoimmune thyroid disease and can be measured as Tg Ab. A positive Tg Ab suggests the presence of an autoimmune response against the thyroid. When the immune response against thyroglobulin is very aggressive, it may promote an active thyroiditis that presents as hyperthyroidism. In many cases, the immune response against Tg is gradual and slow and leads to hypothyroidism. In early stages of the disorder, the patient may have normal thyroid markers (TSH, T3, T4); however, a positive Tg Ab test is a predictor for future thyroid disease. Positive Tg Ab suggests the presence of Hashimoto's disease. In aggressive hyperthyroid responses, it is typically classified as Hashimoto's thyroiditis.

In less aggressive hypothyroid responses, it is classified as Hashimoto's hypothyroidism. Typically, TPO Ab is found positive along with Tg Ab.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician. (If hormone levels are low, treatment may include: thyroid hormone replacement.)

Consider diet, nutrition, and lifestyle support for:

- Regulatory T-cells
- The TH-1 system
- The TH-2 system

8. *Thyroid tissue imbalance related to thyroid-stimulating hormone autoimmune response*[120]

Thyroid-stimulating hormone is also known as thyrotropin and is released by the pituitary to stimulate thyroid hormone production. There is a negative feedback loop between TSH and T4. When thyroid levels drop, TSH is released. When T4 levels elevate, TSH is depressed. An elevated TSH is the indicator for hypothyroidism indicating that the pituitary is attempting to stimulate dysfunctional thyroid cells. An autoimmune response can occur, indicated by TSH receptor antibodies (TR Ab). It is also known as thyroid-stimulating immunoglobulin (TSI). Elevations of these antibodies typically suggest Graves' disease, but they may also be elevated in Hashimoto's.

Elevations of TR Ab may promote either hypothyroid or hyperthyroid responses because they have the ability to stimulate, bind, or block the TSH receptor site. If the TR Ab stimulates the receptor site, it may mimic TSH and produce excess thyroid hormones. If the TR Ab blocks the receptor site, it prevents TSH from acting on the receptor site and it may lead to diminished thyroid hormones. Lastly, TR Ab may bind to the thyroid receptors and take up space on the receptor, leading to subtle responses of the TSH receptor sites.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician. (Typical treatment may include: propylthiouracil, methimuzole, or iodine.)

9. *Autoimmune response to thyroxine and triiodothyronine*[278-282]

Thyroxine (T4) and triiodothyronine (T3) are the end result of thyroid metabolism. It is these hormones that bind to thyroid receptor sites in tissues and increase metabolic rate. Although not typically involved, severe autoimmune diseases may lead to the actual thyroid hormones and not the typical autoimmune response to TPO, Tg Ab, and thyroid-stimulating hormone receptors (TR Ab). Testing specifically for T4 and T3 antibodies (T4 and T3 Abs) can be conducted by special immunological laboratories.

Individuals with this pattern typically do not suffer from an isolated autoimmune thyroid disorder. These patterns may be found with polyglandular autoimmune syndrome, systemic lupus erythematosus, and other multiorgan autoimmune diseases. The most clinically relevant subjective finding is that these individuals do worse with bioidentical increase their systemic inflammation and promote greater symptoms of thyroid disease.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician. (If hormone levels are low, treatment may include: thyroid hormone replacement.)

Consider diet, nutrition, and lifestyle support for:

- Regulatory T-cells
- The TH-1 system
- The TH-2 system

10. *Downregulated thyroid peroxidase activity related to progesterone deficiency*[278-287]

The thyroid hormone-producing enzyme, thyroid peroxidase (TPO) is up-regulated with progesterone. That is why when women ovulate, their temperature becomes elevated. During ovulation, a surge of progesterone takes place that stimulates TPO. There is a surge of increased thyroid hormones produced that increases the metabolic rate and temperature. When patients suffer from progesterone insufficiency, they may develop fewer thyroid hormones than their bodies require, but these levels may not be outside the laboratory ranges of TSH, T4, or T3. These patients typically complain of both low thyroid and low progesterone symptoms. Low progesterone symptoms and signs include heavy menstrual bleeding, migraines in the second half of the menstrual cycle, infertility, depression, early miscarriage, breast tenderness, irregular periods, menstrual cramps, water retention before their period, and tendencies for ovarian cysts.

The most common cause for progesterone deficiency in a female of menstruating age is suppressed LH from stress and dysglycemia. In menopausal women, healthy sugar metabolism is also important.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician. (Typical treatment may include: progesterone.)

Consider diet, nutrition, and lifestyle support for:

- The pituitary and female hormone levels
- Sugar metabolism
- The adrenals



11. *Downregulated thyroid peroxidase activity from deficiency of cofactors*[121-123]

Thyroid peroxidase (TPO) is the enzyme in the thyroid that is responsible for the production of thyroid hormones. It liberates iodine for addition onto tyrosine for the production of T4 and T3. Many cofactors are used in the process of thyroid hormone production, including selenium, copper, magnesium, niacin, riboflavin, pyridoxal 5'-phosphate, and zinc.

The elevation of epinephrine and norepinephrine suppresses TPO activity. Studies have estimated that more than 50% of hypothyroid patients have positive TPO autoantibodies. Iodine supplementation in these individuals will stimulate their autoimmune thyroid response.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- TPO cofactors (no iodine/tyrosine)
- TPO cofactors (iodine/tyrosine)

12. *Downregulated 5'-deiodinase activity from deficiency of cofactors*[14-18]

Once the thyroid is stimulated by TSH from the pituitary, it produces T4 and T3 by transporting iodine into the thyroid and by stimulating TPO activity. Thyroid peroxidase is involved in the formation of T4 and T3 as it catalyzes the oxidation of iodine using hydrogen peroxide. The thyroid will produce 94% of the available T4 and 7% of the available T3.

As we know, T4 is inactive and T3 is active thyroid hormone. Therefore, the majority of hormone production at the thyroid is inactive T4. Once the thyroid has produced T4, it is metabolized peripherally from the thyroid into a combination of T3 hormones by the enzyme 5'-deiodinase, mostly at the liver. Poor diet and gut function can lead to a deficiency in the nutrients responsible for adequate function of this enzyme.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- 5'-deiodinase mechanism

13. *Downregulated 5'-deidionase from gastrointestinal dysbiosis and lipopolysaccharides*[14-18]

Once the thyroid has produced T4, it is metabolized peripherally from the thyroid into a combination of T3 hormones by the enzyme 5'-deiodinase, mostly at the liver. Under normal circumstances, about 40% of the available T4 is converted into T3, 20% is converted into reverse T3 that is irreversibly inactive, and 20% is converted into T3 sulfate (T3S) and triiodothyroacetic acid (T3AC). T3 sulfate and T3AC are inactive thyroid hormones until they circulate into the gastrointestinal tract and are acted upon by intestinal sulfatase into active T3. Gastrointestinal sulfatase activity is dependent upon a healthy gut microflora.[121-123]

14. *Downregulated 5'-deiodinase activity from elevated cytokines*[74]

Cytokines have the ability to disrupt peripheral conversion of inactive T4 into active T3. The enzyme 5'-deiodinase catalyzes the conversion of T4 into T3 in peripheral tissues away from the thyroid. Both TH-1 and TH-2 cytokines, IL-6, TNF-alpha, INF-gamma, and IL-1beta have demonstrated the ability to downregulate the thyroid conversion potentials of T3.[14-18] Patients with NTI have demonstrated an inverse relationship between serum T3 and IL-6 levels.[19] The administration of cytokines to healthy human subjects has been shown to lead to immediate changes in thyroid hormone levels. Exposure to IL-6, TNF-alpha, and IFN-alpha to healthy subjects resulted in a rapid reduction of serum T3 and TSH levels and an increase in reverse T3 concentrations, while T4 and free T4 levels were minimally altered.[20-22]

Clinically, it is important to note that T3 levels have very little influence on the TSH negative feedback loop. Therefore, when T3 levels decline from a cytokine 5'deiodinase mechanism the TSH will typically not

be elevated. If TSH is used as an isolated marker to screen for thyroid disorders, the T3 underconversion pattern promoted by an elevation of cytokines may be overlooked. It should be standard to measure T3 when evaluating the thyroid. A pattern of normal TSH, normal T4, and a depressed T3 strongly suggests down-regulation of the 5'-deiodinase enzyme.

15. *Downregulated 5'-deiodinase activity from elevated cortisol*[74]

Elevated cortisol has a suppressive impact on the enzyme 5'-deiodinase, which converts inactive T4 into active T3. Many sources have identified the pattern associated with elevated cortisol and its influence of low T3. The mechanism of lowered T3 may be due to multiple biochemical influences. Typically, T4 is converted to T3, reverse T3, T3 sulfate (T3S), and triiodothyroacetic acid (T3AC) peripherally by the enzyme 5'-deiodinase. In the past, it was believed that the reason for the low T3 state was because it was shunted to the production of irreversible and inactive reverse T3. This explanation has recently proven to be incorrect since reverse T3 production rates measured *in vivo* are generally unaltered in stress patterns. Thus, it has been shown that reverse T3 concentrations result from diminished reverse T3 clearance rather than increased production. Current evidence published by LoPresti and Nicoloff have found that increased T3S and T3AC at the expense of T3 are most likely responsible for the lowered T3 state. T3S and T3AC are inactive forms of T3, but they have the potential to become activated into T3 by sulfation and acetylation. The bottom line is that elevated cortisol will decrease active T3 levels and therefore reduce the potential of optimal gene expression by thyroid hormones.[121-123]

16. *Downregulated 5'-deiodinase activity from peripheral deficiencies of serotonin*[124]

The neurotransmitter serotonin exerts some influence on the hypothalamus-pituitary-thyroid axis and T3 conversion. A study examined simultaneous samples of serum thyroid hormones (TSH, total T4, free T4, total T3, free T3) and cerebrospinal measurements of 5-hydroxyindolacetic acid, the major metabolite of serotonin. Serotonin metabolite concentrations were correlated with plasma TSH and T3 (free and total), but not with T4 (free or total). CSF 5-hydroxyindolacetic acid levels were correlated with plasma TSH and total T3, but not with free T3 or T4 (free or total).[74] It appears that serotonin influences the rate of T3 conversion, possibly either from peripheral metabolism or from central messenger signals to the peripheral metabolism of T4 into T3.

17. *Downregulated 5'-deiodinase activity from peripheral deficiencies of dopamine*[74]

The neurotransmitter dopamine exerts some influence on the hypothalamic-pituitary-thyroid axis. A study examined simultaneous samples of serum thyroid hormones (TSH, total T4, free T4, total T3, free T3) and cerebrospinal measurements of homovanillic acid, the major metabolites of dopamine. CSF homovanillic acid concentrations were correlated with plasma TSH and T3 (free and total), but not with T4 (free or total).[74] It appears that serotonin influences the rate of T3 conversion, possibly either from peripheral metabolism or from central messenger signals to the peripheral metabolism of T4 into T3.

18. *Upregulated 5'-deiodinase activity from elevated testosterone*[124]

Elevated levels of testosterone in women create this pattern of too much T4 being converted into T3. The excess production of T3 overwhelms the cells, and they develop a resistance to the thyroid hormone. This pattern is most often found in women with insulin resistance and PCOS, or in men who use testosterone creams.

19. *Elevations of thyroid-binding globulin leading to decreased percentage of free-fraction thyroid hormones*[125-128]

When thyroid-binding globulin levels increase, the percentage of free-fraction thyroid hormones drop. This may be observed by a depressed T3 uptake or a depression of free thyroid hormones and a normal TSH. This elevation is typically due to a source of exogenous estrogens. The most common cause of this pattern is from oral contraceptives and Premarin®.[125-128]

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Estrogen levels



Please see registered trademark ownership in Appendix 1.

20. *Depression of thyroid-binding globulin leading to increased percentage of free-fraction thyroid hormones*[129-131,232]

When thyroid-binding globulin levels drop, the percentage of free thyroid hormones elevates. This may be observed by an elevated T3 uptake or elevated free thyroid hormones with a normal TSH. This elevation is typically subtle and does not create symptoms of hyperthyroidism. Instead, the individual develops symptoms of thyroid hypofunction due to thyroid receptor-site resistance. This pattern is typically caused by elevations of estrogens. Endogenous elevation in females is typically associated with metabolic syndrome. The constant unrelenting surges of insulin in metabolic syndrome stimulate the activity of 17,20-lyase in the theca cells of the ovaries, which lead to elevations of testosterone. This pattern may also be promoted by exogenous overload of testosterone.[129-131]

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Sugar metabolism
- The testosterone clearance mechanism

21. *Thyroid resistance promoted by an elevation of cytokines*[28-31]

Inflammatory cytokines alter thyroid function and thyroid receptor-site proteomic responses.[28-30] Proinflammatory cytokines have been associated with NTI in view of their capability to decrease D1 TR-beta-1 mRNA expression. Proinflammatory cytokine induction leads to activation of the inflammatory pathways nuclear factor-kappa-B (NF-kB) and activator protein-1. The proinflammatory cytokine interleukin-1 beta decreases TR-beta-1 mRNA in an NF-kB–dependent way, leading to decreased post-translational receptor activity.[31]

Clinically, the amount of receptor-site expression cannot be measured in private practice and assumptions about thyroid receptor-site activity must be made with elevated cytokines. Active immune-related patterns may hinder performance at levels of thyroid metabolism and a careful evaluation of immune-related disorders must be provided to all individuals who present with thyroid symptomatology. A perfect example of this clinically is the autoimmune thyroid patient or a patient with an inflammatory disease that is taking replacement hormones but is still suffering from thyroid symptoms. In these scenarios, the elevation of the inflammatory immune cytokines are depressing thyroid receptor-site responsiveness and the replacement hormones are creating a normal appearance of thyroid laboratory markers (TSH, T4, and T3). The concept of cytokines suppressing thyroid receptor-site responsiveness is not clinically acknowledged and the patient ends up without proper management.

22. *Thyroid resistance promoted by elevations of cortisol*[132-134]

Once thyroid hormones bind to receptor sites, a series of biochemical reactions called intercellular transduction is initiated. This intercellular transduction response carries the message of binding to nuclear receptors. Once the nuclear receptor has been activated, it will respond by producing proteins that express enhanced metabolic rate and energy production. Elevations in cortisol have been shown to decrease the expression of both alpha-1 and -2 thyroid receptor-site responses.

| For treatment of this condition:<br>Seek appropriate conventional medical management and drug administration by a physician. |
|---|

Consider diet, nutrition, and lifestyle support for:

- Circadian rhythms

23. *Thyroid resistance promoted by deficiencies of vitamin A*[6]

Once thyroid hormones bind to receptor sites, a series of biochemical reactions called "intercellular transduction" is initiated. This intercellular transduction response carries the message of binding to nuclear receptors. Once the nuclear receptor has been activated, it will respond by producing proteins that express enhanced metabolic rate and energy production. Vitamin A appears to influence thyroid hormone nuclear receptors. Thyroid hormone transcription activation involves vitamin A-dependent, retinoic acid–specific receptors.[6]

Vitamin A deficiency should be suspected in iron-deficient and iron-anemic patients, individuals with liver disease and cirrhosis, and alcoholics. The key clinical symptom of a vitamin A deficiency is difficulty with night vision not related to astigmatism. The mechanism of vitamin deficiency needs to be clinically addressed.

For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Vitamin A levels (with thyroid cofactors)

24. *Thyroid resistance promoted by elevations of homocysteine*[288]

Once thyroid hormones bind to receptor sites, a series of biochemical reactions called intercellular transduction is initiated. This intercellular transduction response carries the message of binding to nuclear receptors. Once the nuclear receptor has been activated, it will respond by producing proteins that express enhanced metabolic rate and energy production. Elevations in homocysteine have demonstrated the ability to dampen the expression of thyroid hormones.

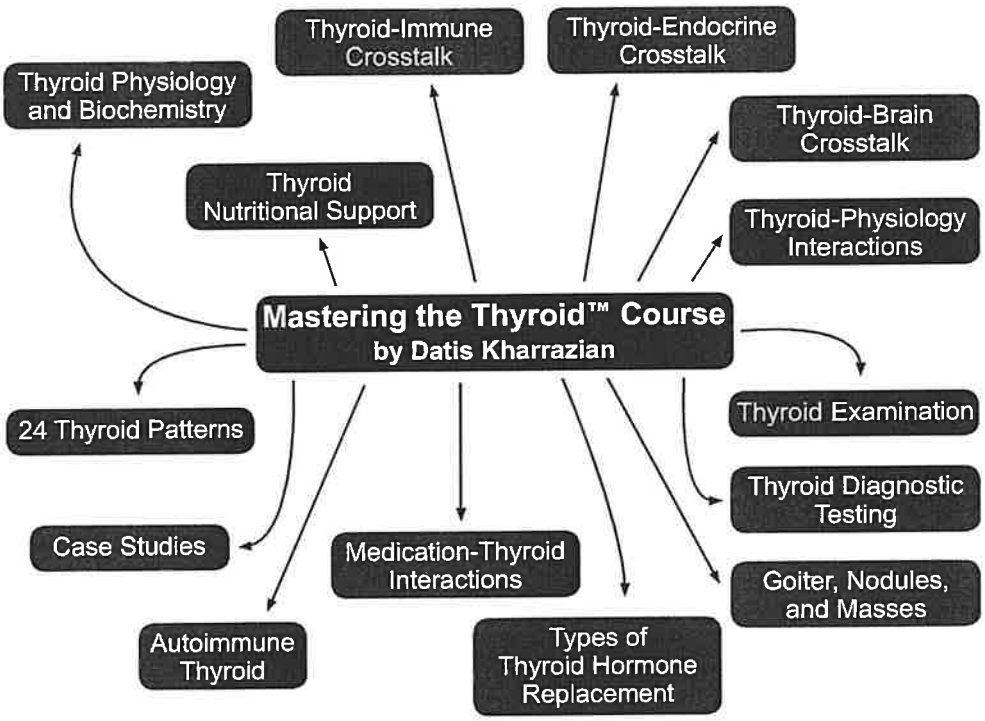
Elevations in homocysteine should always be considered in individuals taking antacid medications, oral contraceptives, and estrogens. Individuals with hypochlorhydria and *H. pylori* infections are also susceptible to elevations in homocysteine. These patterns are difficult to identify, since the patient has hypothyroid symptoms and normal thyroid panels. Elevated homocysteine produces no subjective indication, and therefore, objective serum testing must be conducted to identify this pattern. A history of risk factors related to homocysteine should warrant testing. Nutritional support with a high dose of methyl donors, such as homocysteine, is warranted.

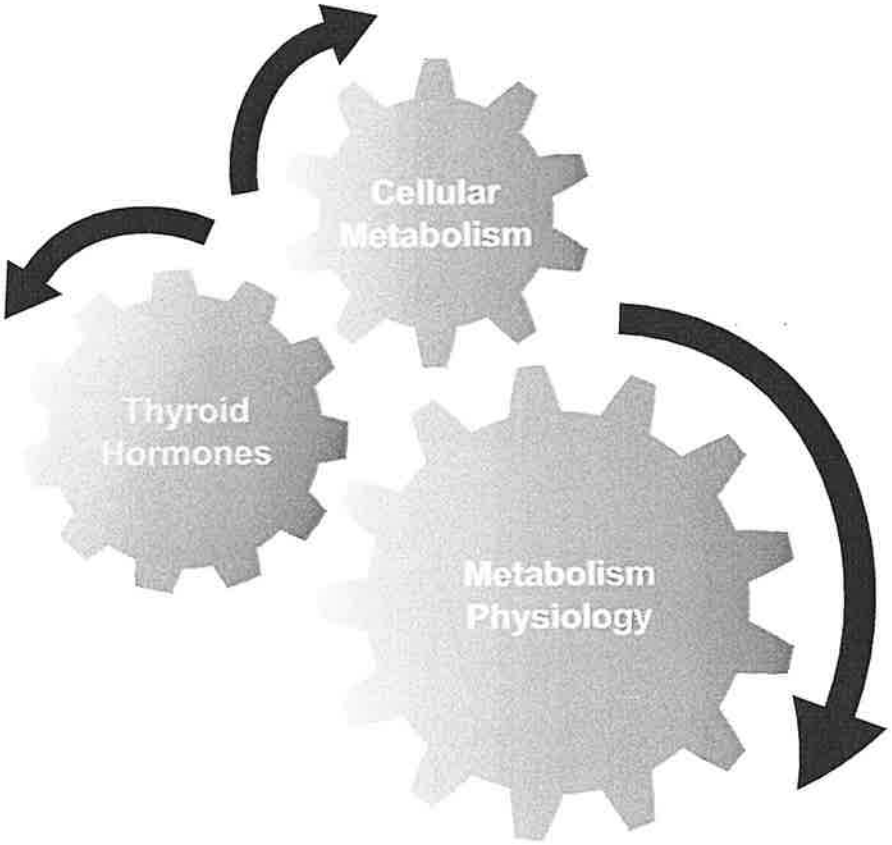
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

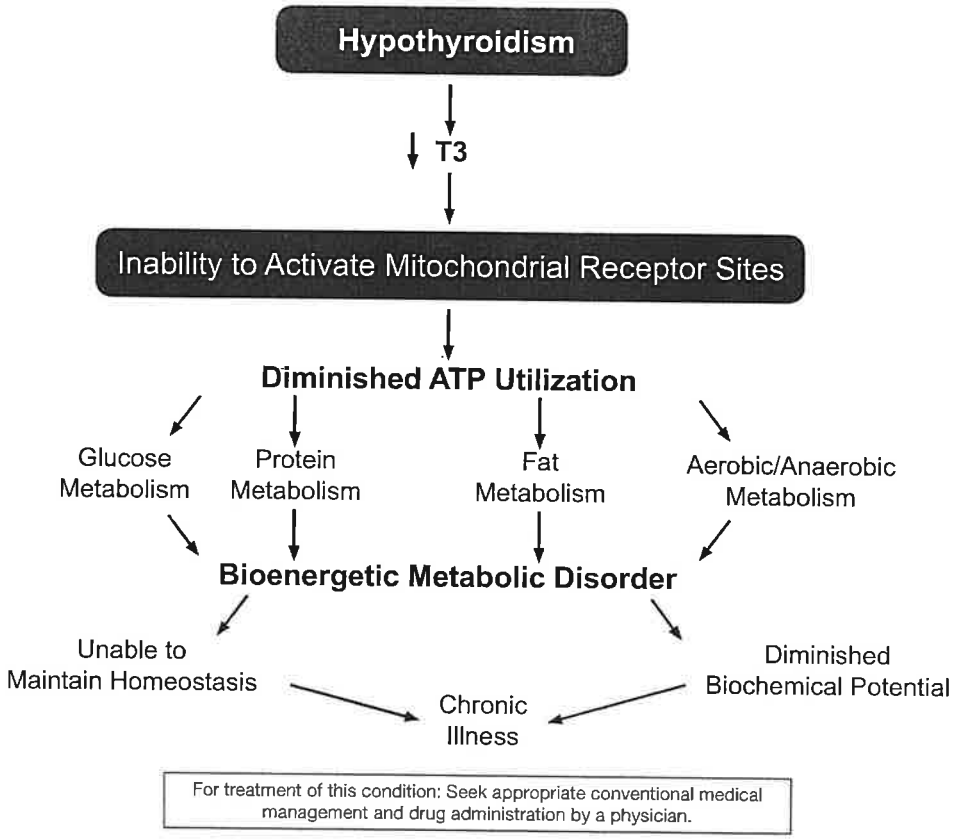
Consider diet, nutrition, and lifestyle support for:

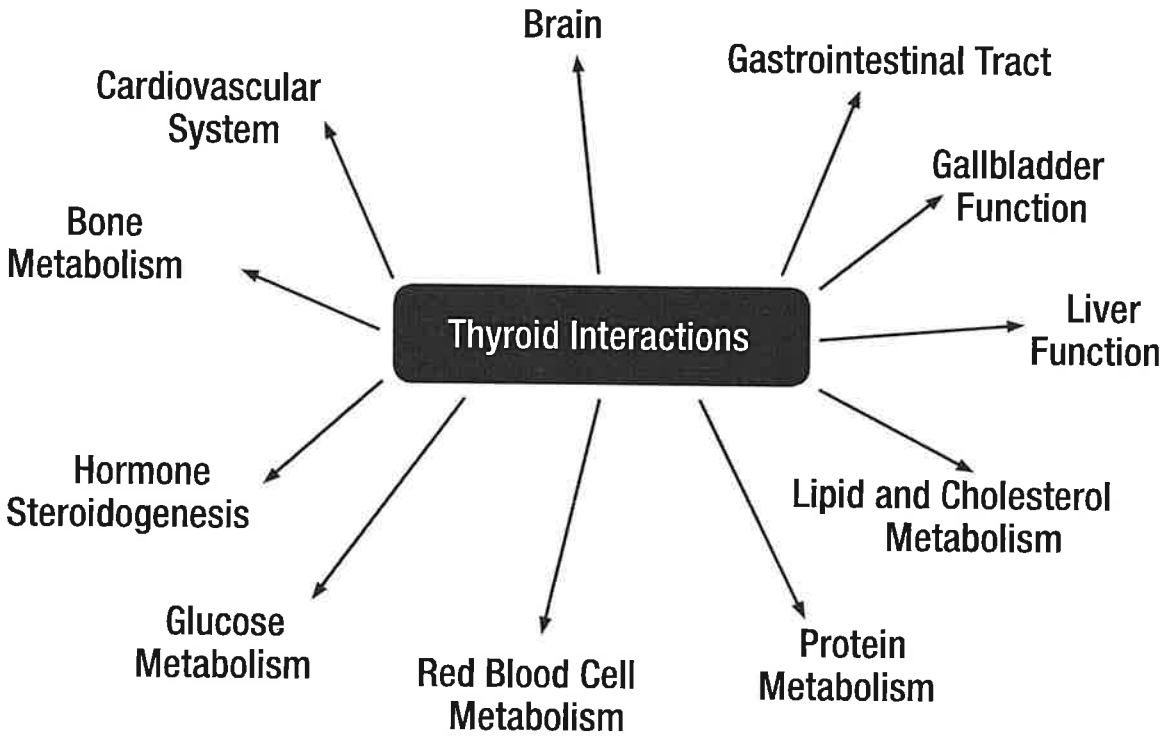
- Methyl donors

# Charts


Thyroid-Immune Crosstalk
Thyroid-Endocrine Crosstalk
Thyroid Physiology and Biochemistry
Thyroid-Brain Crosstalk
Thyroid Nutritional Support
Thyroid-Physiology Interactions
Mastering the Thyroid™ Course
by Datis Kharrazian
24 Thyroid Patterns
Thyroid Examination
Thyroid Diagnostic Testing
Case Studies
Medication-Thyroid Interactions
Goiter, Nodules, and Masses
Autoimmune Thyroid
Types of Thyroid Hormone Replacement


Cellular Metabolism
Thyroid Hormones
Metabolism Physiology

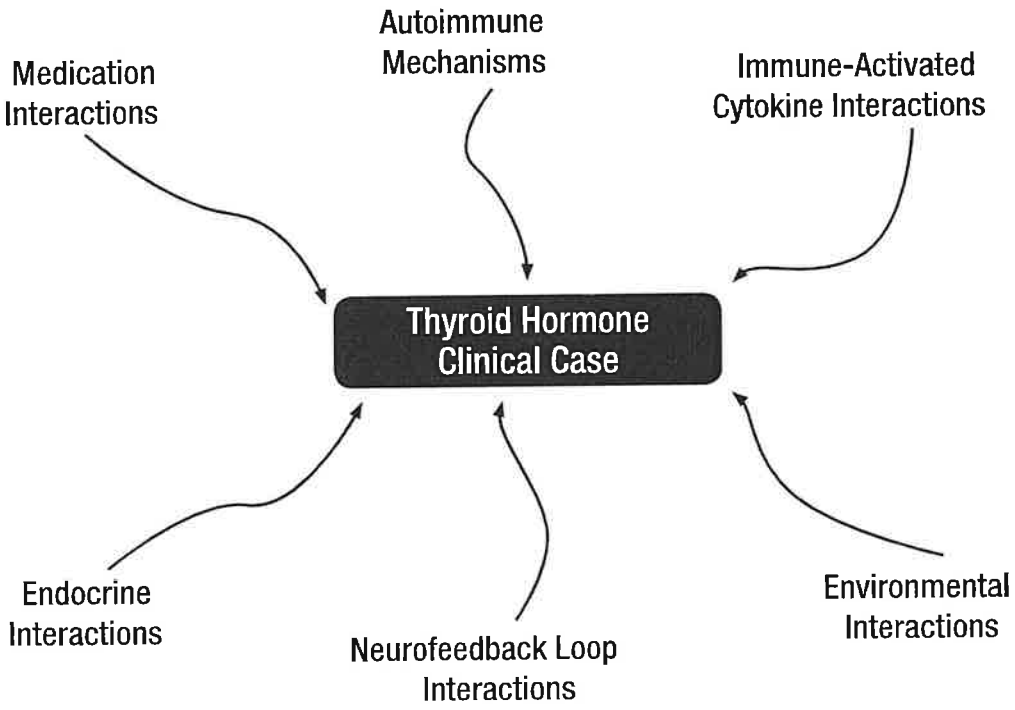

Hypothyroidism
T3
Inability to Activate Mitochondrial Receptor Sites
Diminished ATP Utilization
Glucose Metabolism
Protein Metabolism
Fat Metabolism
Aerobic/Anaerobic Metabolism
Bioenergetic Metabolic Disorder
Unable to Maintain Homeostasis
Diminished Biochemical Potential
Chronic Illness
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.


Thyroid Interactions
Brain
Gastrointestinal Tract
Gallbladder Function
Liver Function
Lipid and Cholesterol Metabolism
Protein Metabolism
Red Blood Cell Metabolism
Glucose Metabolism
Hormone Steroidogenesis
Bone Metabolism
Cardiovascular System

## TOP 50 MOST PRESCRIBED MEDICATIONS

1. Hydrocodone (pain)
2. Lisinopril (hypertension)
3. Simvastatin (high cholesterol)
4. Levothyroxine (hypothyroidism)
5. Amoxicillin (bacterial infection)
6. Azithromycin (bacterial infection)
7. Lipitor® (high cholesterol)
8. Hydrochlorothiazide (edema/hypertension)
9. Alprazolam (anxiety/depression)
10. Atenolol (hypertension)
11. Metformin (type 2 diabetes)
12. Metoprolol Succinate (hypertension)
13. Furosemide, oral (edema/hypertension)
14. Metoprolol tartrate (hypertension)
15. Sertraline (depression)
16. Omeprazole (ulcers/reflux)
17. Zolpidem tartrate (insomnia)
18. Nexium® (ulcers/reflux)
19. Lexapro® (depression)
20. Oxycodone (pain)
21. Singulair® (asthma)
22. Ibuprofen (pain/inflammation)
23. Plavix® (blood clotting)
24. Prednisone, oral (allergies/inflammation)
25. Fluoxetine (depression)
26. Synthroid® (hypothyroidism)
27. Warfarin (blood clotting)
28. Cephalexin (bacterial infection)
29. Lorazepam (anxiety)
30. Clonazepam (epilepsy/anxiety)
31. Citalopram HBR (depression)
32. Tramadol (pain)
33. Gabapentin (epilepsy/pain)
34. Ciprofloxacin HCl (bacterial infection)
35. Propoxyphene-N (pain)
36. Lisinopril (hypertension)
37. Triamterene (edema/hypertension)
38. Amoxicillin (bacterial infection)
39. Cyclobenzaprine (muscle injury/spasm)
40. Prevacid® (ulcers/reflux)
41. Advair® (asthma)
42. Effexor XR® (depression)
43. Trazodone HCl (depression)
44. Fexofenadine (allergy)
45. Fluticasone nasal spray (allergy)
46. Diovan® (hypertension)
47. Paroxetine (depression/anxiety)
48. Lovastatin (high cholesterol)
49. Crestor® (high cholesterol)
50. Trimethoprim (bacterial infection)

AARP. The 50 Most Prescribed Drugs. 2009 Oct. http://pubs.aarp.org/aarpbulletin/200910?pg=55#pg55.




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Evaluating a Thyroid Patient

**History**

- Symptoms
- Past History
- Family History
- Medications
- Hormone Replacement
- Supplements
- Outcomes

→

**Physical Examination**

- General Inspection
- Hand Inspection
- Neck Inspection
- Neck Palpation
- Eye Examination
- Neurological Examination
- Tibial Examination
- Auscultation

→

**Serum Laboratory Test**

- TSH
- Total T4
- Total T3
- Free T4
- Free T3
- TPO Ab
- TGB Ab
- TSI Ab
- Reverse T3
- Thyroglobulin
- T3 Uptake
- Free Thyroxine Index

→

**Diagnostic Studies**

- Ultrasound
- Radioactive Iodine Uptake
- Fine-Needle Aspiration

↙ ↘

**Pathological Disorder** ⟷ **Functional Disorder**

- Thyroid Metabolic Disorder
- Autoimmune Disorder
- Thyroiditis
- Euthyroid Sick Syndrome
- Goiter (Toxic or Nontoxic)
- Thyroglossal Cyst
- Chondroma
- Cancer

- 24 Patterns

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Medications

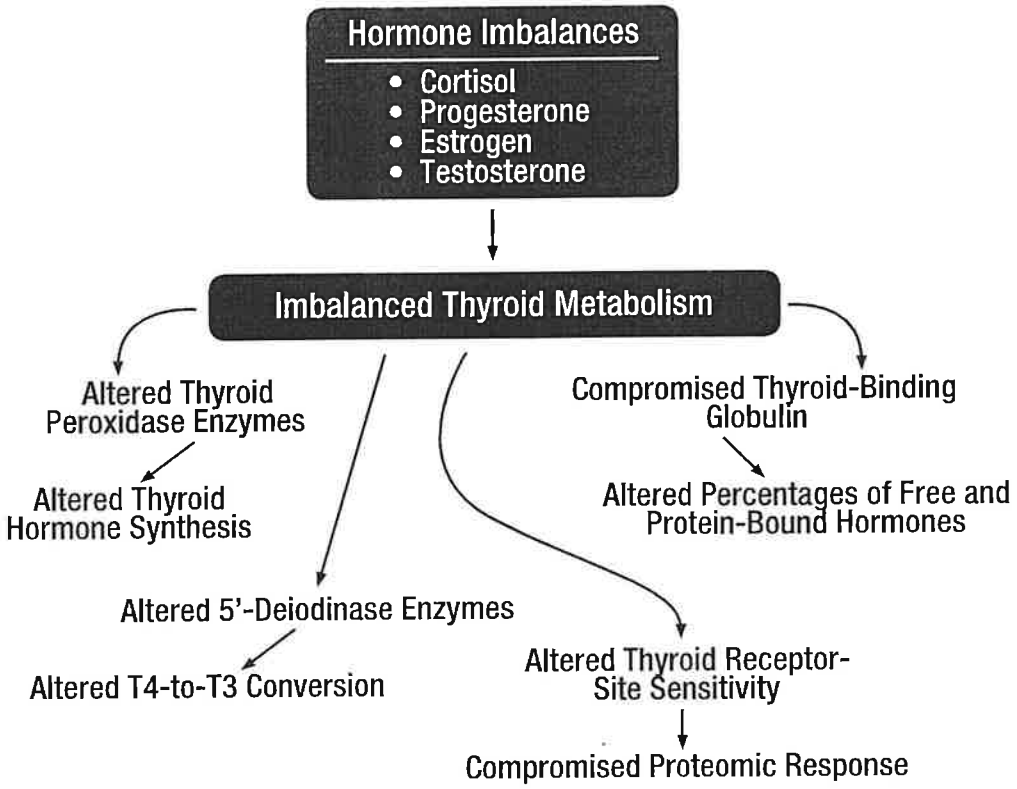
Anti-Inflammatory Agents
Antibiotics
Antidepressants
Diabetic Medication
Hypertensive Medication
Pain Medication
Antacids
Cholesterol-Lowering Medications

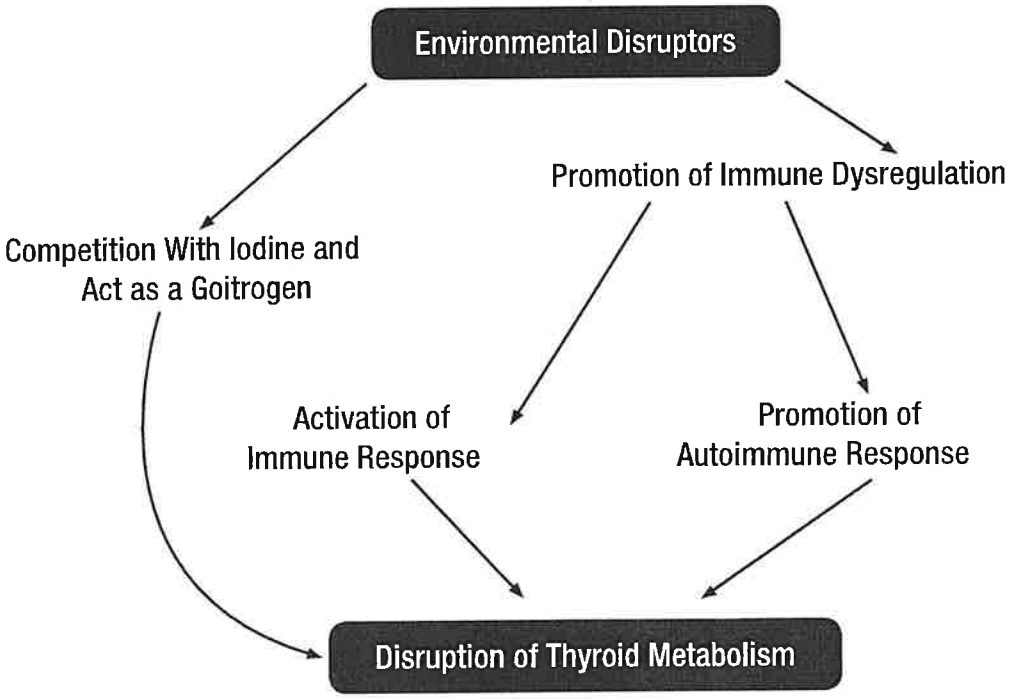
→ No Change in TSH → Undiagnosed Hypothyroid

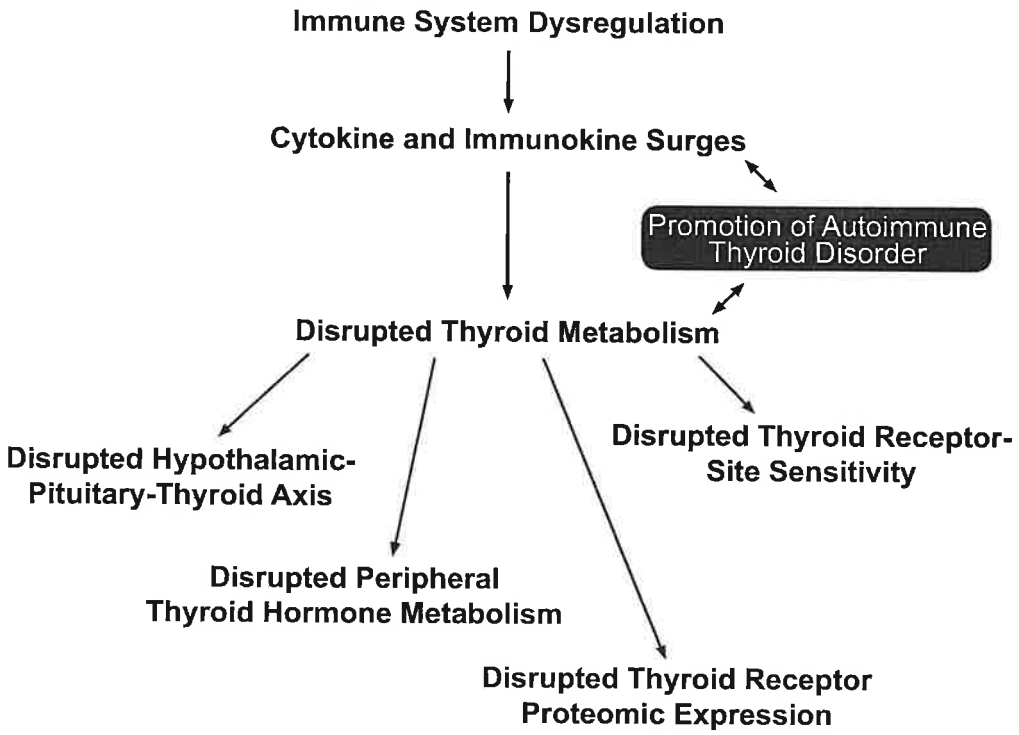
↘ Disrupted Thyroid Hormone Metabolism

- ↙ Altered Thyroid Hormone Secretion
- ↙ Decreased Absorption of Hormones
- ↙ Altered Thyroid-Binding Globulin Levels
- ↓ Altered Thyroid Hormone Clearance
- ↙ Altered TSH Responsiveness
- ↙ Altered Thyroid Receptor Responsiveness
- ↘ Dysregulated HPT Feedback Loop
- ↘ Decreased T4-to-T3 Production

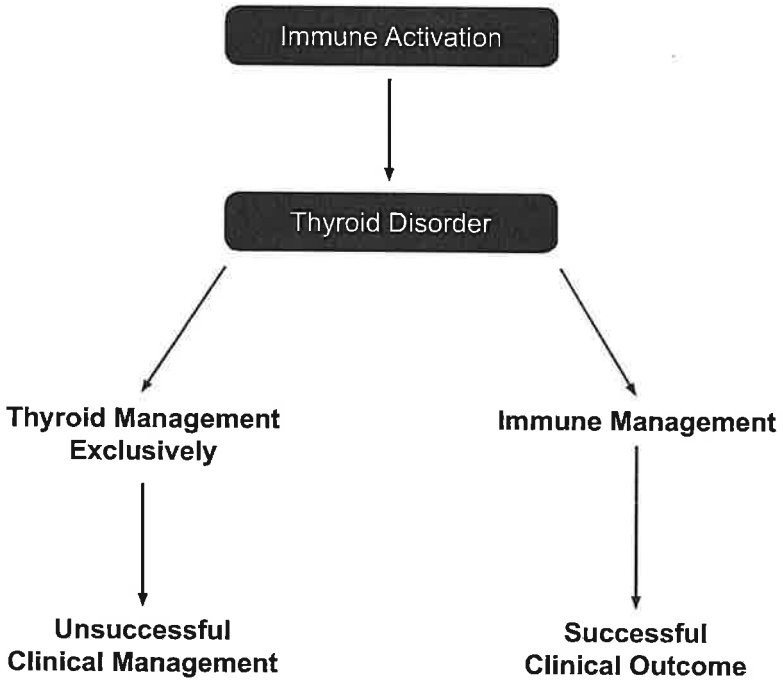
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.


Hormone Imbalances
• Cortisol
• Progesterone
• Estrogen
• Testosterone
Imbalanced Thyroid Metabolism
Altered Thyroid Peroxidase Enzymes
Altered Thyroid Hormone Synthesis
Altered 5'-Deiodinase Enzymes
Altered T4-to-T3 Conversion
Compromised Thyroid-Binding Globulin
Altered Percentages of Free and Protein-Bound Hormones
Altered Thyroid Receptor-Site Sensitivity
Compromised Proteomic Response


Environmental Disruptors
Competition With Iodine and Act as a Goitrogen
Promotion of Immune Dysregulation
Activation of Immune Response
Promotion of Autoimmune Response
Disruption of Thyroid Metabolism

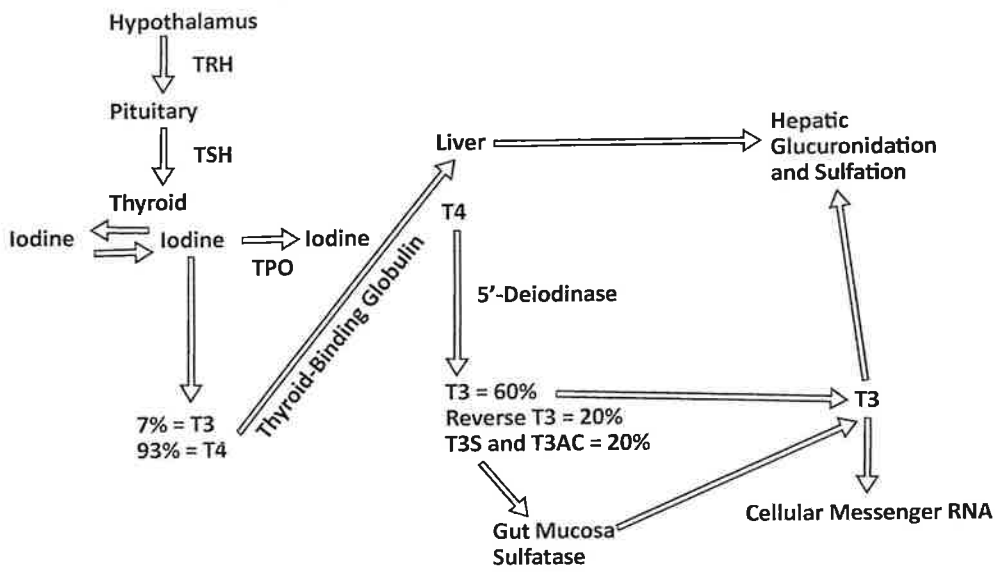


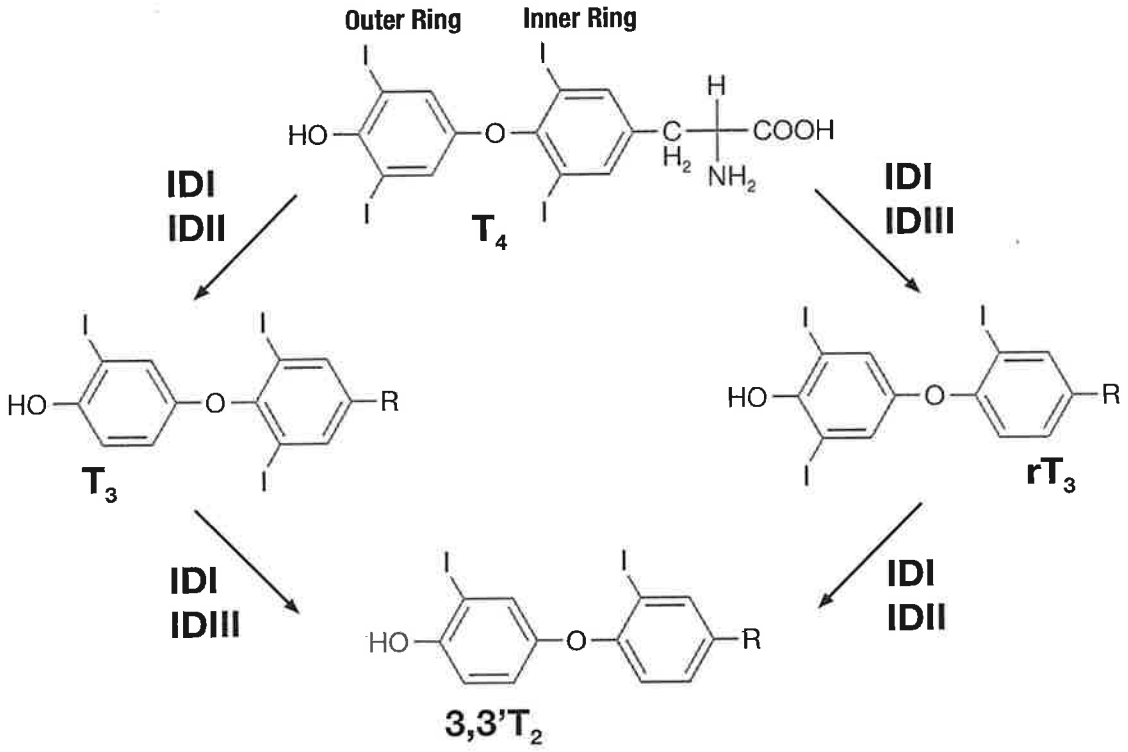

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



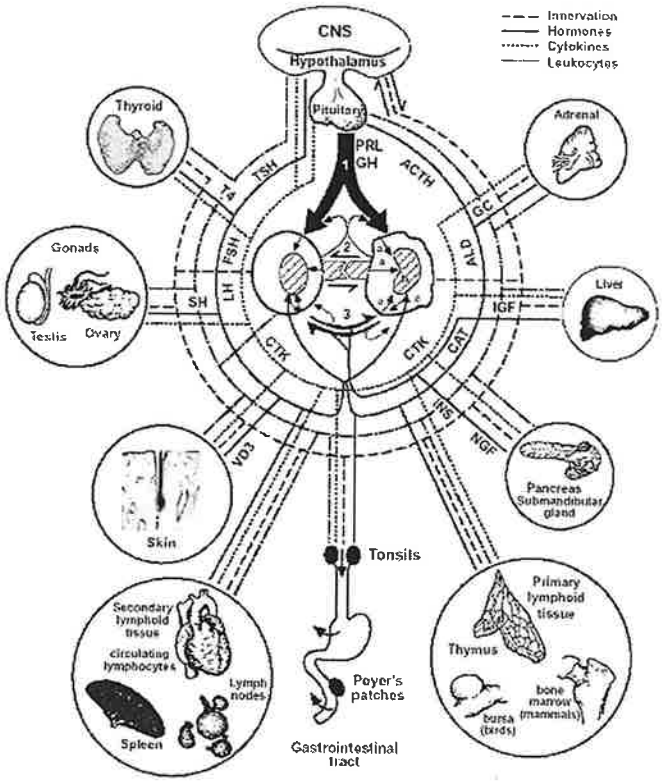

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

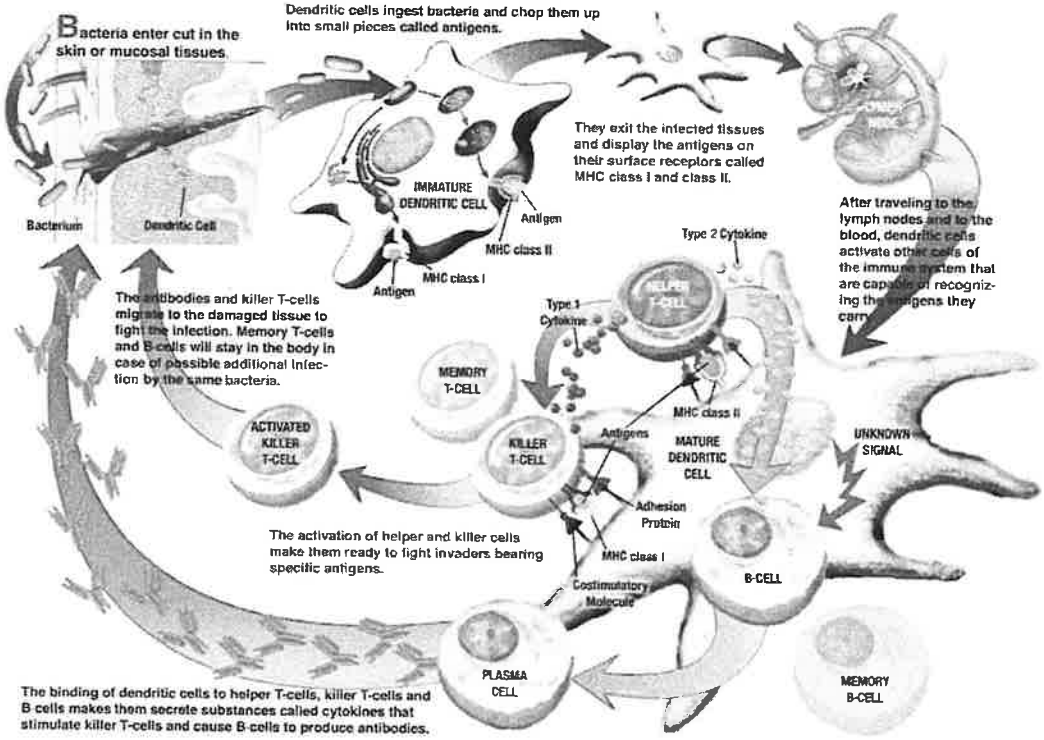
# THYROID METABOLISM



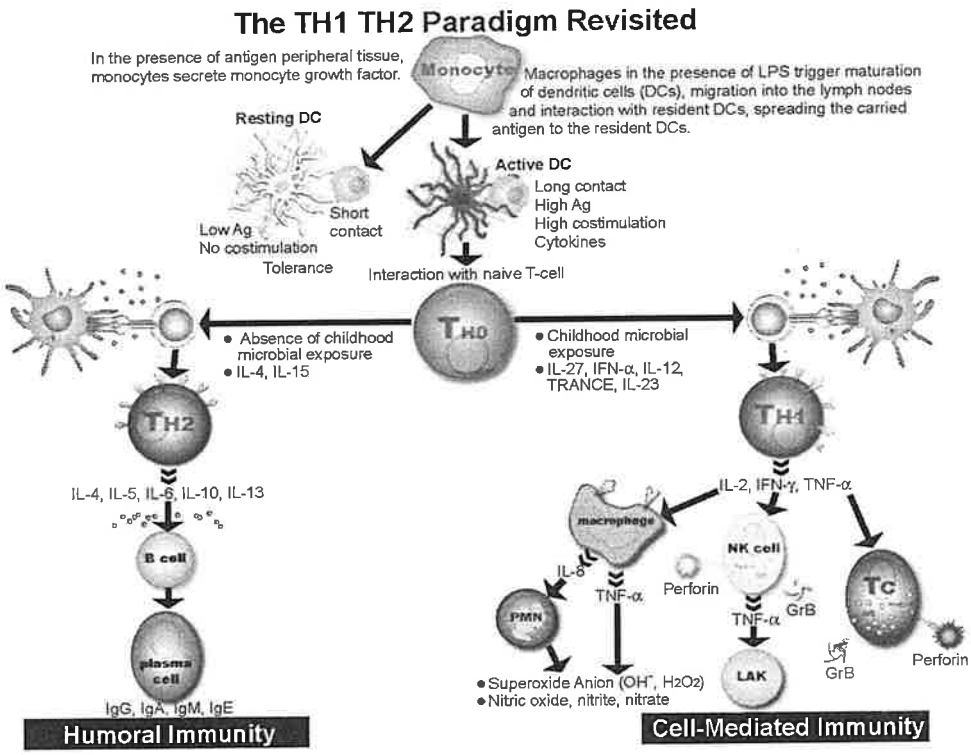



## The Neuroimmune Regulatory Network




*Used with permission of Aristo Vojdani, PhD, MSc, CLS and Cyrex Laboratories, LLC.*




*Used with permission of Arista Vojdani, PhD, MSc; CLS and Cyrex Laboratories, LLC.*

# Cytokines and the HPT Axis

Immune Cell Activation

↓

IL-1beta, IL-6, TNF-alpha

↓

Altered Signaling of the Hypothalamic Paraventricular Cells

↓

Altered HPT Axis

↓

Depressed TSH

↓

Thyroid Hormone Response Due to Preexisting Hormone Receptor-Site Sensitivity

↓

Patient Demonstrates Thyroid Symptoms – Euthyroid

## Cytokines and Peripheral Thyroid Hormone Conversion

Immune Cell Activation

↓

↑IL-6, TNF-alpha, IFN-gamma, IL-1beta

↓

Downregulate the Expression of 5'-deiodinase

↓

Diminished T3, Increased Reverse T3, Normal TSH

↓

Patient Demonstrates Thyroid Symptoms – Euthyroid

## Lipopolysaccharides and Thyroid Metabolism

Gut-Associated Lymphoid Tissue Activation

↓

Release of Lipopolysaccharides (LPS) From Gut Bacteria

↓

LPS Binds to Immune Complex

↓

Release of Inflammatory Cytokines

↙ ↓ ↘

T3 Receptor-Site Resistance

↓T3
↑Reverse T3

TSH

# Cytokine and Thyroid Receptor-Site Expression

Immune Cell Activation

↓

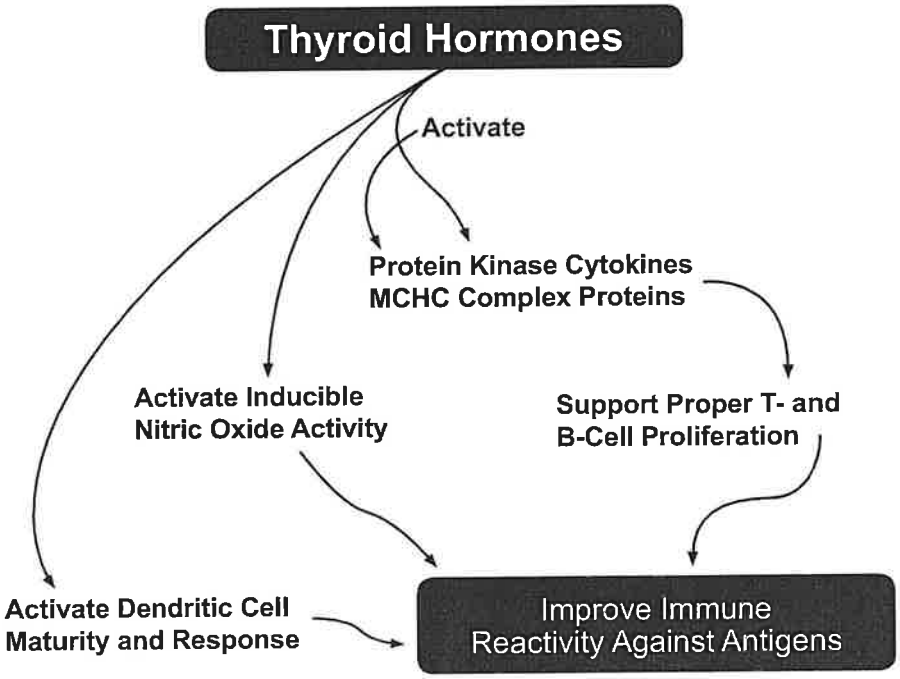
Increased Production of Inflammatory Cytokines

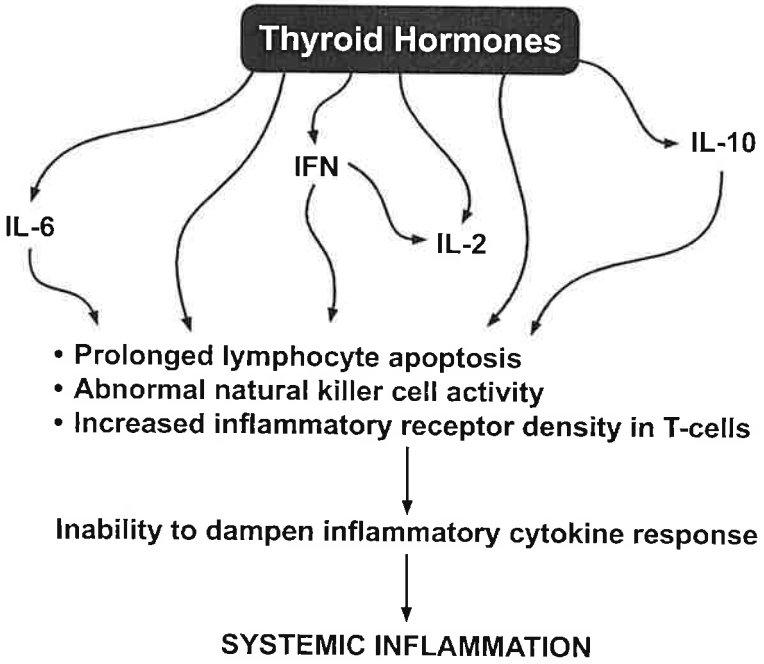
↓

Decreased Thyroid Hormone Receptor mRNA Expression

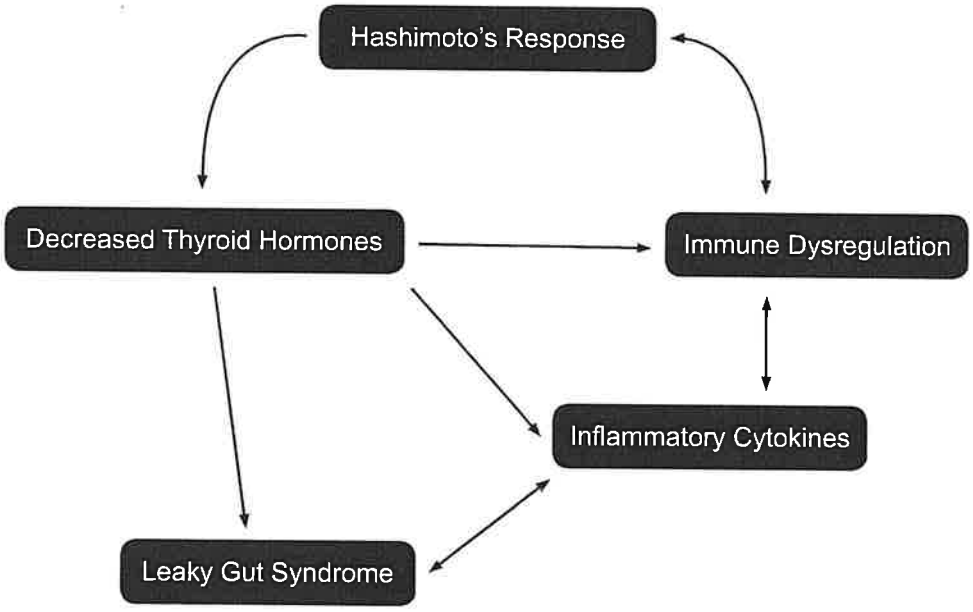
↓

Patient Demonstrates Thyroid Symptoms – Euthyroid






For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

**Thyroid Hormones**

**Modulate CD4 and CD8 Activity**

**Modulate TH-1 and TH-2 Expression**

**Influence the Potential to Regulate Abnormal Autoimmune Expression**

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

**Hashimoto's Autoimmune Thyroid Attack**

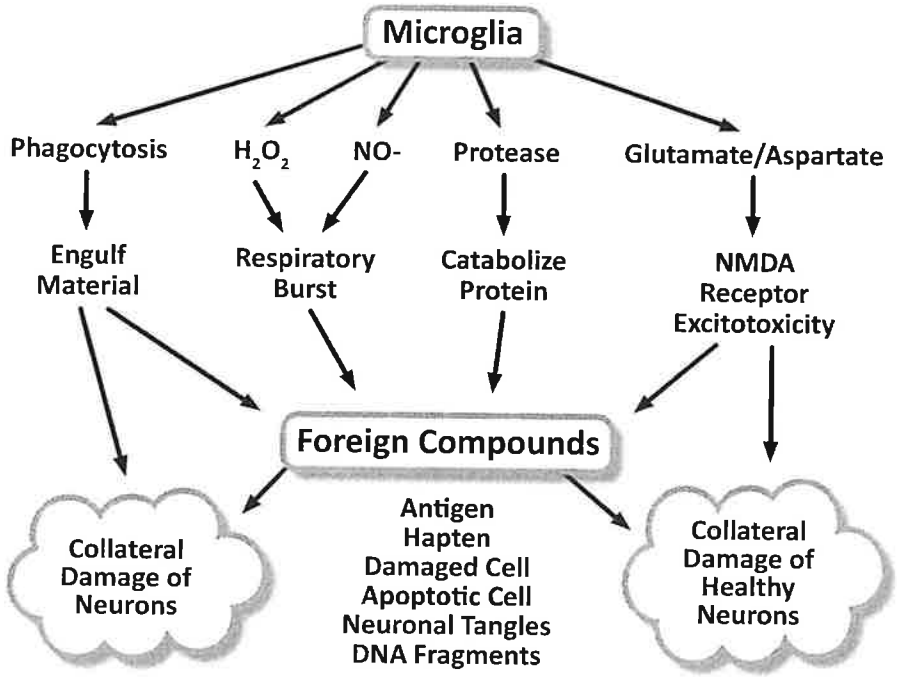
**Destroys Thyroid Gland**

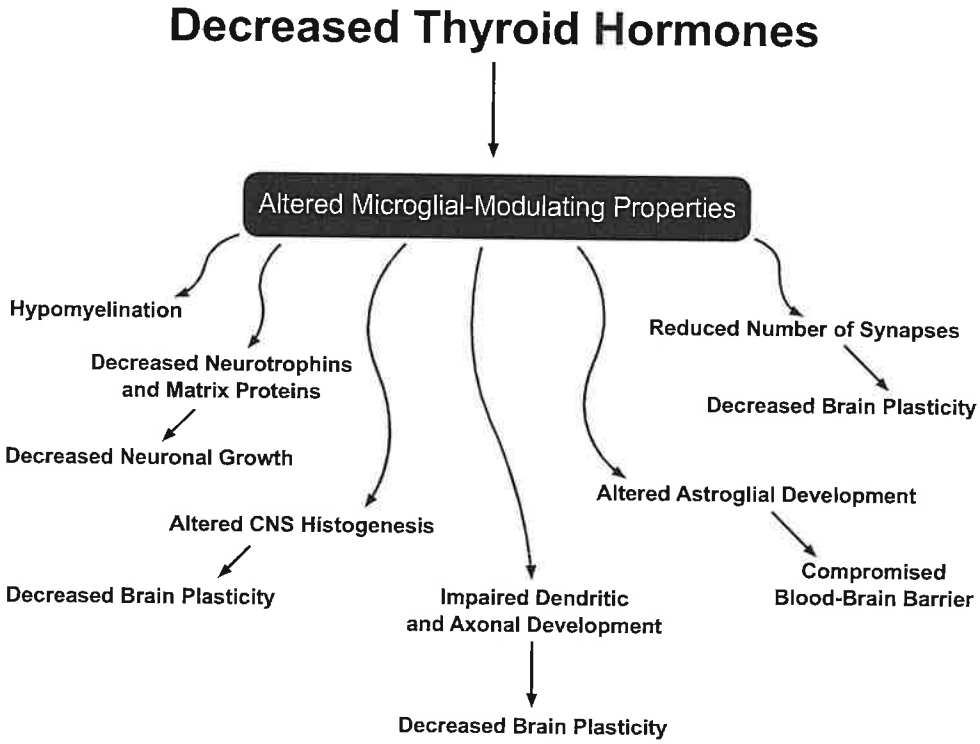
**Decreases Thyroid Hormone Production**

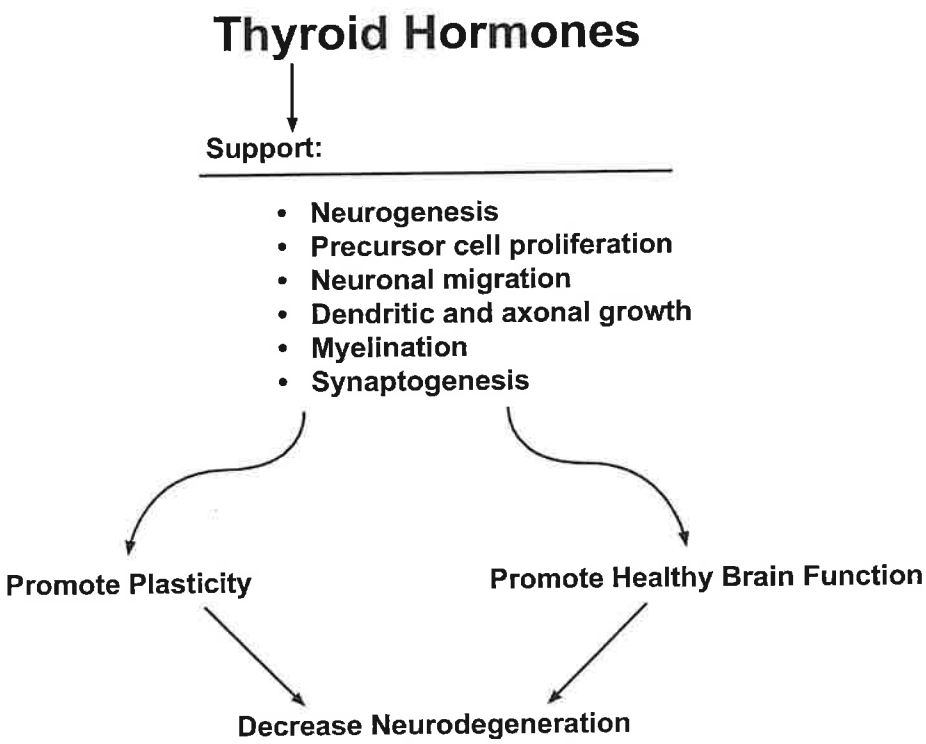
**Inability to Modulate TH-1/TH-2 Activity**

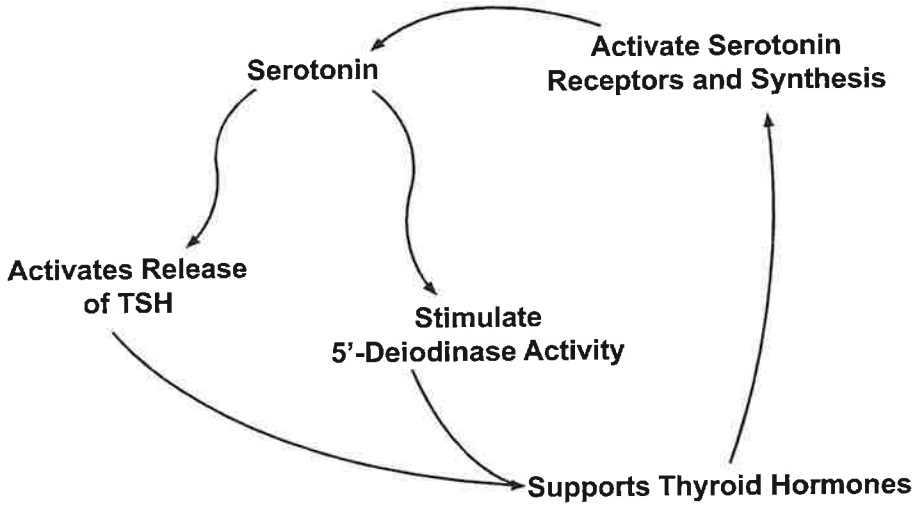
**Promotion and Flare-up of Hashimoto's Response**

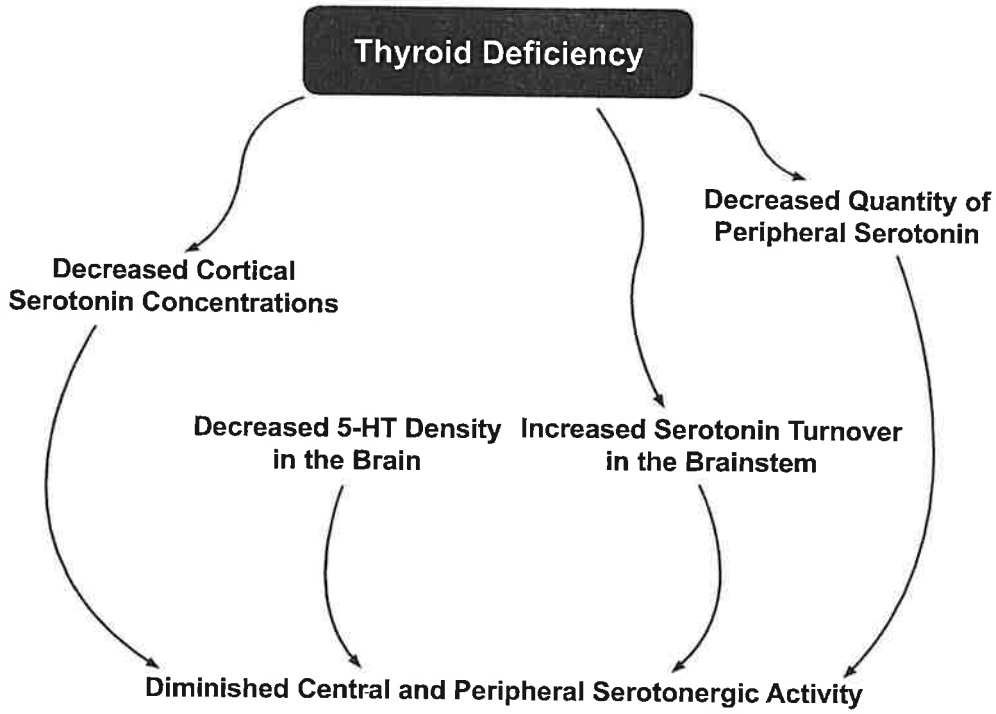
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

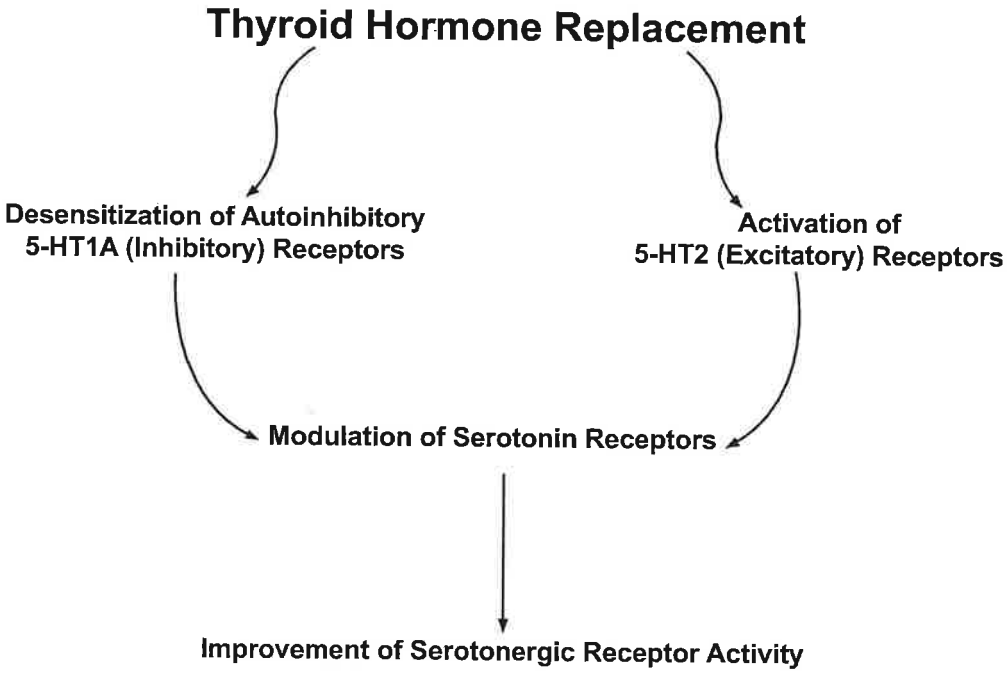

Microglia
Phagocytosis
$H_2O_2$
NO-
Protease
Glutamate/Aspartate
Engulf Material
Respiratory Burst
Catabolize Protein
NMDA Receptor Excitotoxicity
Foreign Compounds
Collateral Damage of Neurons
Antigen
Hapten
Damaged Cell
Apoptotic Cell
Neuronal Tangles
DNA Fragments
Collateral Damage of Healthy Neurons

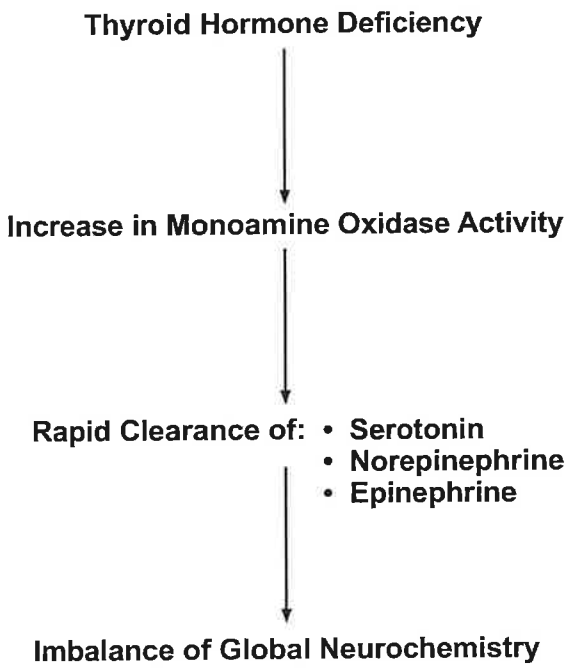

Decreased Thyroid Hormones
Altered Microglial-Modulating Properties
Hypomyelination
Decreased Neurotrophins and Matrix Proteins
Decreased Neuronal Growth
Altered CNS Histogenesis
Decreased Brain Plasticity
Impaired Dendritic and Axonal Development
Decreased Brain Plasticity
Altered Astroglial Development
Compromised Blood-Brain Barrier
Reduced Number of Synapses
Decreased Brain Plasticity


Thyroid Hormones
Support:
• Neurogenesis
• Precursor cell proliferation
• Neuronal migration
• Dendritic and axonal growth
• Myelination
• Synaptogenesis
Promote Plasticity
Promote Healthy Brain Function
Decrease Neurodegeneration

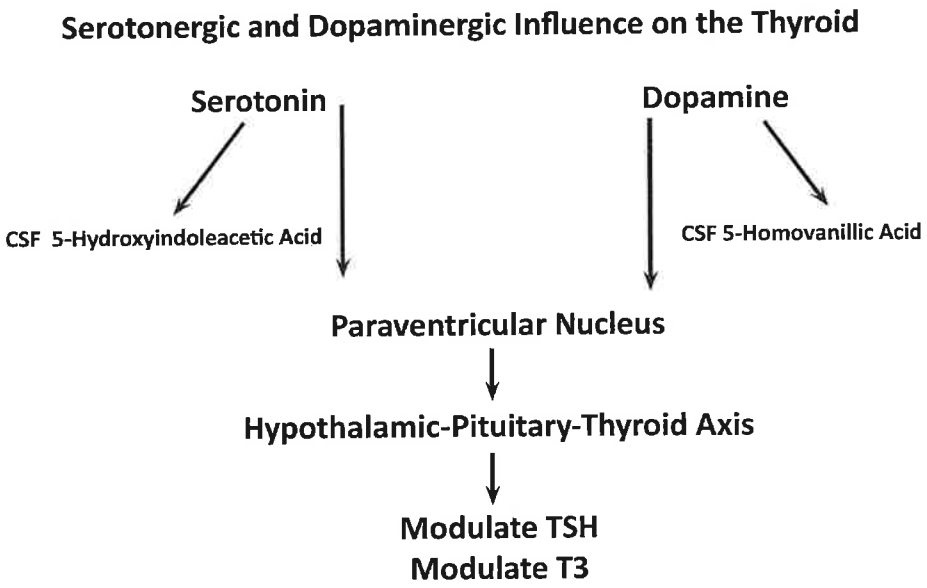

Serotonin
Activate Serotonin Receptors and Synthesis
Activates Release of TSH
Stimulate 5'-Deiodinase Activity
Supports Thyroid Hormones


Thyroid Deficiency
Decreased Quantity of Peripheral Serotonin
Decreased Cortical Serotonin Concentrations
Decreased 5-HT Density in the Brain
Increased Serotonin Turnover in the Brainstem
Diminished Central and Peripheral Serotonergic Activity

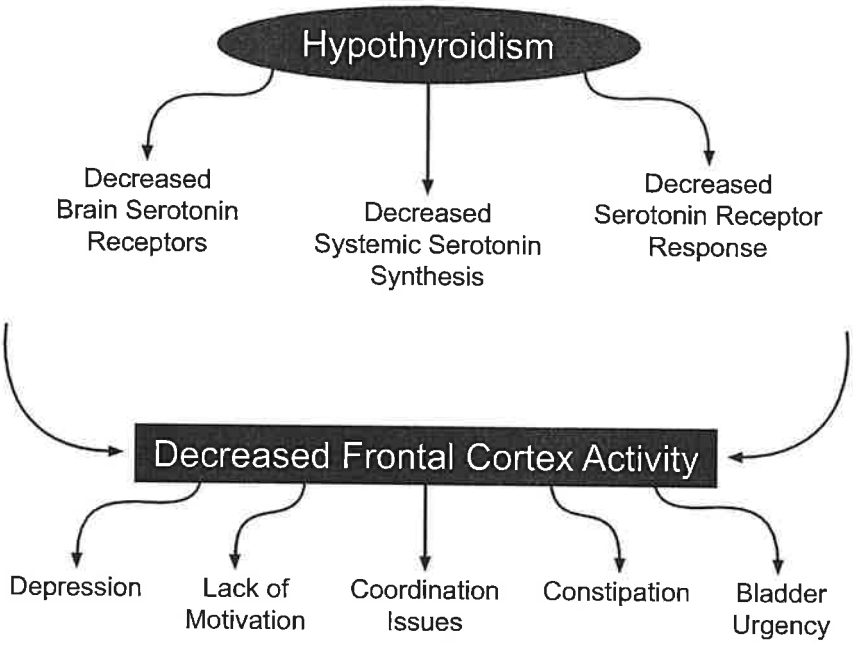

Thyroid Hormone Replacement
Desensitization of Autoinhibitory 5-HT1A (Inhibitory) Receptors
Activation of 5-HT2 (Excitatory) Receptors
Modulation of Serotonin Receptors
Improvement of Serotonergic Receptor Activity





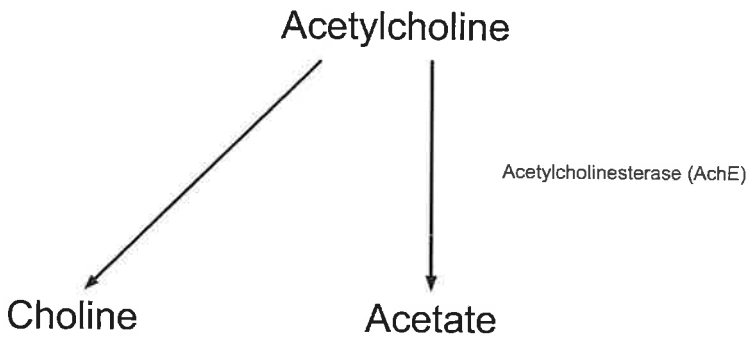

**No influence on T4*
**No influence on TBG*

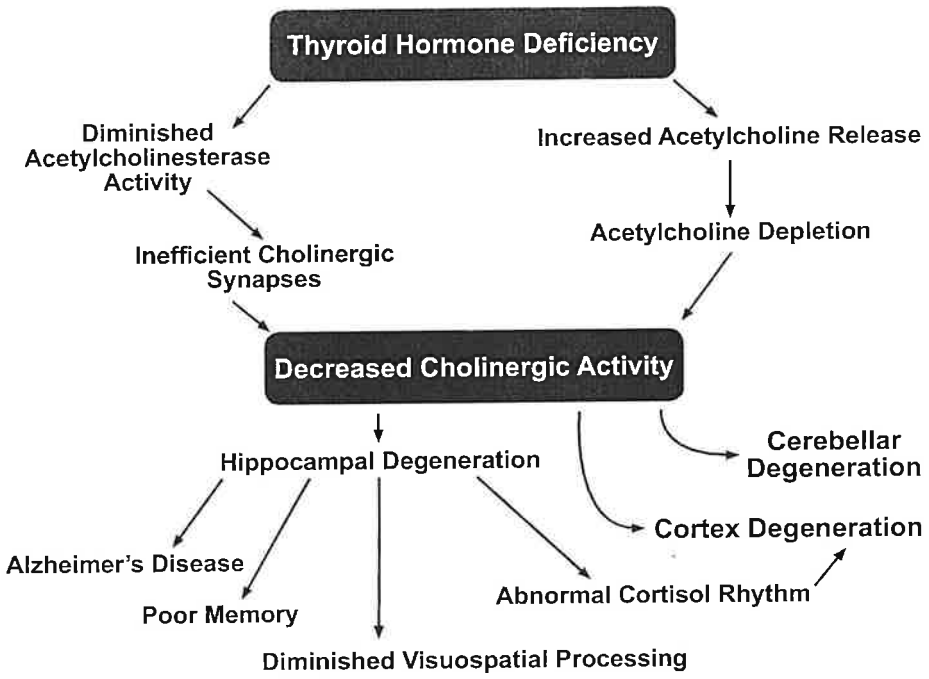



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

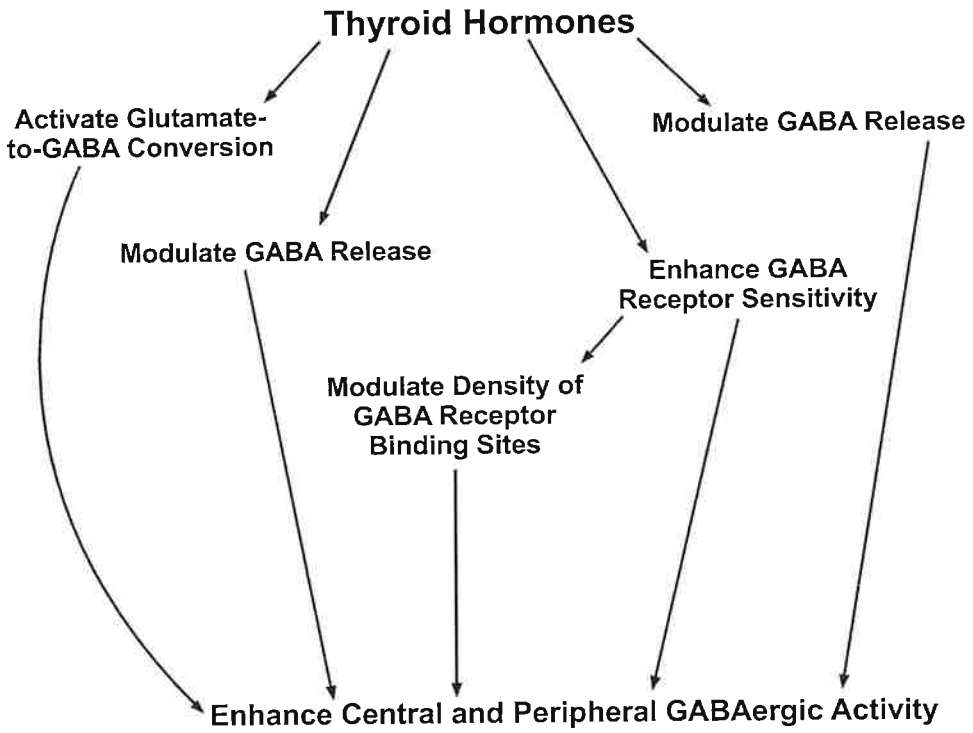


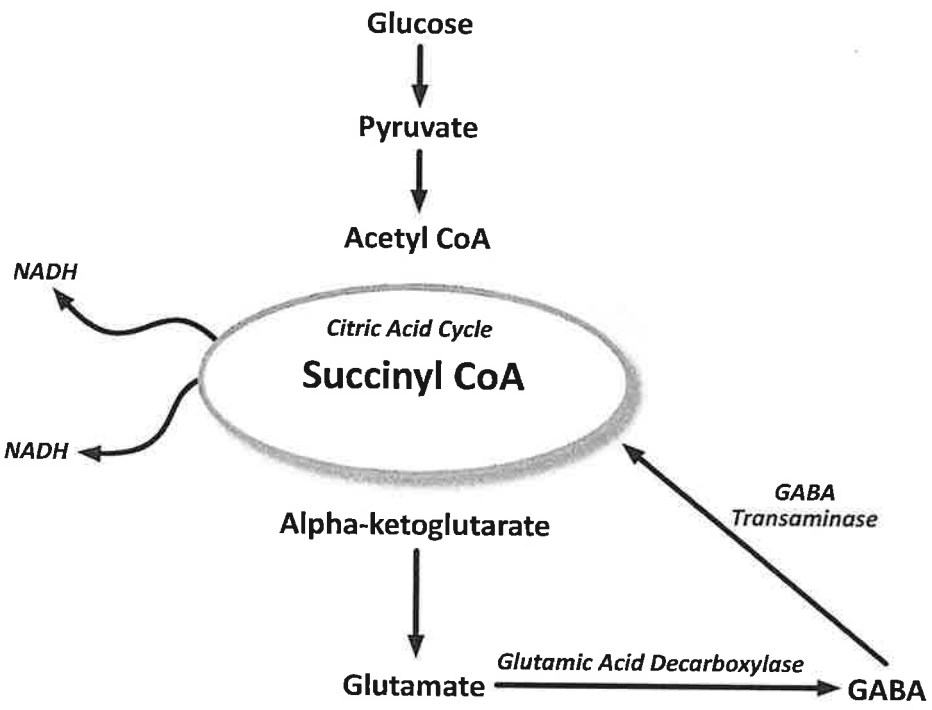
# Acetylcholine Catabolism

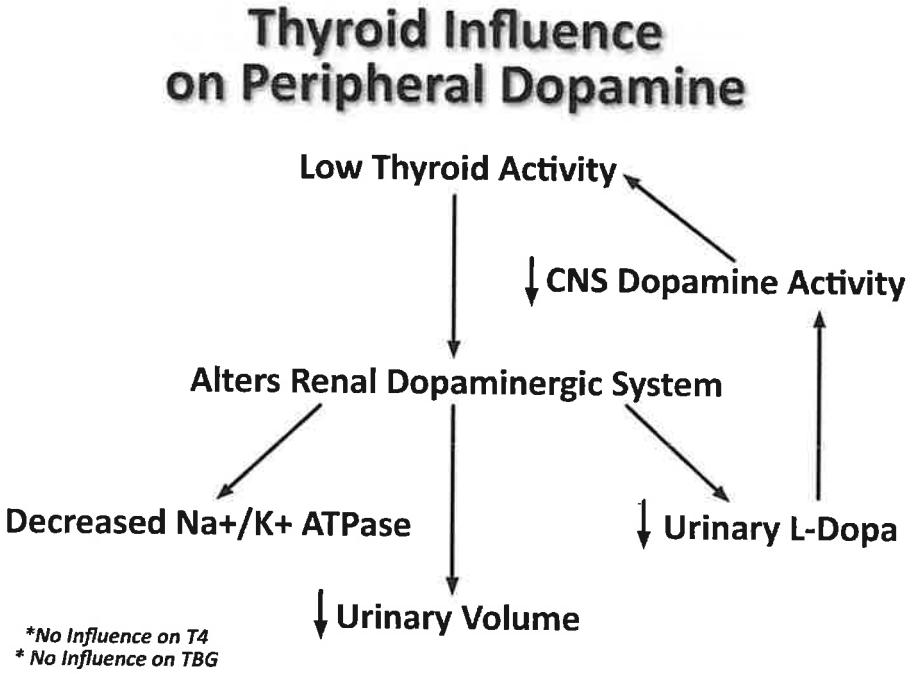




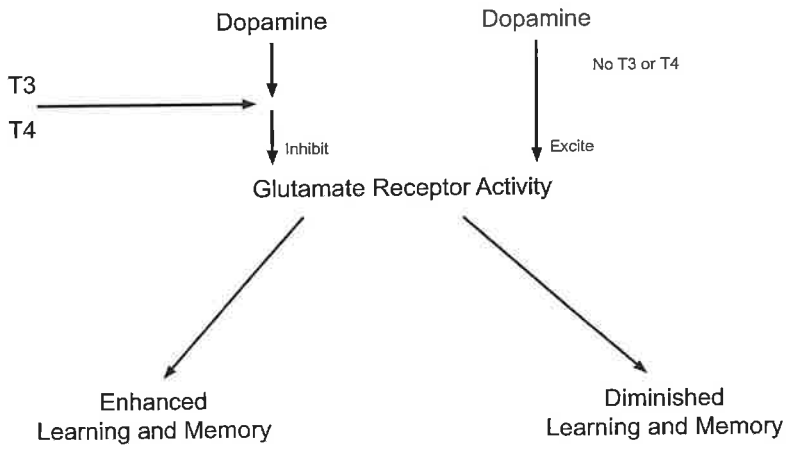

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



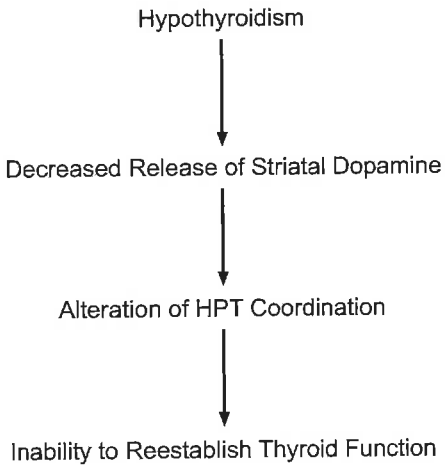

Glucose
Pyruvate
Acetyl CoA
NADH
Citric Acid Cycle
Succinyl CoA
NADH
GABA
Transaminase
Alpha-ketoglutarate
Glutamate
Glutamic Acid Decarboxylase
GABA


Thyroid Influence
on Peripheral Dopamine
Low Thyroid Activity
↓CNS Dopamine Activity
Alters Renal Dopaminergic System
Decreased Na+/K+ ATPase
↓Urinary L-Dopa
↓Urinary Volume
*No Influence on T4
* No Influence on TBG

# Thyroid and CNS Dopamine Response

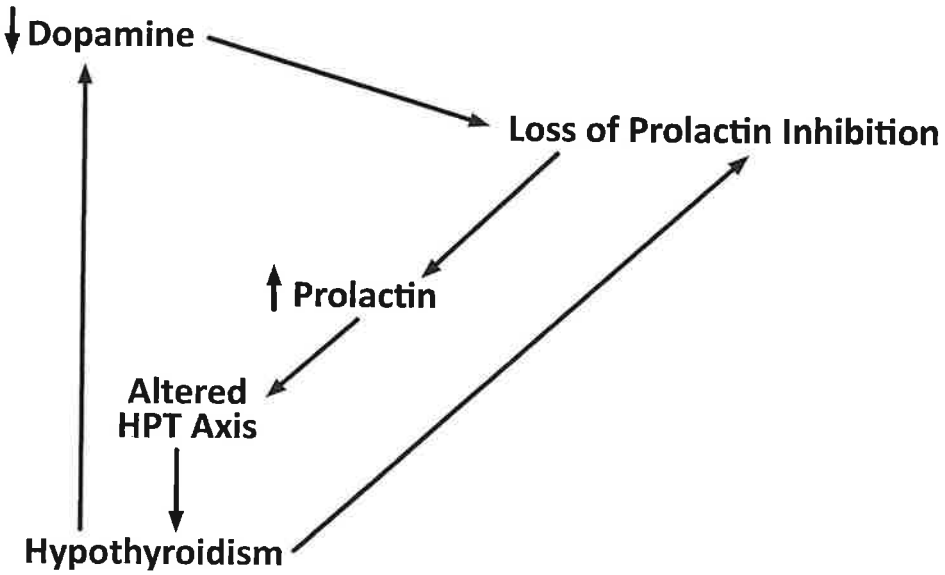


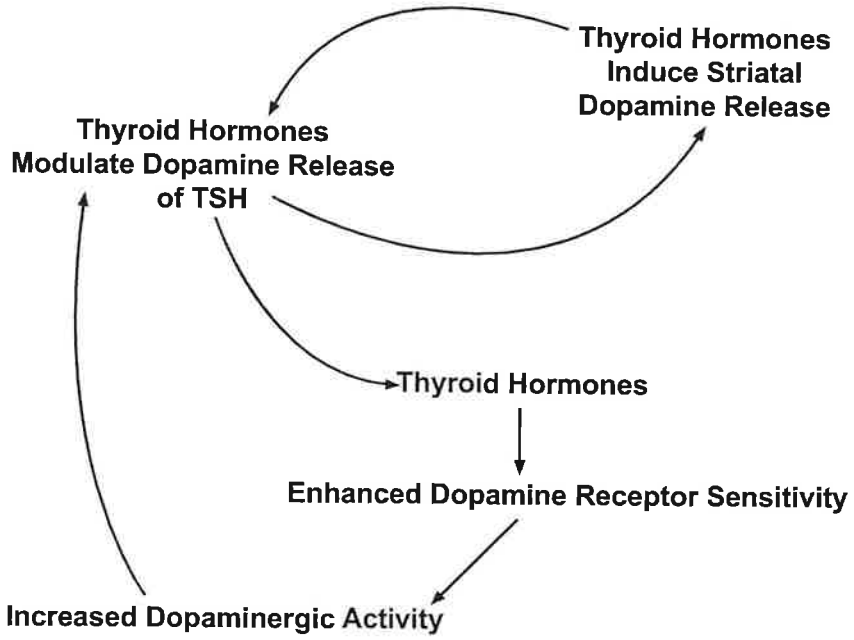

# Hypothyroidism and Dopamine Release



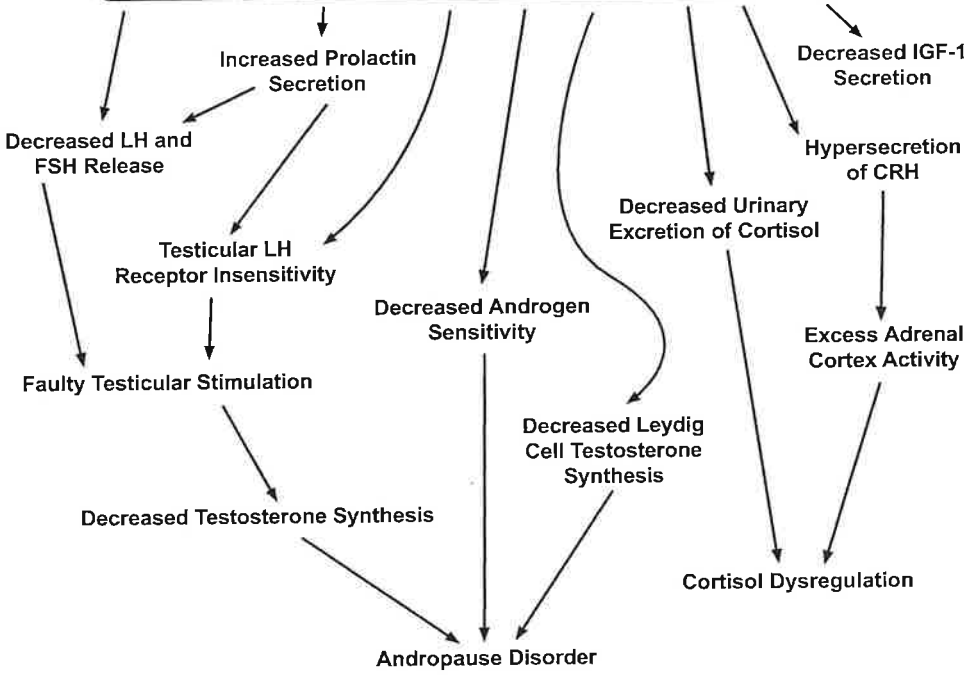

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Thyroid-Dopamine-Prolactin Loop



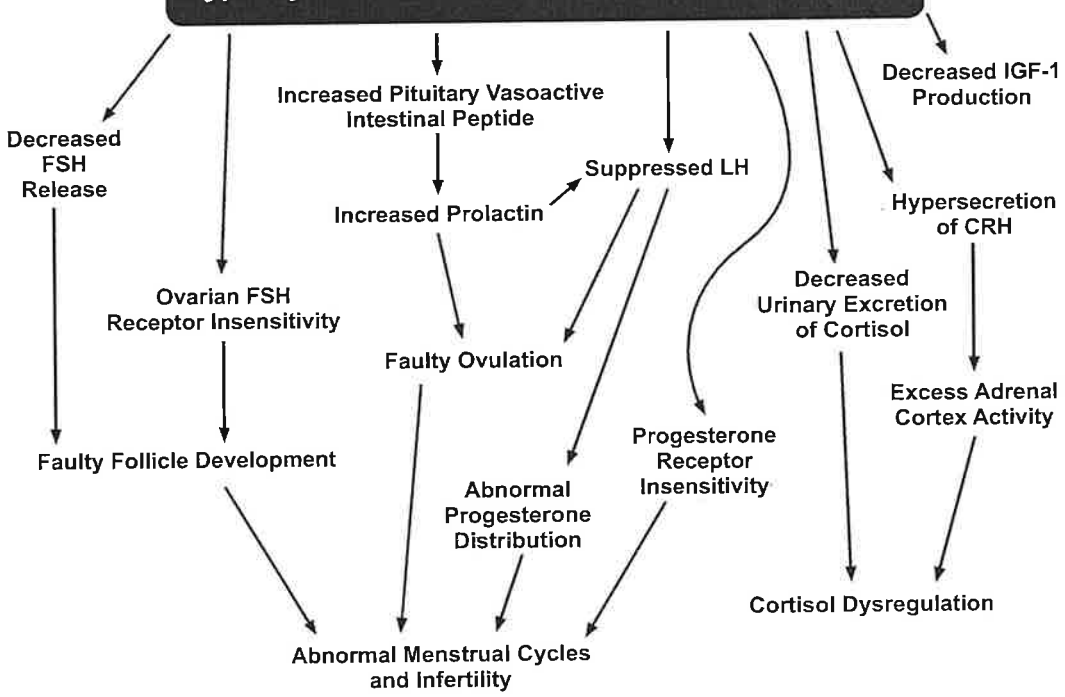



## Hypothyroidism and Male Hormone Influences




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Hypothyroidism and Female Hormone Influences




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

**Low Thyroid Status**

↓

**Reduced Sex Hormone-Binding Levels**

↓

**Increased Free-Fraction Hormones**
- **Estrogen**
- **Testosterone**

↓

**Receptor-Site Resistance**

↓

**Symptoms of Reproductive Hormone Imbalance**



## Hypothyroidism and Estrogen Metabolite Metabolism

Hypothyroidism

↓

Diminished thyroid hormone influence on P450 1A1 and 1A2 cytochrome P450 enzymes

↓

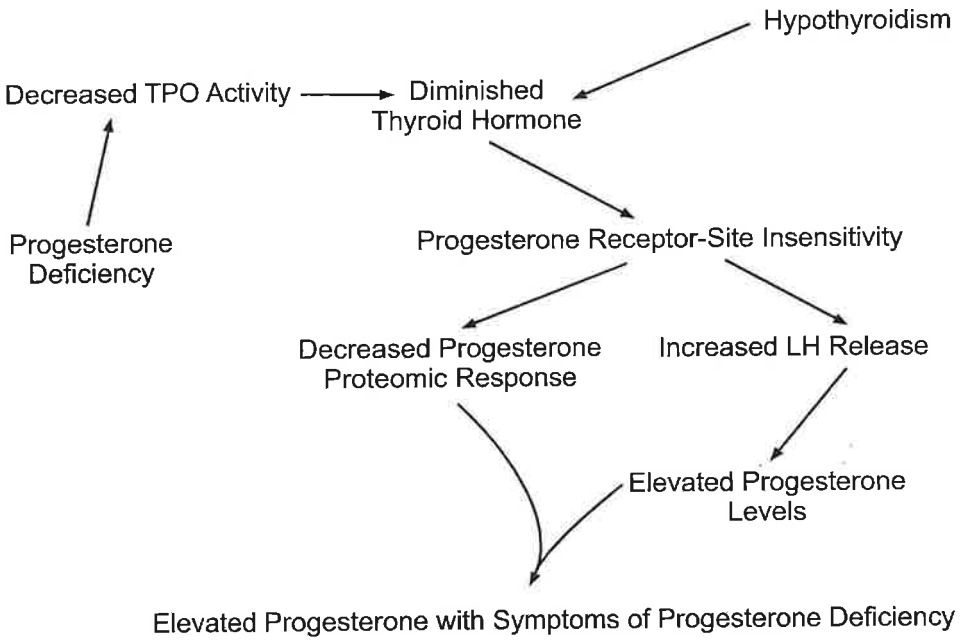
Decreased 2:16 estrogen/metabolite ratio

↓

Promoted estrogenic tissue proliferation

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Progesterone-Thyroid Interplay

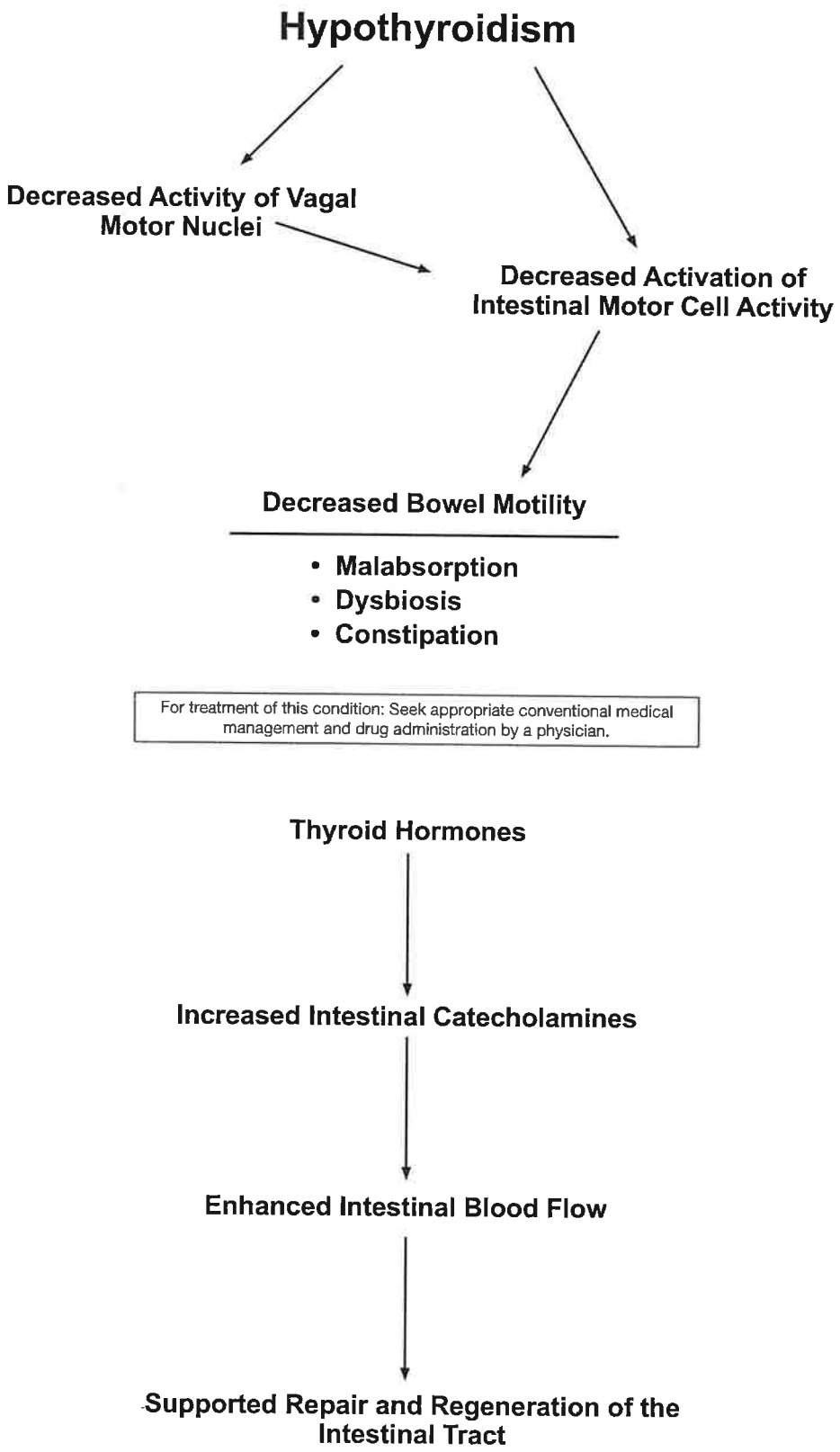



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Hypothyroidism

- **Microarchitecture**
- **Accumulated microscopic damage**
- **Collagen quality**
- **Mineral crystal size**
- **Bone turnover**

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.


Hypothyroidism
Decreased Activity of Vagal Motor Nuclei
Decreased Activation of Intestinal Motor Cell Activity
Decreased Bowel Motility
• Malabsorption
• Dysbiosis
• Constipation
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.
Thyroid Hormones
Increased Intestinal Catecholamines
Enhanced Intestinal Blood Flow
Supported Repair and Regeneration of the Intestinal Tract

Low Thyroid Status

↓

Maturation of Phase II Conjugating Enzyme Pathways Is Hindered

↓

Impaired Phase II Detoxification



## Thyroid-Cardiovascular Health

Low Thyroid Status in Vascular Endothelium

↓ Activity of eNos

↓ Activity of nNos

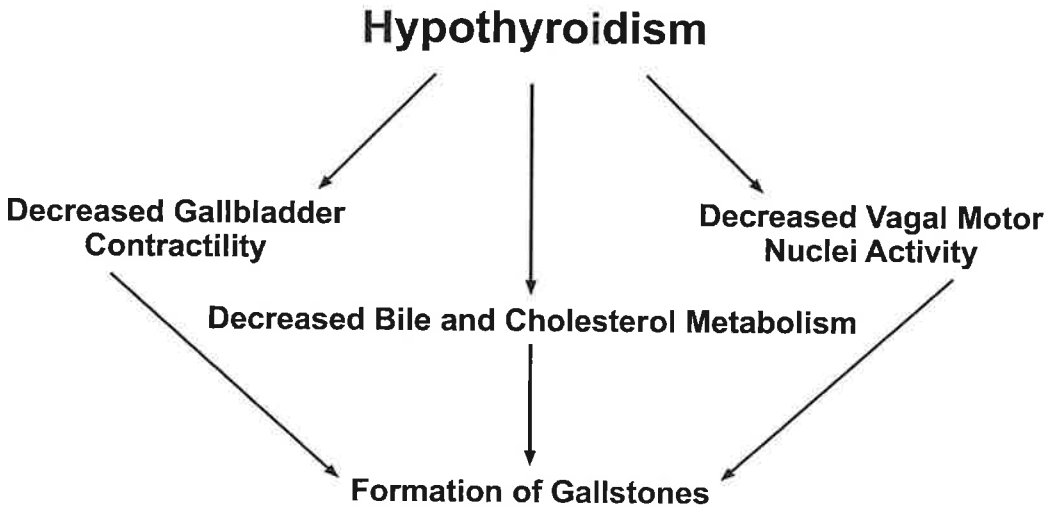
↑ Activity of iNos

Homocysteine

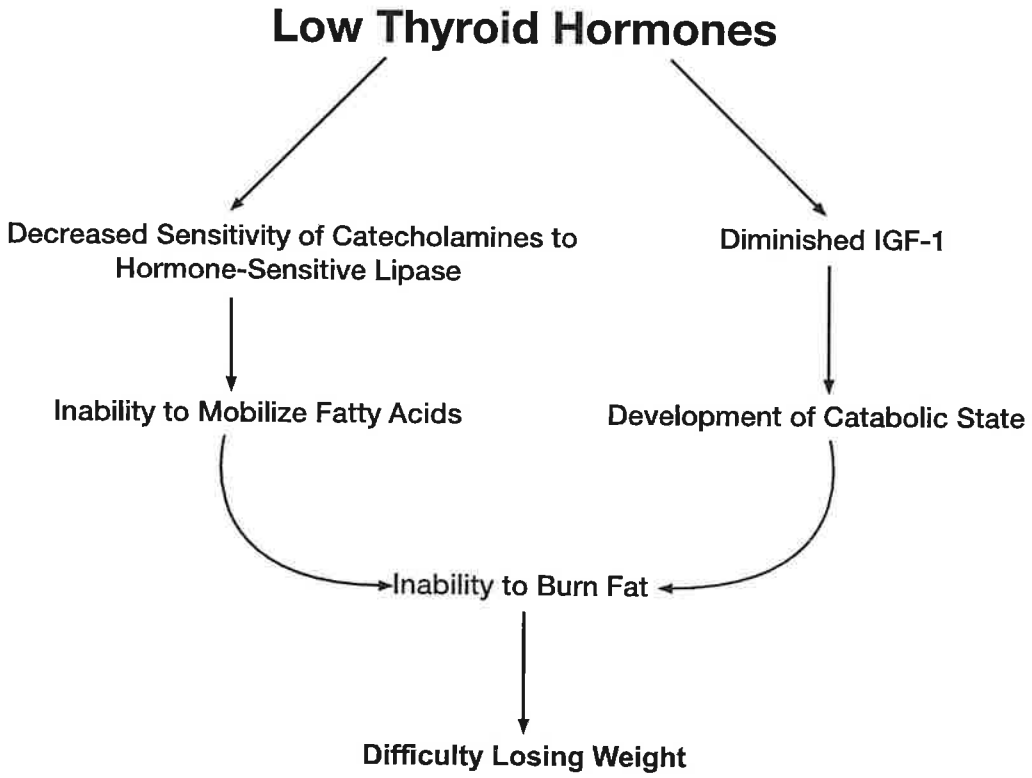
Severe Vascular Endothelial Destruction

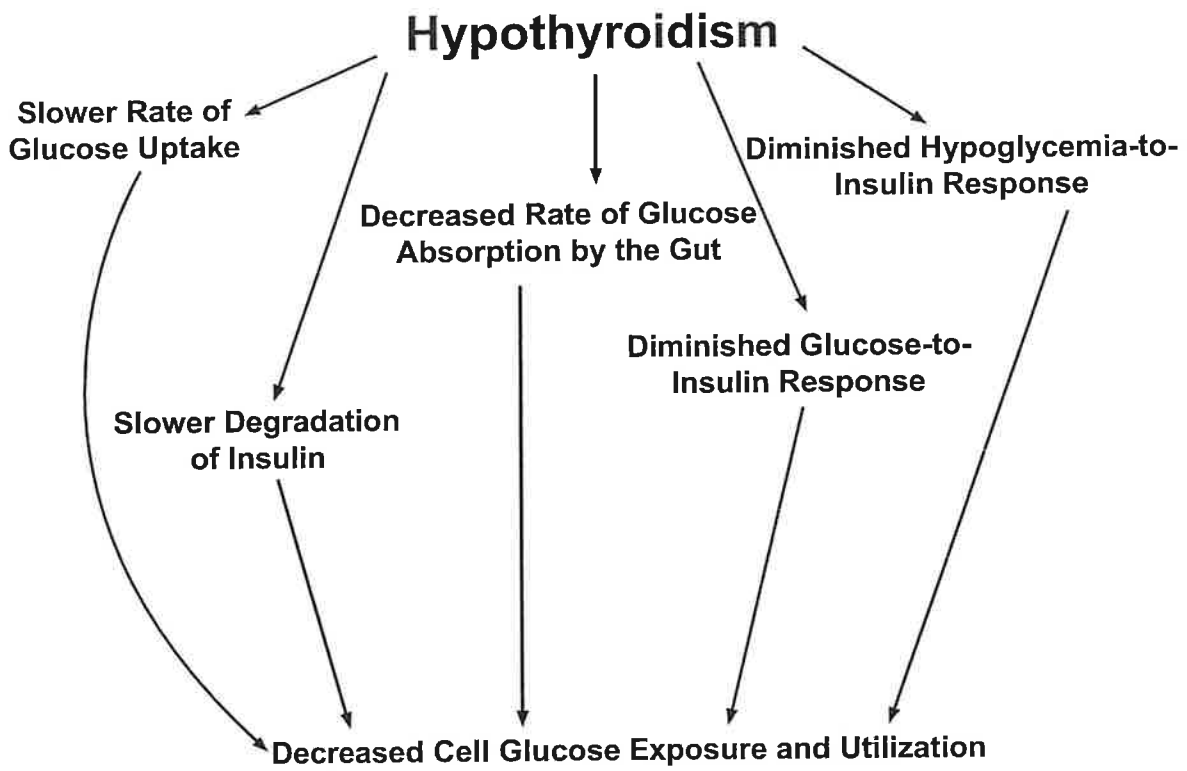
↓

Arterial Plaquing

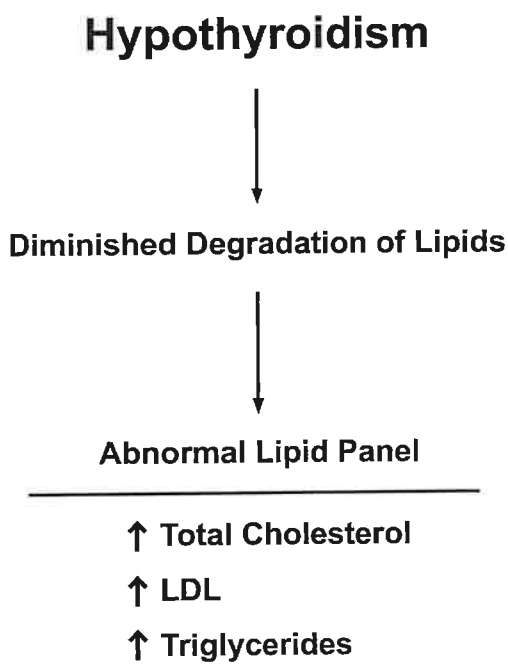



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

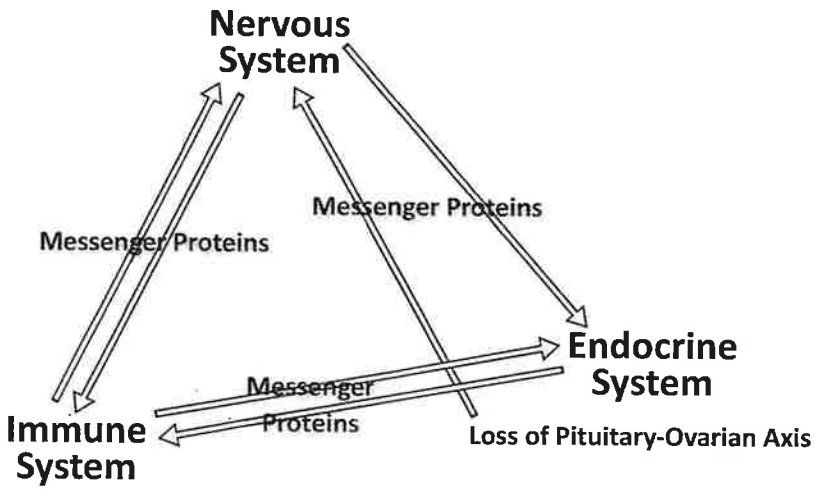





For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



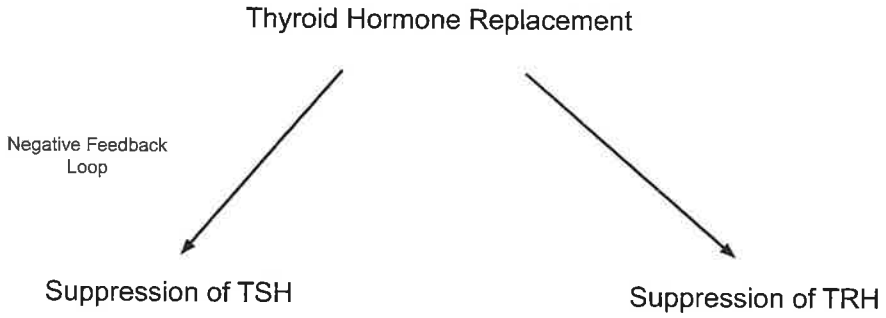

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

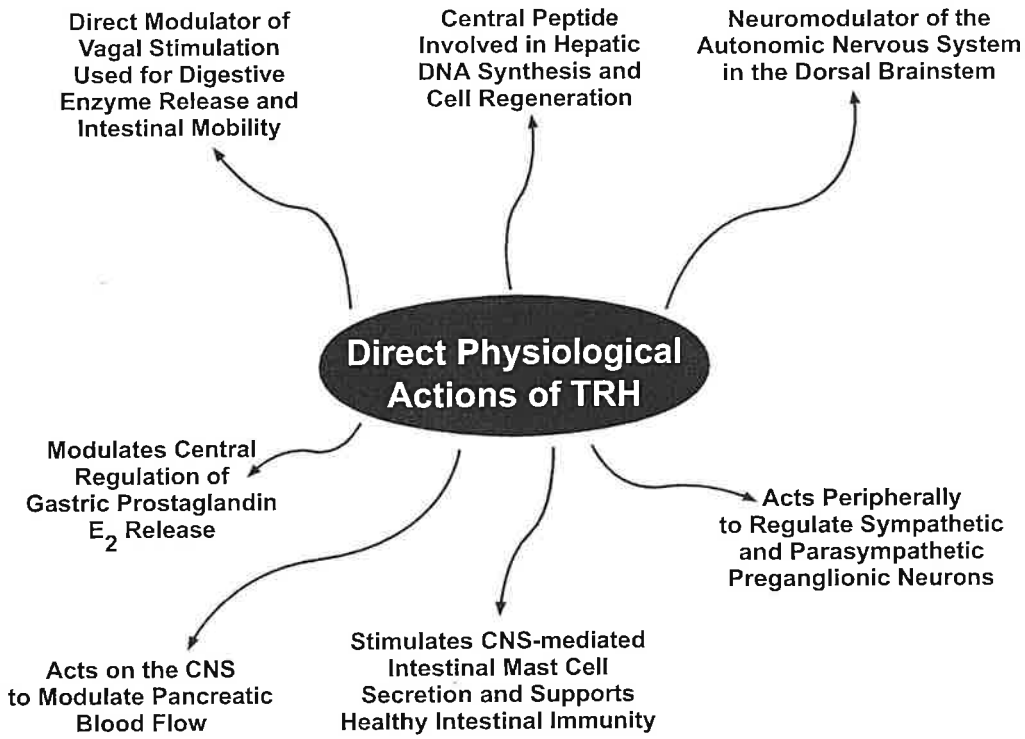


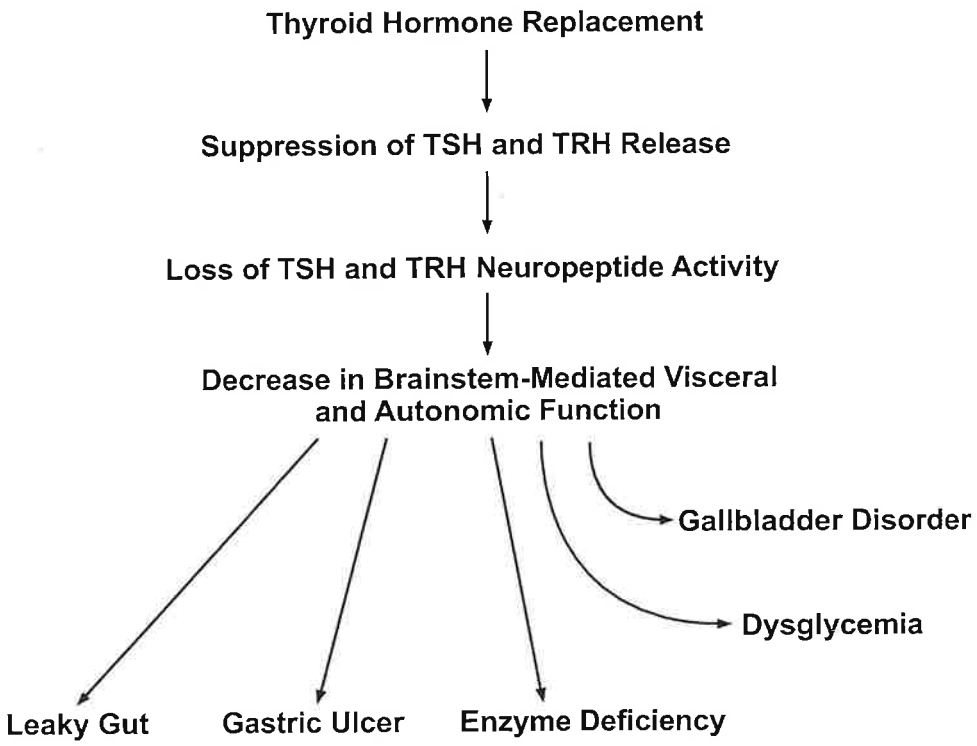



# Negative Feedback Loop

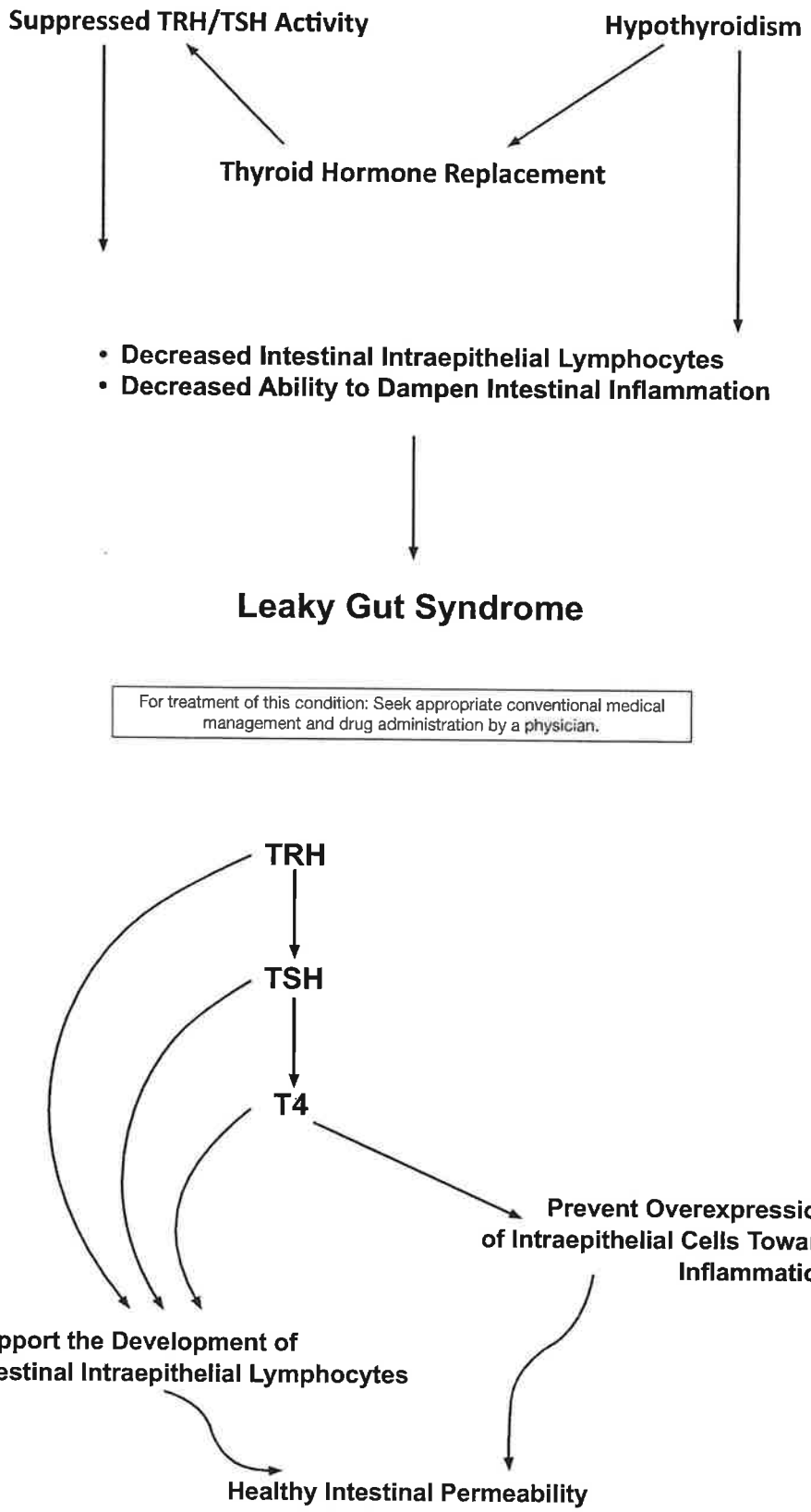
Thyroid Replacement with TSH and TRH



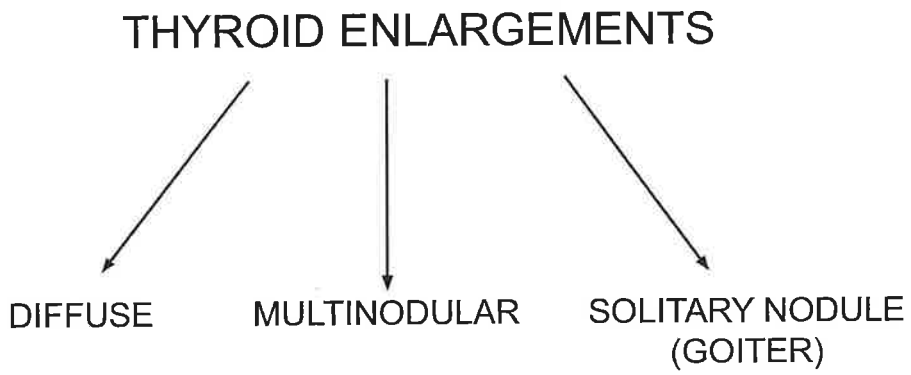



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

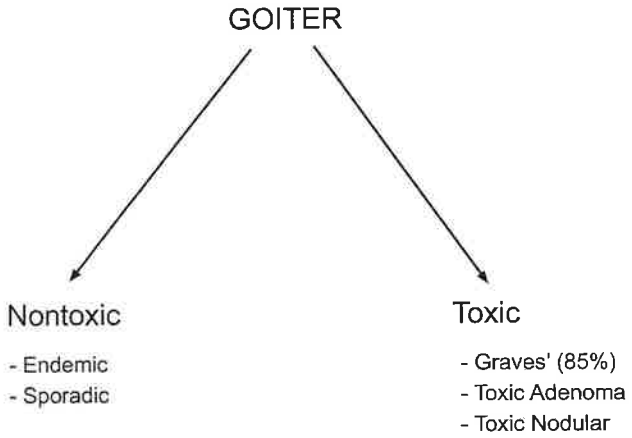

Suppressed TRH/TSH Activity
Hypothyroidism
Thyroid Hormone Replacement
• Decreased Intestinal Intraepithelial Lymphocytes
• Decreased Ability to Dampen Intestinal Inflammation
Leaky Gut Syndrome
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.
TRH
TSH
T4
Prevent Overexpression of Intraepithelial Cells Toward Inflammation
Support the Development of Intestinal Intraepithelial Lymphocytes
Healthy Intestinal Permeability

# Three Classifications of Enlargements

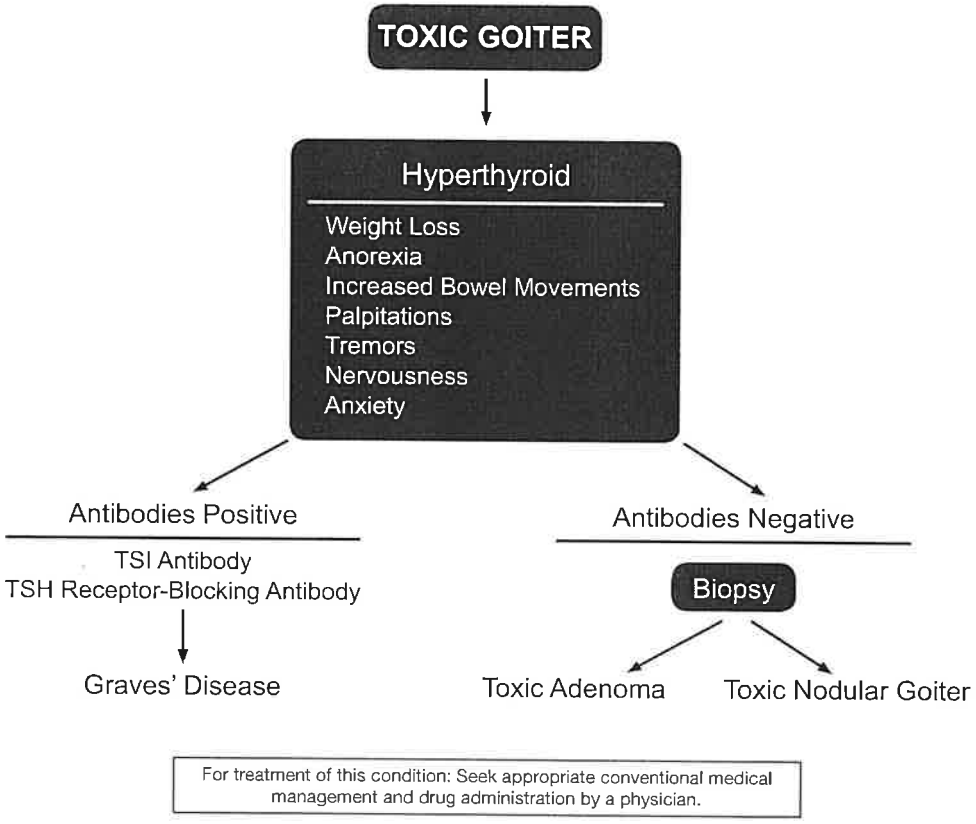


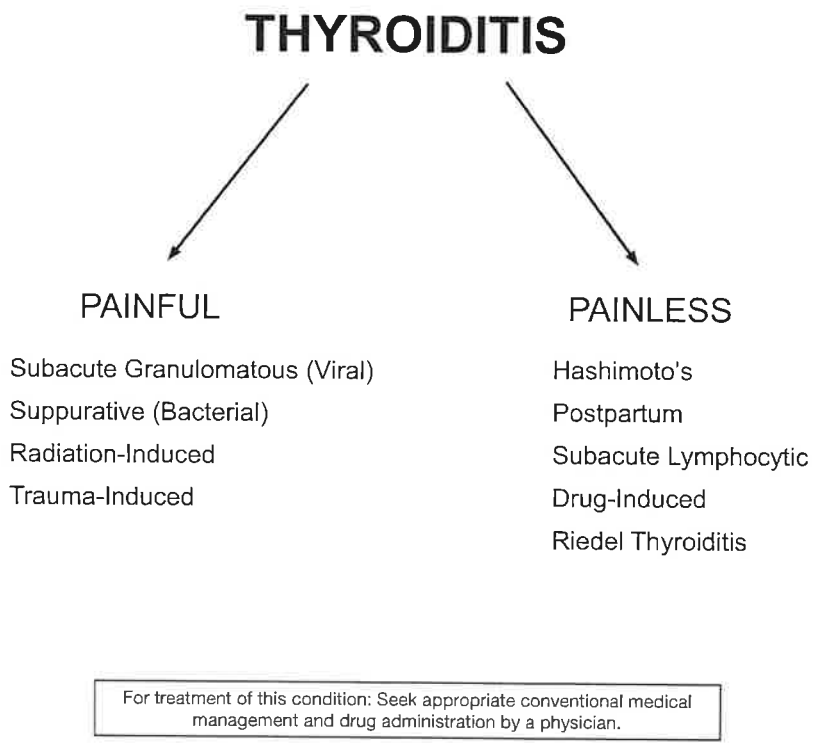


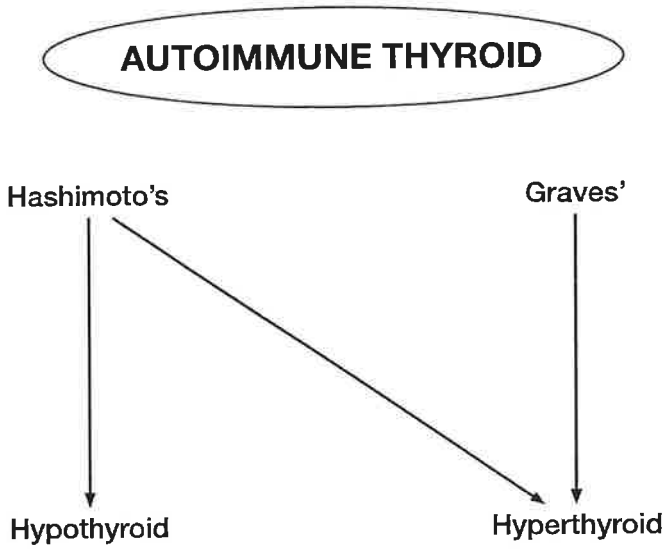

# Diffuse Enlargement




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

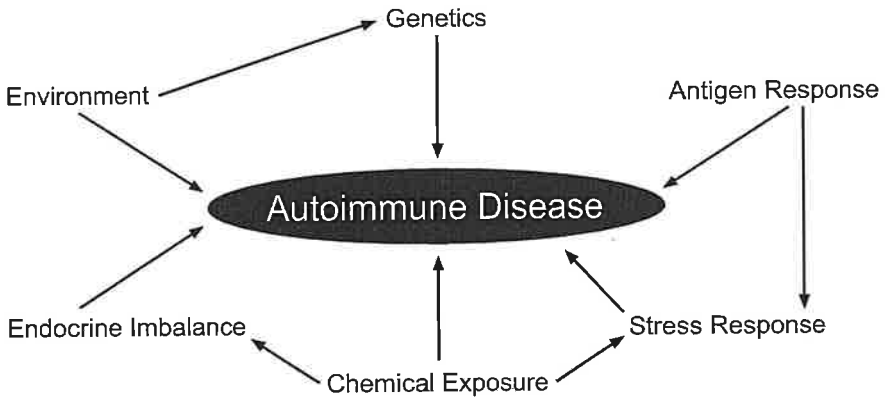

TOXIC GOITER
Hyperthyroid
Weight Loss
Anorexia
Increased Bowel Movements
Palpitations
Tremors
Nervousness
Anxiety
Antibodies Positive
TSI Antibody
TSH Receptor-Blocking Antibody
Graves' Disease
Antibodies Negative
Biopsy
Toxic Adenoma
Toxic Nodular Goiter
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.


THYROIDITIS
PAINFUL
Subacute Granulomatous (Viral)
Suppurative (Bacterial)
Radiation-Induced
Trauma-Induced
PAINLESS
Hashimoto's
Postpartum
Subacute Lymphocytic
Drug-Induced
Riedel Thyroiditis
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

# Mechanism of Autoimmune Disease




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

# Hashimoto's Process

Genotypic Susceptibility

↓

Phenotypic Expressions for Hashimoto's

↓

Immune System Slowly Attacks Thyroid Over Several Years

↓

Loss of Thyroid Tissue, Leading to Decreased Thyroid Production and Elevation of TSH

↓

Diagnosis of Hypothyroidism

↓

TSH Managed by Replacement, but No Management for Immune Response, Unless Hyperactive Thyroiditis

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



## Hashimoto's Polyendocrine-Autoimmune Pattern

→ Antibodies to TPO → Thyroid Gland Destruction

→ Antibodies to Transglutaminase → Celiac Disease

→ Antibodies to the Cerebellum → Neurological Disorder

→ Antibodies to Intrinsic Factor → Pernicious Anemia

→ Antibodies to Glutamic Acid Decarboxylase

↘ Islet Cell Dysfunction

↘ Type 1 Diabetes

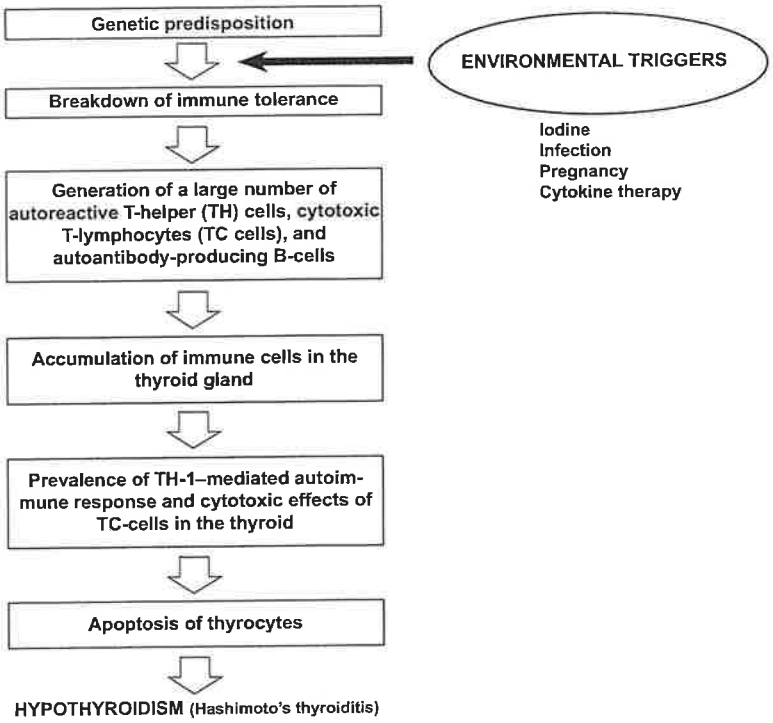
↓

Inability to Convert Glutamate to GABA

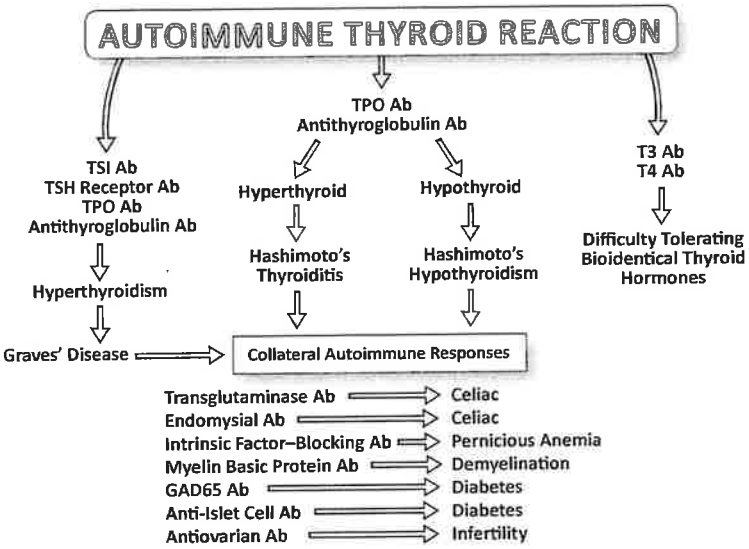
↓

Anxiety, Panic, Ataxia, Limbic Windup

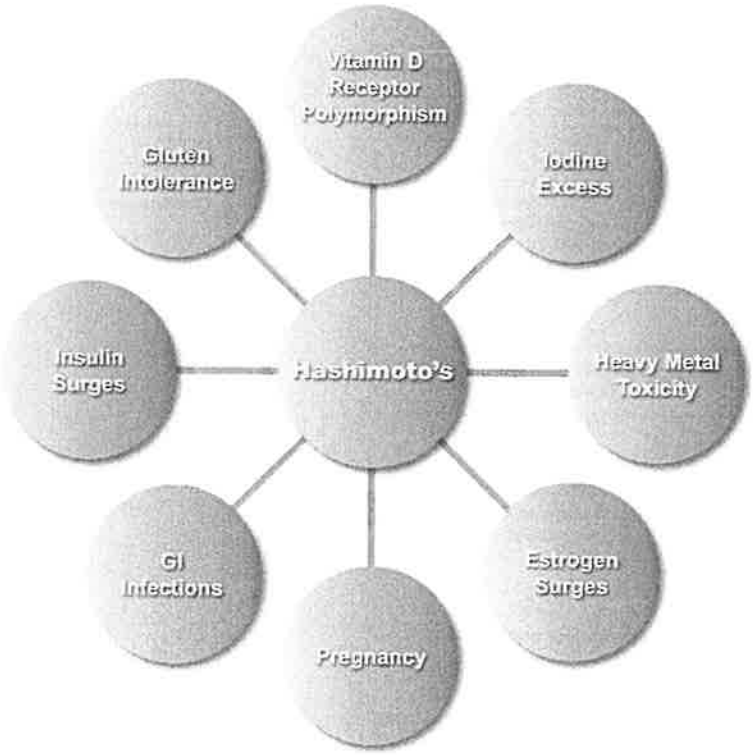
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

## Two Mechanisms for Insulin Surges

**INSULIN RESISTANCE**

- Feels fatigue after meals
- Craves sugar after meals
- Has difficulty falling asleep

**HYPOGLYCEMIC**

- Feels energetic after meals
- Craves sugar before meals
- Has inability to stay asleep
- Crashes and craves sweets between 3:00 p.m. and 5:00 p.m.

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

# Perimenopause and Autoimmunity

Loss of Pituitary-Ovarian Feedback Loop

↓

Surges of Estrogens

↓

Phenotypic Expression of Hashimoto's

↓

Increased Attack of Thyroid Tissue

↓

Abnormal Release of Thyroid Hormones, Increasing Metabolic Rate

↓

Symptoms of Hot Flashes, Insomnia, and Irritability

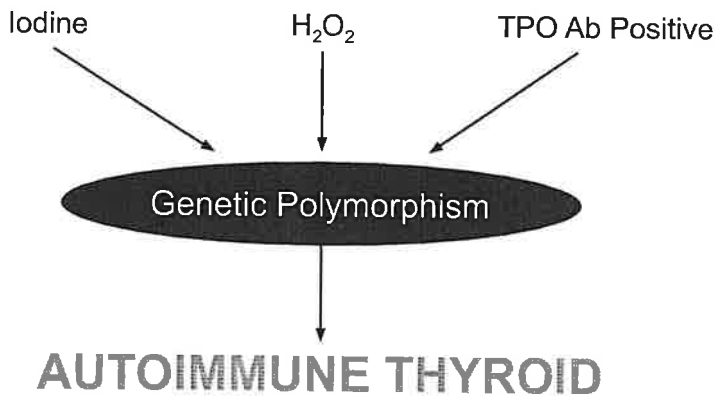
For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

# Pregnancy and Autoimmunity

Third Trimester → TH-2 Dominance

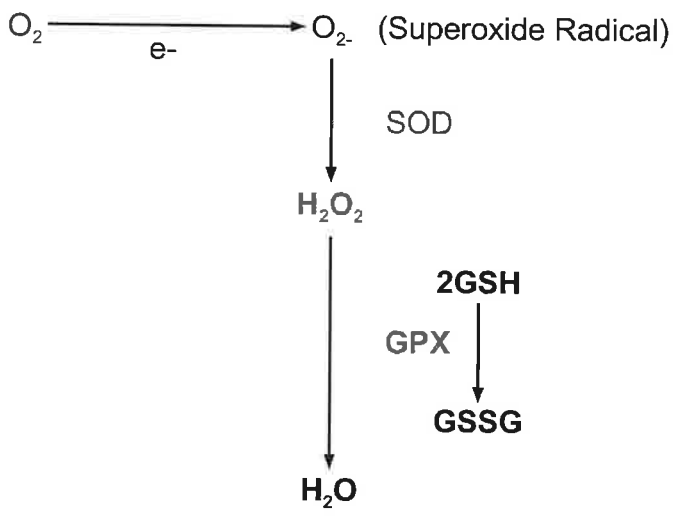
Postpartum → TH-1 Dominance

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

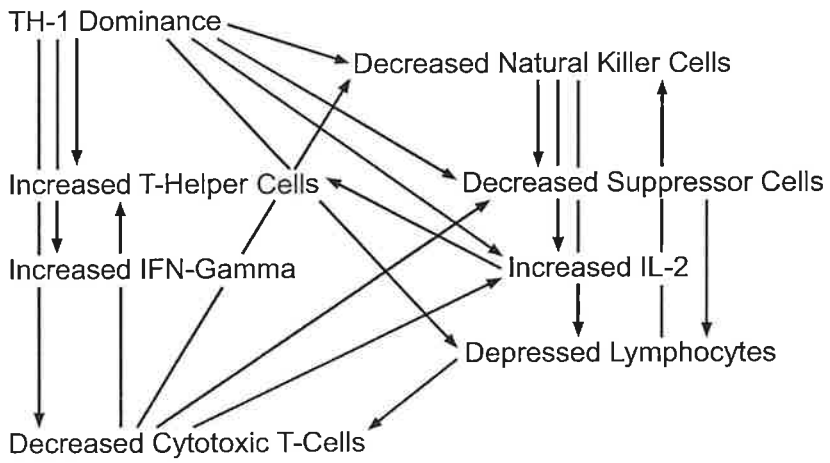



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

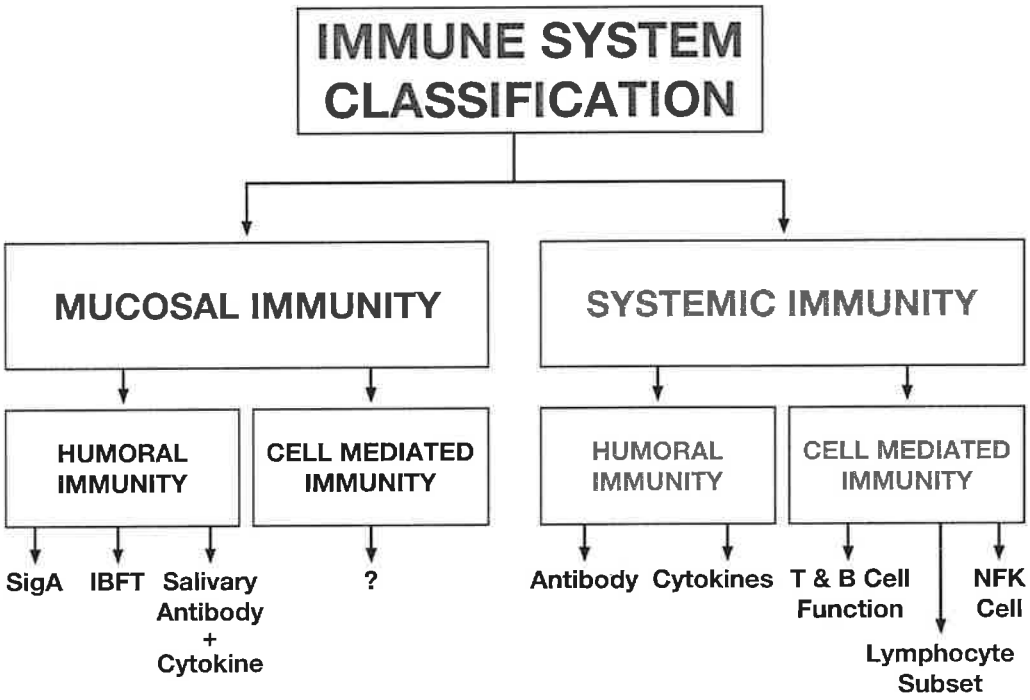
## SOD and GSH Free Radical Quenching



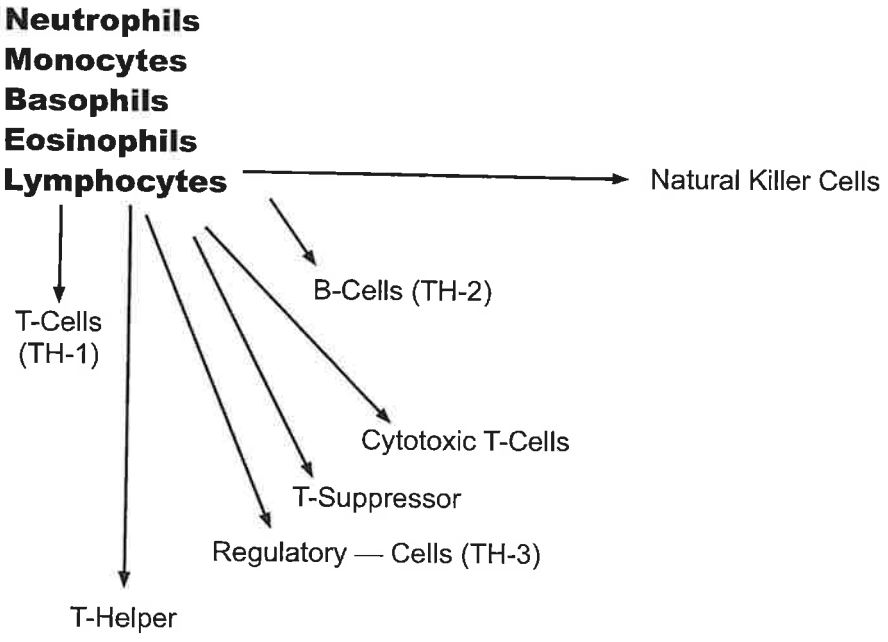
# Hashimoto's



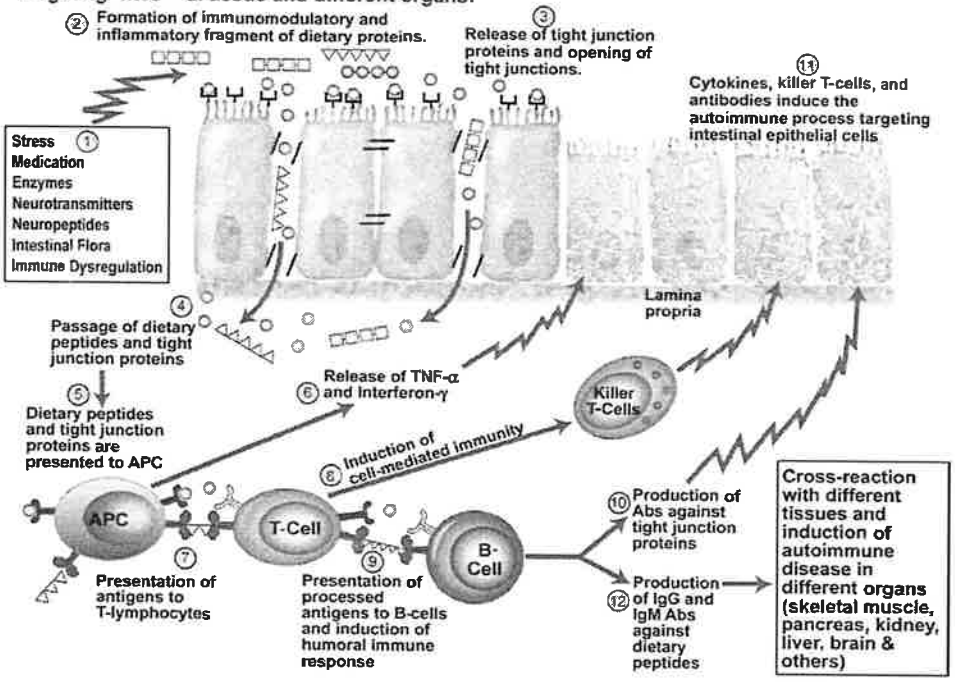

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



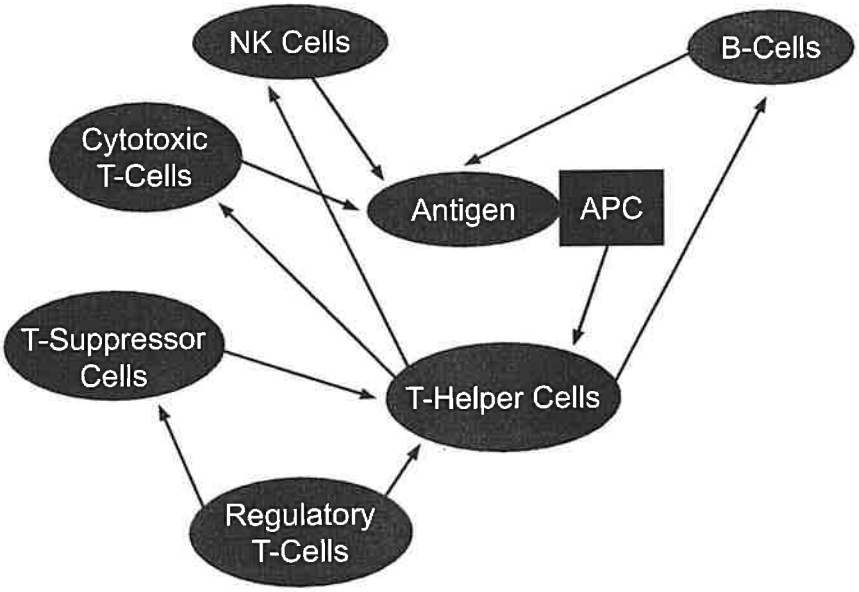
# White Blood Cells Organizations

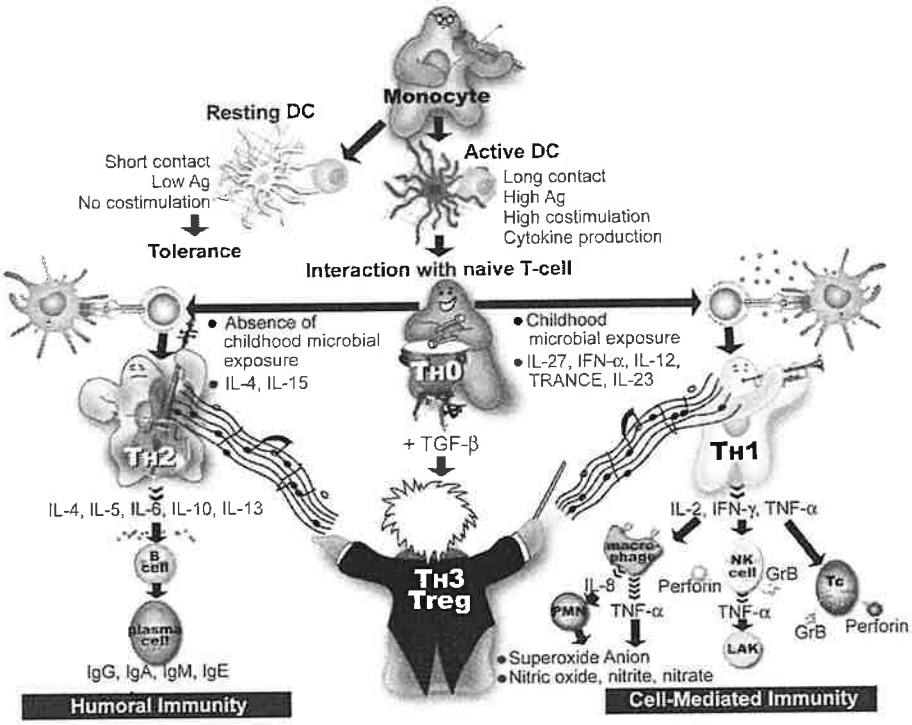



Proposed role of abnormal intestinal permeability in the pathogenesis of autoimmune disease targeting intestinal tissue and different organs.




*Used with permission of Aristo Vojdani, PhD, MSc, CLS and Cyrex Laboratories, LLC.*




*Used with permission of Arista Vojdani, PhD, MSc, CLS and Cyrex Laboratories, LLC.*

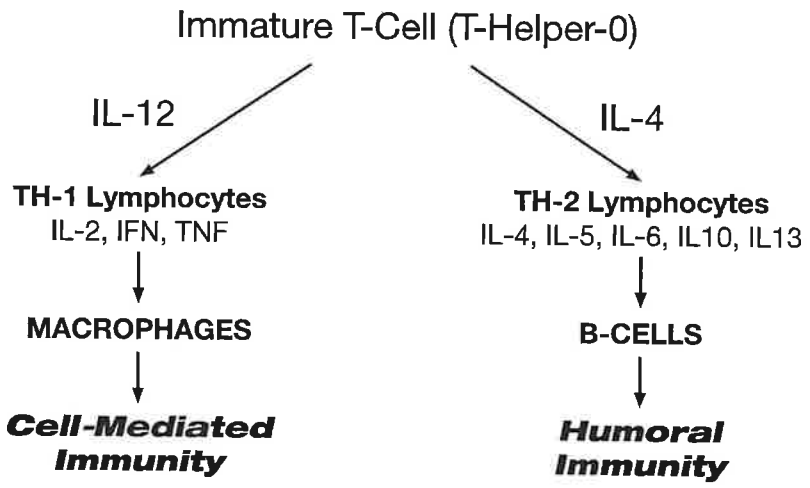
# Infectious Agents Associated with Thyroid Disease Mechanism:

## Cross-reaction between thyroid-stimulating hormone receptors and infectious agents

- Rubella
- Rubeola
- EBV
- Retrovirus
- Influenza B virus
- Coxsackievirus
- *Yersinia*

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

# Development of T-Helper Cells



# TH-1 Cytokines and Thyroid Response

Hashimoto's TH-1 Response

↓

Increased IL-1, IL-2, and TNF-Alpha

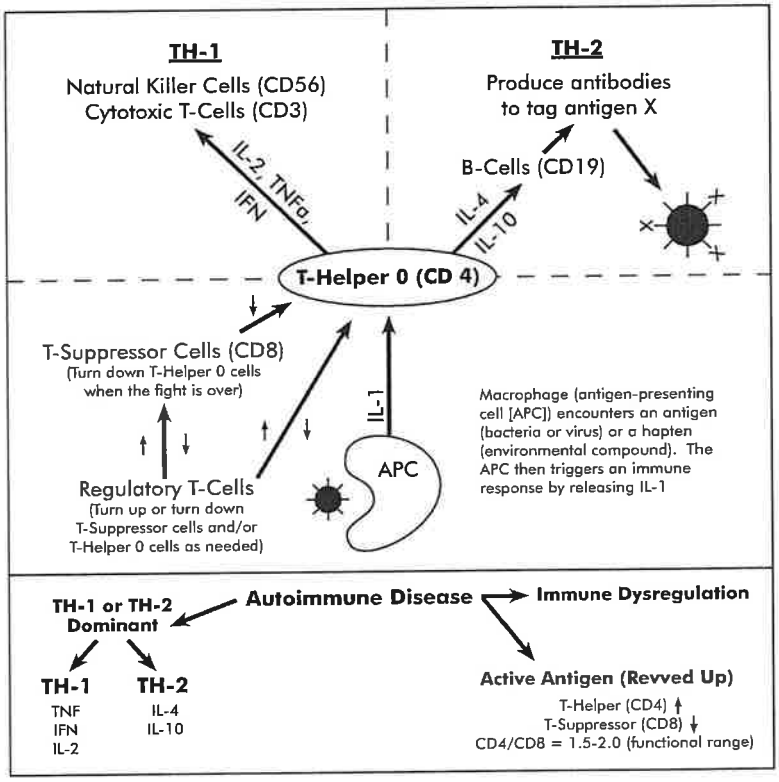
↓

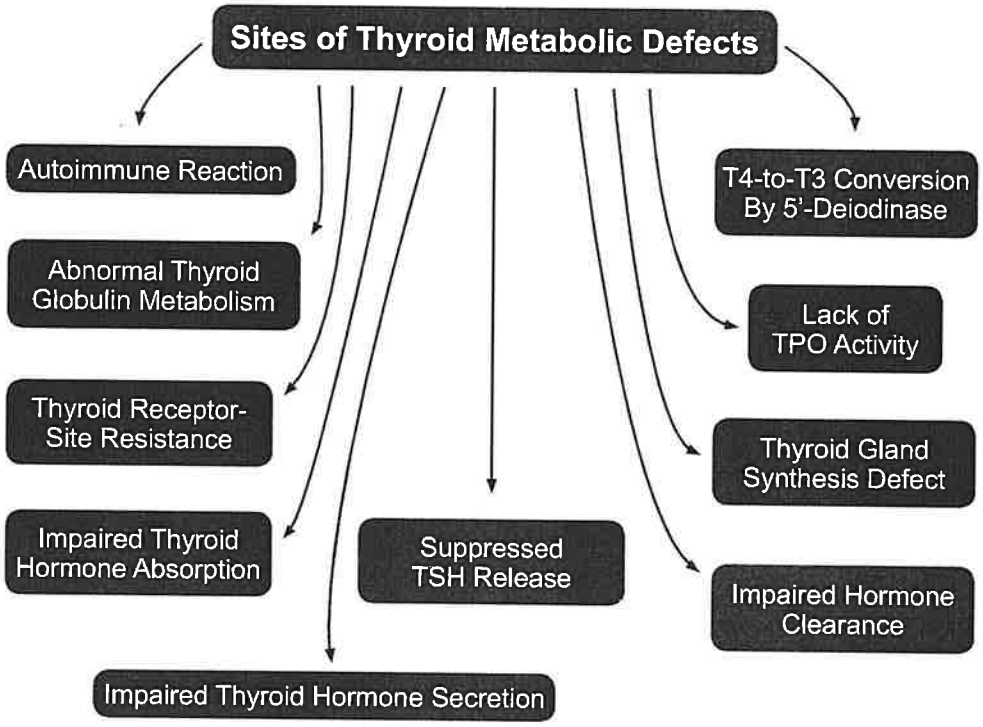
Diminished Thyroid Receptor-Site Expression

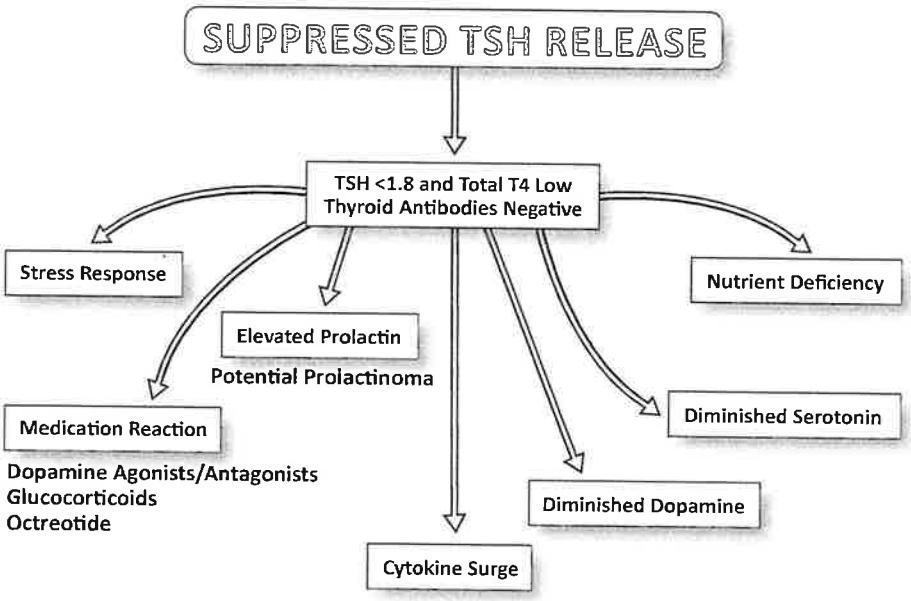
↓

**Hyperthyroid Symptoms Despite Normal TSH**

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.






For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Reading labs

1) go to Antibodies first

1) Autoimmune Rxn — Hashimotos Reactivity

2) Production problem yes or no

3) Is Conversion problem

T4 – T3 blocked

4) Binding protein problems

Hashimotos — TBG ↓

Free Fraction ↑↑

– dysgeusia

↑ Antibodies — ↓ TBG

↑ FSH

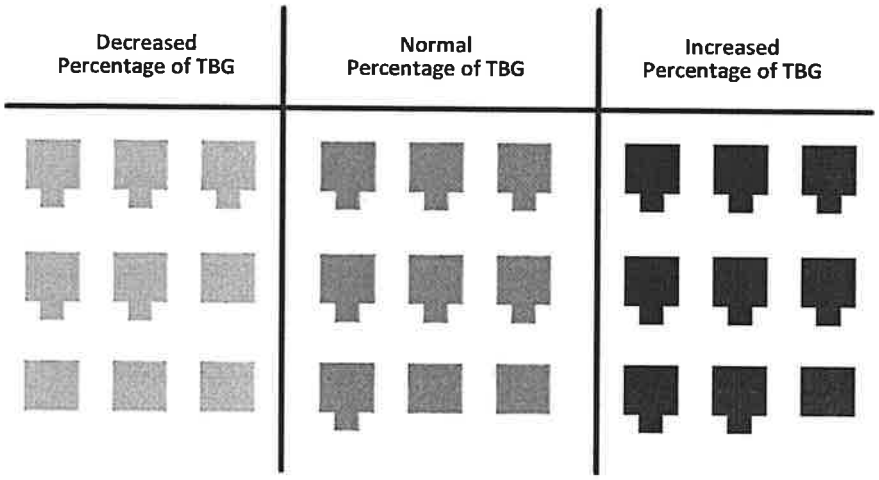
– pbs

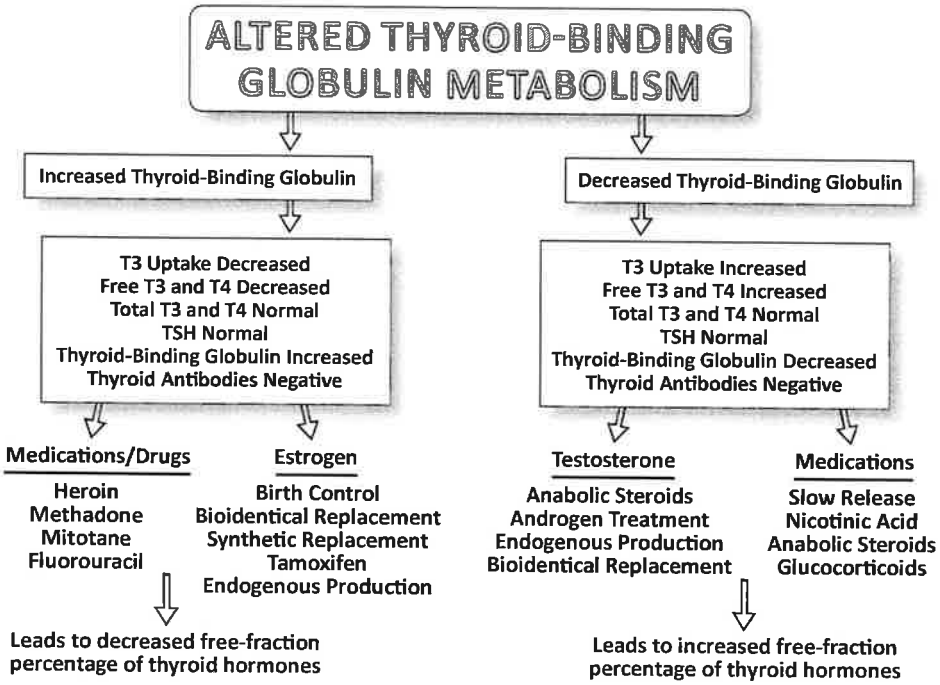
– dysgeusia

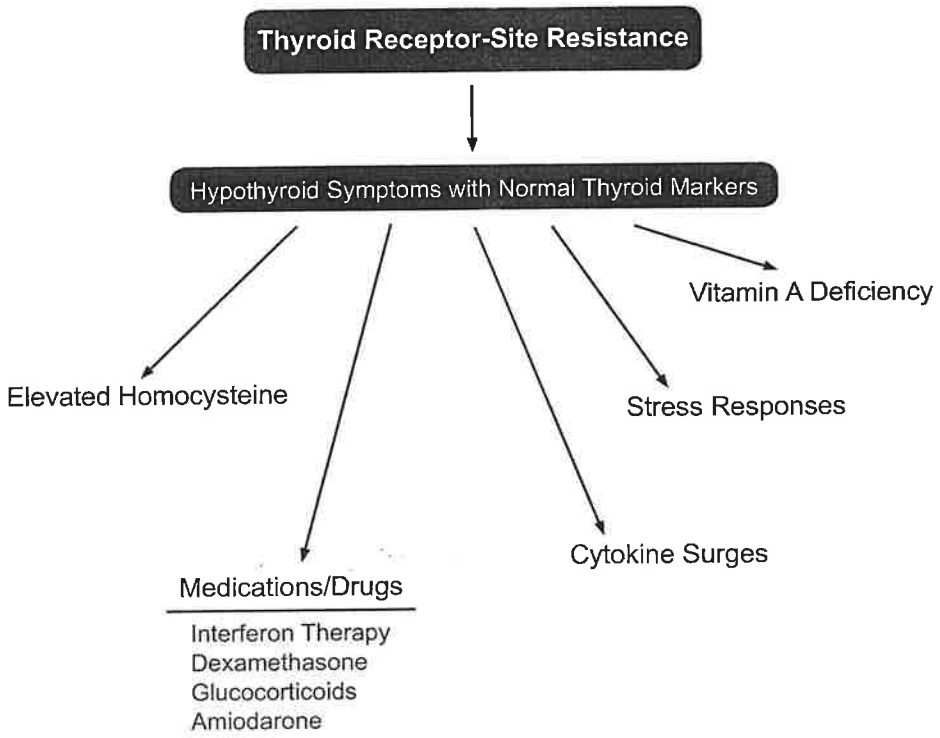
NO cytokines or insulin

Deficiency
ncy
sm
SE
d Serotonin
d Dopamine
Response
MARRIOTT

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.


Decreased Percentage of TBG
Normal Percentage of TBG
Increased Percentage of TBG


ALTERED THYROID-BINDING GLOBULIN METABOLISM
Increased Thyroid-Binding Globulin
T3 Uptake Decreased
Free T3 and T4 Decreased
Total T3 and T4 Normal
TSH Normal
Thyroid-Binding Globulin Increased
Thyroid Antibodies Negative
Medications/Drugs
Heroin
Methadone
Mitotane
Fluorouracil
Estrogen
Birth Control
Bioidentical Replacement
Synthetic Replacement
Tamoxifen
Endogenous Production
Leads to decreased free-fraction percentage of thyroid hormones
Decreased Thyroid-Binding Globulin
T3 Uptake Increased
Free T3 and T4 Increased
Total T3 and T4 Normal
TSH Normal
Thyroid-Binding Globulin Decreased
Thyroid Antibodies Negative
Testosterone
Anabolic Steroids
Androgen Treatment
Endogenous Production
Bioidentical Replacement
Medications
Slow Release
Nicotinic Acid
Anabolic Steroids
Glucocorticoids
Leads to increased free-fraction percentage of thyroid hormones




For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.